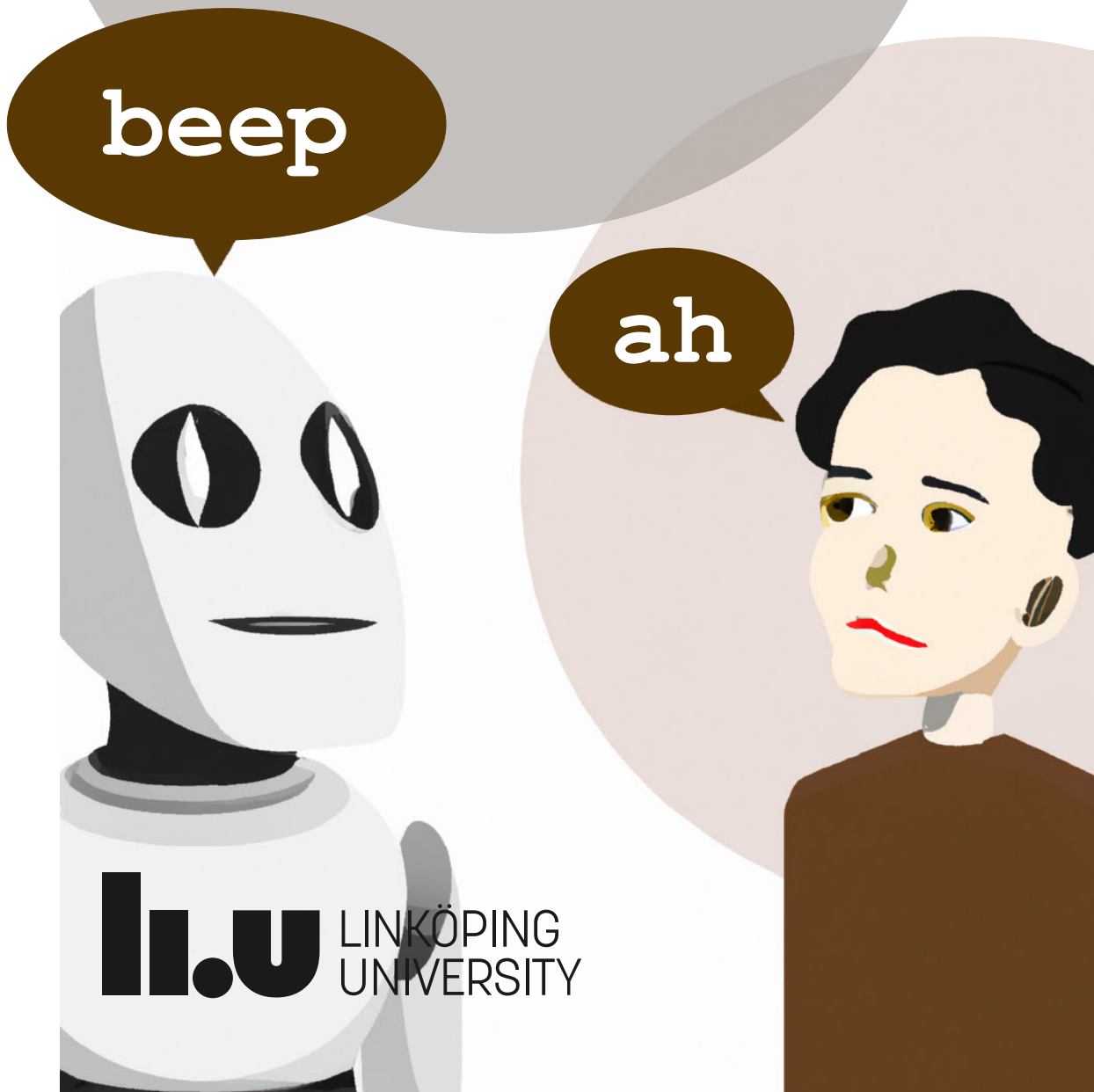
Linköping Studies in Arts and Sciences No. 853
Studies in Language and Culture No. 36

Robot Sound in Interaction

Analyzing and Designing Sound for Human-Robot Coordination

Hannah Pelikan



LINKÖPING UNIVERSITY

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Linköping 2023

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beeep ... beeep beeep beeep ..... beeep beeep ... beeep bip beeep beeep .....
beeep bip bip bip ... bip ... bip beeep bip bip ... beeep beeep beeep ... bip
bip bip beeep ... bip ... beeep bip bip ..... bip bip beeep bip ... bip beeep ...
beeep beeep ... bip bip ... bip beeep bip bip ... beeep bip beeep beeep

POPULÄRVETENSKAPLIG SAMMANFATTNING

Ljud har en viktig roll i det mänskliga samspelet. Förutom att vi talar och lyssnar producerar vi en mängd olika ljud som kan anses ligga i språkets utkant: vi suckar och grymtar, vi delar sinnesintryck och känslor, exempelvis genom att säga “mmh” när vi vill meddela en kompis att maten smakar riktigt gott eller uppmuntra barnen att äta grönsaker. Förutom rösten använder vi oss även av andra kroppsdelar eller instrument: Vi klappar eller tutar för att synkronisera rörelser under en danslektion eller koordinera trafiken. I denna avhandling utforskar jag hur insikter om mänsklig samordning kan bidra till att designa robotljud som är förståeliga för människor. Metodologiskt kombinerar jag etnometodologi och multimodal interaktionsanalys med interaktionsdesign och använder mig därmed av videoinspelade interaktioner både som forskningsdata och designmaterial.

För att förstå hur människor tolkar befintliga robotljud spelade jag in leksaksroboten Cozmo hemma hos barnfamiljer och två autonoma bussar i trafik i stadsmiljö. Tidigare forskning har försökt verifiera att ett ljud verkligen förmedlar den betydelse som designern har avsett, oftast genom enkätfrågor. Jag studerade istället hur människor tolkar robotljud i realtid genom att närstudera videoinspelningar som bl.a. analyserades med hjälp av detaljerade multimodala transkriptioner. Jag visar att människor inte automatiskt behandlar robotljud som meningsfulla. Vad en robot gör, inklusive ljudet den spelar, tolkas i ett sammanhang av tidigare handlingar – och detta skapar i sin tur förväntningar på och möjligheter för nästkommande handlingar. Cozmos mekaniska ljud, vars design inspirerats av roterande kuggar, behandlas till exempel som en indikator på att roboten arbetar – och inte är trasig – när den inte omedelbart slutför en uppgift. Människan kan då understödja robotens arbete genom att vänta tills den är klar. Pling-ljud på en autonom buss kan tolkas som en uppmaning att flytta sig ur vägen. I avhandlingens olika studier visar jag hur robotljud kan uppfattas som sociala ifall de spelas vid en tidpunkt då de upplevs som relevanta för den pågående interaktionen.

Vidare har jag tillämpat ovanstående resultat om robotljudets roll för koordination med människor i två designstudier. Jag undersökte hur en robotprototyp ämnad för grupparbete kan samarbeta på ett minimalt sätt och utvecklade ljud för olika trafiksituationer till de autonoma bussarna. Jag visar att den exakta tidpunkten för produktionen av ett ljud är minst lika viktig som hur ljudet låter. Designers bör utforska när ett ljud över huvud taget är relevant i ett interaktionsflöde innan de bestämmer sig för hur det ska låta. Mitt arbete belyser även hur ljud hör ihop med rörelse. Designers kan med fördel använda sina egna kroppar för att utveckla ljud, och ska då vara uppmärksamma på hur röst, gester och ansiktsuttryck används tillsammans för att kommunicera. Jag visar också hur videoinspelningar av naturligt förekommande interaktioner kan användas för att designa ljud i konkreta interaktionssekvenser. Designers kan dra nytta av denna metod för att identifiera vilka specifika handlingar en robot behöver kunna utföra i ett visst sammanhang och för att utforma robotljud som är anpassade för interaktion i realtid.

ABSTRACT

Robots naturally emit sound, but we still know little about how sound can serve as an interface that makes a robot's behavior explainable to humans. This dissertation draws on insights about human practices for coordinating bodily activities through sound, investigating how they could inform robot design. My work builds on three video corpora, involving i) a Cozmo robot in ten family homes, ii) autonomous public shuttle buses in an urban environment, and iii) a teamwork robot prototype controlled by a researcher and interacting with study participants in an experimental setting. I approached the data from two methodological angles, exploring how they can speak to each other: I first carried out an empirical analysis of the video data from an Ethnomethodology and Conversation Analysis (EMCA) perspective, focusing on how humans make sense of robot sound on a moment-by-moment basis in naturally occurring interaction. Subsequently, taking an Interaction Design perspective, I used my video recordings as a design material for exploring how robot sound could be designed in and for real-time interaction. My work contributes to Human-Robot Interaction through detailed studies of robots in the world (rather than in the lab), focusing on how participants make sense of robot sounds. I present a novel framework for designing sound in and for interaction and a prototyping practice that allows practitioners to embed an EMCA stance into their designs. The dissertation contributes to EMCA by describing how members embed autonomous machines into the social organization of activities and how humans treat robots as participants in the interaction. I make a contribution to the development of EMCA hybrid studies by seeking a synthesis between EMCA and robot interaction design.

Acknowledgments

Many of my ideas emerged in dialogue with others, and I am incredibly thankful for the many formal and informal conversations that have contributed to this work. Most importantly, I would like to thank my supervisors for embarking on this journey with me. Leelo Keevallik, I cannot thank you enough for your wise guidance, and for leaving me so much freedom to explore within your project. Your curiosity and leadership continue to inspire me. Mathias Broth, you have sparked my initial interest in interaction analysis, and I am so grateful to have you as a co-supervisor. Malte Jung, while not formally registered as a supervisor, you have been an important mentor far beyond the months that I spent in your lab. I have tremendously benefited from the discussions with my opponents Stuart Reeves and Karola Pitsch, and readers Brian Due, Roberto Bresin, Malte Jung and Jakob Cromdal at the PhD seminars. Thank you for encouraging me to explore new avenues and pushing me to better articulate my ideas.

To my amazing colleagues in Leelo's project: Sally Wiggins, Emily Hofstetter, Adrian Kerrison, Agens Löfgren and Kätlin Aare, thank you for welcoming me to the world of non-lexical vocalizations. I would like to thank the members of the Robots in Groups Lab at Cornell University for their enthusiasm for developing video interaction design, in particular Jenny Fu, Yoyo Tsung-Yu Hou, Qiyu Hu, Andy Ricci, and Asher Lipman. My collaborators on various (side) projects deserve my thanks: Frederic Robinson for joining efforts towards building a Sound in HRI community, Katie Winkle and David Porfirio for exploring better HRI design methods, Sofia Thunberg and Ericka Johnson for our explorations of gender in robotics, and Dimos Kontogiorgos for discussions about dialogue modeling.

Throughout the time of my PhD, several people have shaped my thinking. I would like to thank Kia Höök for encouraging my explorations towards a more holistic view of body and mind. Thank you, Niklas Rönnberg for teaching me important lessons about sound design. My thanks to Nils Dahlbäck for helping me reconcile my identity as a cognitive scientist, and to Sam Thellman, Oscar Bjurling, Franzi Babel and Philipp Hock for insightful discussions about humans and robots. Thanks, Ericka Johnson, for creating teaching and learning opportunities related to AI. Asta Čekaitė, thank you for your tips on recording in family homes. Thanks, Cristina Zaga, for encouraging my engagement with RtD. Thank you, Matti Arvola for pointing me to video literature. Thanks, Ahmet Börütecene for welcoming me to LiU's design community. I benefited considerably from discussions with like-minded EMCA scholars at the EMCAI-network meetings, which I co-organized with Saul Albert, Lynn de Rijk, Kati Cyra and Andreas Liesenfeld, and the AI in Interaction

data session series that I led with Philippe Sormani. I would like to thank the attendees of the SiS data sessions and the Mobsin workshops, as well as the participants of the AI in Society seminar, particularly Katarina Sperling and Dominika Lisy, for stimulating discussions.

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A PhD journey has many ups and downs and I am lucky to have incredibly supportive friends. Thank you, Klervie Toczé, for the writing retreats and for taking me by the hand when I was at my lowest. Arvind Balachandran for always being up for adventures, and for reminding me that a part of me is an engineer. Sofia Thunberg for your honesty, I continue to learn so much from you. Alex Kapeller for the parties and the hikes, and for always bringing food for thought. Héctor Rodriguez Déniz for sharing your PhD wisdom - you were always right in the end. Tabea Schroth for our conversations about financial stability and leadership. Rodrigo Saar de Moraes and Karol Wojtulewicz for discussing artificial intelligence with me. Swati Pandita for being in the right place at the right time. I would also like to thank my fellow board members at LiUPhD for the pub nights and Nobel dinners, and for our joint work for increasing communication between the departments at LiU. Thank you, to the members of the German-Swedish language café for welcoming me in Sweden, in particular Albin Petterson for helping me learn to understand Östgötska. Annerose Willemsen, thank you for making sure I don't forget my Dutch.

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Hannah Pelikan, Linköping, May 2023

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C Consent Form for Bus Study **113**

List of Articles

This dissertation is based on five publications, which are referred to with their Roman numerals throughout the text.

I Pelikan, H. R. M., Broth, M., & Keevallik, L. (2020). "Are you sad, Cozmo?": How humans make sense of a home robot's emotion displays. *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, 461–470. https://doi.org/10.1145/3319502.3374814

II Pelikan, H., & Hofstetter, E. (2022). Managing delays in human-robot interaction [Just Accepted]. *ACM Transactions on Computer-Human Interaction*. https://doi.org/10.1145/3569890

III Pelikan, H. R. M. (2021). Why autonomous driving is so hard: The social dimension of traffic. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 81–85. https://doi.org/10.1145/3434074.3447133

IV Pelikan, H., Fu, J., Hou, Y. T.-Y., Hu, Q., & Jung, M. (submitted). Minimal sound for maximal interaction: Sequential timing of expressive robot behavior. *ACM Transactions on Human-Robot Interaction*

V Pelikan, H. R. M., & Jung, M. F. (2023). Designing robot sound-in-interaction: The case of autonomous public transport shuttle buses. *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, 172–182. https://doi.org/10.1145/3568162.3576979

In the following, I provide details on my contribution on the articles with joint authorship.

For **Article I**, the whole team jointly worked on the analysis. I was responsible for writing the related work and formulating implications. Broth and Keevallik contributed connections to literature on human-human interaction and by streamlining the arguments.

For **Article II**, I contributed the analysis of human-robot interaction and Hofstetter contributed the analysis of human-human interaction. The arguments were developed through close joint discussion. I managed large portions of writing and editing the first drafts and took the main responsibility for a major revision, formulating the framing of the paper as centered on delays, as well as creating a framework with design implications.

For **Article IV**, the team jointly developed the methods, the argument and presentation format. I led the project, prepared data sessions and wrote large portions of the paper. Fu helped prepare the sound design sessions and contributed

related work. Hou participated in the sound design and data sessions, and supported formulating takeaways for designers. Hu contributed through transcription and participation in the data sessions and commented on several drafts. Jung contributed by writing parts of the related work and helped articulate the findings.

For **Article V**, I collected the data, developed designs and evaluated them, and wrote a full first draft of the paper. Jung contributed literature on video for design, helped to develop the argument and design framework, and made minor edits to the draft.

All five articles are appended to the print version of this dissertation. Articles I, II, III, and V are reproduced in line with ACM's policy for reuse: "Authors can include partial or complete papers of their own (and no fee is expected) in a dissertation as long as citations and DOI pointers to the Versions of Record in the ACM Digital Library are included." Article IV has been submitted to *ACM Transactions on Human-Robot Interaction* for consideration. Its contents may not be further disseminated until a final decision regarding publication has been made and further permission has been granted.

Additional Articles

The following articles are not formally included in this dissertation but have been written and published during my time as a PhD student in the project "Vocal practices for coordinating human action" (VR-2016-00827). While not forming the central part of the thesis, they further illustrate my work and are listed here for completeness.

VI Pelikan, H. & Keevallik, L. (submitted). How humans make sense of robot sound in interaction. In R. Savery (Ed.), *Sound in Robotics*.

VII Pelikan, H. (2024). Transcribing human-robot interaction: Methodological implications of participating machines. In P. Haddington, T. Eilittä, A. Kamunen, L. Kohonen-Aho, T. Oittinen, I. Rautiainen, & A. Vatanen (Eds.), *Ethnomethodological Conversation Analysis in Motion: Emerging Methods and New Technologies*. Routledge

VIII Pelikan, H. R. M. (2023). Robot sound-in-interaction. *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, 754–756. https://doi.org/10.1145/3568294.3579975

IX Pelikan, H., Broth, M., & Keevallik, L. (2022). When a robot comes to life: The interactional achievement of agency as a transient phenomenon. *Social Interaction. Video-Based Studies of Human Sociality*, *5*(3). https://doi.org/10.7146/si.v5i3.129915

X Pelikan, H. R. M., Hou, Y. T.-Y., Fu, X. J., Keevallik, L., Broth, M., & Jung, M. F. (2022). Interaction prototyping with video: Bridging video interaction analysis & design. *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, 1–4. https://doi.org/10.1145/3491101.3503765

XI Pelikan, H., Robinson, F. A., Keevallik, L., Velonaki, M., Broth, M., & Bown, O. (2021). Sound in human-robot interaction. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 706–708. https://doi.org/10.1145/3434074.3444871

XII Pelikan, H., Keevallik, L., & Broth, M. (2020). Människa-robot-interaktion. In M. Broth & L. Keevallik (Eds.), *Multimodal Interaktionsanalys* (pp. 395–415). Studentlitteratur

XIII Kontogiorgos, D., & Pelikan, H. R. M. (2020). Towards adaptive and least-collaborative-effort social robots. *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, 311–313. https://doi.org/10.1145/3371382.3378249

XIV Pelikan, H. (2019). "A stubborn child" - How robot sounds are oriented to in everyday situated interaction at home. *Mensch und Computer 2019 Workshopband*. https://doi.org/10.18420/muc2019-ws-659

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List of Abbreviations

CA	Conversation Analysis
CSCW	Computer-Supported Cooperative Work
EMCA	Ethnomethodology and Conversation Analysis
HCI	Human-Computer Interaction
HRI	Human-Robot Interaction
PD	Participatory Design
RtD	Research through Design
TCU	Turn-constructional unit
TRP	Transition-relevance place
WoZ	Wizard-of-Oz

"if we try to imagine it [interaction],
we often imagine things that rarely happen,
or conversely we fail to imagine things
that actually happen all the time"
(Enfield, 2013, p. 1)

"the listened to sound
is recognized and identified as
a sounded doing"
(Garfinkel & Wieder, 1992, p. 197)

"Good design requires good communication,
especially from machine to person,
indicating what actions are possible,
what is happening, and what is about to happen."
(Norman, 2013, p. 8)

1 Introduction

"Da-beep", "drrrrrrrrrrr", "ding!" Robots give notice of their presence even when one is not looking at them (Cha et al., 2018). The humming of servo motors, the rustling of fans, and the cracking of cogs are a natural consequence of their material existence (Frid & Bresin, 2022; Moore et al., 2017; Tennent et al., 2017). Beeps (Read & Belpaeme, 2014) and vocalizations (Park, Gelsomini, Lee, Zhu, et al., 2017) can further increase a robot's expressiveness. This work explores robot sound as a resource for supporting human-robot coordination. I combine moment-by-moment analyses informed by Ethnomethodology and Conversation Analysis (EMCA) with interaction design rooted in Human-Computer Interaction (HCI) and Human-Robot Interaction (HRI).

Robots are machines that can sense their environment and move with a degree of autonomy (Bekey, 2005; International Organization for Standardization [ISO], 2021; Matarić, 2007). They may contribute recognizable actions at relevant moments and can thereby become potential participants in social interaction (c.f. Alač, 2016; Pelikan, 2024; Pitsch, 2016; Tuncer et al., 2023). Designing a robot's behavior to be immediately intelligible for any user is particularly relevant in public spaces and homes, where robots repeatedly meet humans who neither have time nor a professional interest in reading instruction manuals to make the robot work (Ju, 2015; Norman, 2013). Instead, robots need to smoothly coordinate with humans.

Originating from Latin, *co-ordination* can be translated as *putting in order together* or *making order together*[1]. Taking inspiration from others who have used the term "coordination" to highlight the skillful ways in which humans adapt to

[1]The Swedish translation *samordning*, which literally means *together-order* neatly illustrates this *joint production of order*. I would like to thank my brother Lennard Pelikan for discussions on the Latin translation.

and build on each others' actions in interaction (Heath, 1984; Suchman, 1993), I use this phrasing to highlight that if robots are supposed to engage in interaction with humans for longer stretches of time, they need to participate in this constant mutual adjustment. Human interaction is a joint effort and builds on fine coordination, which is so mundane that it may often go unnoticed: when encountering others on the road, pedestrians and car drivers do not move on a perfectly straight line but they organize their movement in such a way that they do not bump into each other (Goffman, 1971; Ryave & Schenkein, 1974). In conversation, people organize their turns at talk so that their speech is not constantly colliding, while at the same time avoiding overly long pauses (Sacks, Schegloff, & Jefferson, 1974; Stivers et al., 2009). The interplay of embodied and verbal actions is similarly organized. People negotiate a position of mutual orientation at the beginning of interactions, for instance before customers start ordering items with a salesperson (Harjunpää, Mondada, & Svinhufvud, 2018), or before patients present reasons for their visit to a doctor (Heath, 1984). Close study of how humans do things together repeatedly demonstrates that interaction is orderly (Garfinkel, 1967; Goffman, 1971; Sacks, 1984). This work explores how robots currently contribute to and break this order, and how they could be designed to better respect the organization of human interaction[2].

Exploring how robots can coordinate with humans (or at least support humans in adjusting to a robot's limited coordination abilities), I focus on sound as an understudied resource at the margin of language (Keevallik & Ogden, 2020). Sound is interesting because it is relatively independent of one's native language or prior experience with robots. Being closely intertwined with embodied actions, studying sound highlights that social actions occur as Multimodal Gestalts (Mondada, 2019), with different modalities interacting and contributing to the perception of a holistic performance.

1.1 Human-Robot Coordination

Coordination between humans and robots becomes most visible when a robot and a human are involved in a joint activity, in which both parties are pursuing a common goal. I will illustrate how this can look like with an extract from my video recordings of a Cozmo toy robot in Swedish family homes. In Figure 1.1, we join a family who is just teaching Cozmo to recognize their faces, so that the robot will be able to greet them with their names in the future. It is Anna's turn, and her mother has typed Anna's name into the Cozmo app. They just found a configuration that is suitable for the robot after some initial struggles in which the robot accidentally started to scan the face of Anna's brother.

Image 1 shows the formation that the children established, with Anna's face placed in front of the small robot on the table. The father (not in picture) recognizes that finally an opportunity for a successful scan has arisen, and exclaims "Now!", followed by a "maybe" as the robot takes a moment before it finally makes a gibberish sound, which I represent in English letters "oade-e" in my transcripts. In image 2

[2]I am not suggesting that robots should copy human interaction (c.f. Luff, Gilbert, & Frohlich, 1990). Instead, I am interested in how core principles in the human organization could be translated into guiding principles for interaction design.



Figure 1.1: Coordinating face recognition with a Cozmo robot. 1: Anna is waiting for the robot to scan her face, while her brothers duck under the table. 2: Cozmo is scanning Anna's face. 3: Anna turns towards her Dad after Cozmo said her name. 4: Cozmo plays a sound while displaying smiley eyes and Anna pets the robot.

we see what happens when Cozmo moves into the actual face scanning sequence, indicated with extended cog-like "rrr" sounds and squinted eyes on Cozmo's small screen-face. While Anna was initially smiling in anticipation, she is now intensely looking at the robot. When Cozmo reaches the end of the scanning it plays a ding sound, followed by "uh" and then produces Anna's name in a robotic voice. In image 3 we see Anna jump up, turning to her father and exclaiming "Oh, he said Anna!". The robot repeats her name, as if to confirm that she heard correctly. One of her brothers acknowledges that this closes the face learning sequence, by immediately asking that his name (and face) should be next. In image 4, Cozmo plays a happy animation that always follows the facial scanning sequence, which Anna reacts to by turning to the robot and petting its back.

Within just a few seconds, the robot played a range of different, carefully designed animations that feature motor and machine-inspired sounds as well as human-inspired vocalizations. The sounds contribute to making transparent what the robot is doing, and crucially influence the dynamics in the family. Anna is sitting still during the scanning sounds, but jumps up as soon as the robot says her name. The subsequent extended sounds and smiley eyes draw her attention back to the robot and she pets it, acknowledging that the robot successfully accomplished the scan-

ning. Similarly, Anna's brothers sit still during the scan, watching the robot from under the dinner table. As soon as the Cozmo utters Anna's name, they ask for their faces to be scanned (left brother) and get up to touch the robot (right brother).

The coordination work that happens here goes beyond the interaction between the robot and its "user", Anna. The rest of the family is crucially involved: they support the activity, for instance by staying out of the robot's camera angle, coordinating their actions with those of the robot and with Anna so that a successful completion of the scanning can be accomplished. While Cozmo's phone app provides an instruction on how to position oneself (in English, which the parents need to translate for their Swedish-speaking children), it does not provide any instructions on how long one needs to wait. The family members discover with the help of Cozmo's animated displays that the robot can only learn one face at a time and that the person whose name is learnt needs to keep looking at the robot. This dissertation aims to shed light on how sound, as part of such animated displays, can be designed to facilitate this kind of coordination, making the robot's behavior intelligible to humans. Being particularly interested how sound can function as a resource for interaction, I build on recent work that has demonstrated the importance of non-lexical vocalizations, sounds at the margins of language (Keevallik & Ogden, 2020), for the coordination of human activities.

1.2 Aims

The overarching aim of this project is to understand in detail whether robot sound can support coordination with humans, and if so how it can be designed to do so. I study robots in specific local contexts during everyday interaction in homes (Cozmo toy robot), on public roads (autonomous shuttle buses for public transport) and in groupwork settings (Vero teamwork robot). Taking an EMCA approach, I scrutinize how coordination is interactionally accomplished and how coordination problems emerge and influence subsequent interaction. Contributing topically to the emerging field of sound in HRI, my work takes design inspiration from the skillful ways in which humans use sound as a resource for coordination. In developing methods that combine EMCA with interaction design, my work can also be seen as an exploration of how the intersection between EMCA and HRI could form a new, hybrid discipline.

EMCA research has studied robots in a range of settings, including museums (Kuno et al., 2007; Pitsch et al., 2009; Yamazaki et al., 2009), (pre)schools (Alač, Movellan, & Tanaka, 2011; Tuncer, Gillet, & Leite, 2022) and roads (Brown, 2017; Vinkhuyzen & Cefkin, 2016). A relatively large body of work explores interaction with humanoid robots that interact through verbal utterances like Pepper and Nao (Jarske, Raudaskoski, & Kaipainen, 2020; Pelikan & Broth, 2016; Pitsch, 2016; Rudaz et al., 2023; Stommel, de Rijk, & Boumans, 2022; Tuncer et al., 2023). Some papers investigate interaction with robots that primarily rely on sounds and movements like Pleo and Cozmo (Pitsch & Koch, 2010; Tuncer, Gillet, & Leite, 2022). Others specifically focus on robots that could assist humans, in the form of robot guide "dogs" (Due, 2023), robotic wheelchairs (Fukuda et al., 2018), and reminder robots (Krummheuer, Rehm, & Rodil, 2020). EMCA work has started to map specific challenges for human-robot interaction, including turn-taking (Pelikan & Broth,

2016; Stommel, de Rijk, & Boumans, 2022), recipient design (Tuncer et al., 2023), and the establishment of a focused encounter (Gehle et al., 2017; Rudaz et al., 2023). These studies typically take a critical stance, pointing out problems with common experimental paradigms in HRI (Porcheron, Fischer, & Reeves, 2020; Rudaz et al., 2023) or challenging a simplistic view of robots as fully autonomous and independent contributors (Alač, 2016; Pitsch, 2016). EMCA robot research seems to be characterized by two main aims: Some studies use robots to gain insights on human interaction (see e.g. Alač, Movellan, & Tanaka, 2011; Due, 2022), while others seek to improve the design of robots (Gehle et al., 2017; Krummheuer, Rehm, & Rodil, 2020; Yamazaki et al., 2008). The latter seek to primarily make a contribution to the interdisciplinary field of HRI. While my work also has the goal to contribute to HRI, as evidenced by my publications in HRI and HCI venues, I am interested in breaking the duality and exploring further how EMCA and HRI can be synthesized into one program.

HRI is a relatively young field and the majority of studies are still carried out in laboratories where researchers can ensure that robots are safe. Testing robots in real world settings can be difficult, because the technology is complex and expensive. Some robots are too fragile and heavy to take outside the lab, and it is often easier to ask potential users to come and see the robot in a controlled environment. Only within the last decade, the technology has matured enough to start entering homes (often in the form of vacuum cleaning and lawn mowing robots) and public spaces (including autonomous cars and delivery robots). Unaccompanied by their designers that could scaffold the interaction, robots may face unanticipated challenges with maneuvering the complexities of everyday settings (Fernaeus et al., 2010; Forlizzi & DiSalvo, 2006; Šabanović, Michalowski, & Simmons, 2006), often leading to situations that cause interactional trouble (see e.g. Gehle, Pitsch, & Wrede, 2014). In these contexts, robots are usually not interacting with single users, a simplified ideal that studies often draw on, but robots are embedded into groups of people with complex interaction dynamics (Jung & Hinds, 2018). Although the importance of studying interaction in real-world settings has been stressed, a comprehensive body of work that studies robots during longer real-world deployments is still lacking, and robot design still often happens far away from the contexts in which the robots will be used. Robot sound in particular is usually designed and evaluated in quiet lab spaces. In studying robots in everyday interaction in groups of people, my project project contributes to the understanding of how people interact with and interpret robot behavior in *situated* (Suchman, 1987), local and specific contexts. This is a prerequisite for informing design that is a better fit for the realities of life (c.f. Norman, 2013; Šabanović, 2010).

EMCA investigates how people actually interact. This is important, since "if we try to imagine it [interaction], we often imagine things that rarely happen, or conversely we fail to imagine things that actually happen all the time" (Enfield, 2013)[3]. EMCA studies have repeatedly demonstrated that human interaction is systematically organized. While people may say and do all kinds of things to each other, the way in which they do this emerges in a coordinated way. Well-known examples that

[3]This is a condensed summary of an earlier argument made by Harvey Sacks in his 1971 lecture on hypothetical data (Sacks, 1992, p. 419f.). Jefferson (1984) provides a slightly different formulation of the same argument.

appear to be universal across cultures are the organization of turn-taking (Sacks, Schegloff, & Jefferson, 1974; Stivers et al., 2009) and of repair (Dingemanse, Torreira, & Enfield, 2013; Schegloff, 1992). These patterns have been documented in detail for human-human interaction, primarily through conversation analysis (CA), an approach that is rooted in ethnomethodology and with influences from anthropology, interactional linguistics and discursive psychology. To study how coordination is achieved at a fine level of detail, interaction needs to be recorded, so that it can be repeatedly scrutinized as it plays out in real time. Video is the preferred resource, as it captures both hearable and visual behavior (Heath, Hindmarsh, & Luff, 2010; Jordan & Henderson, 1995; Suchman, 1987). In studying how coordination between humans and robots is organized on a moment-by-moment basis, this work sheds light onto human social practices for interacting with robots. The fine-grained analysis enables pinpointing at which moments interactional trouble arises and can inform robot interaction design by its focus on the actions that robot behavior could accomplish.

Sound is one resource that robots can use for interaction with humans. Beeps, melodies and gibberish speech have been explored for this purpose (Yilmazyildiz et al., 2016). Beeps may convey emotion (Read & Belpaeme, 2016) and intent (Fischer, Jensen, & Bodenhagen, 2014). However, even motor sounds may influence how humans perceive a robot (Moore et al., 2017; Tennent et al., 2017) and can indicate that an autonomous vehicle will be stopping at a crossing (Moore, Currano, & Sirkin, 2020). I want to refrain from *a priori* claims about whether sounds are actually communicative or not, and what their specific semantic content may be (Yilmazyildiz et al., 2016). Therefore, I am using the general term "sound" in this dissertation, including any audio signal produced by a robot that humans can hear. While robot sound has been mostly designed to communicate fixed messages, interactional studies highlight the role of sound in organizing interaction: human sounds have been studied in the form of so-called "nature sounds" (Naturlaute, Wundt, 1900), response cries (Goffman, 1978), sound objects (Reber, 2012), liminal signs (Dingemanse, 2020) and non-lexical vocalizations (Keevallik & Ogden, 2020). Recent work demonstrates that such vocalizations are not random outbursts but accomplish a range of social functions. They can convey bodily preoccupation, such as strain grunts in the context of physical activity (Hofstetter, Keevallik, & Löfgren, 2021; Keevallik, 2018). Vocalizations are closely tied to emotion expression. "Mm"s can be used to display gustatory pleasure (Wiggins, 2002) and clicks serve to express "unsayable" delicate matters (Ogden, 2020). Non-lexical vocalizations are also important for organizing the temporality of events, and can help to instruct sequences of actions such as in dance education (Keevallik, 2015, 2021). Being ambiguous by design they can be particularly versatile (Dingemanse, 2020). Non-lexical vocalizations play a vital role in coordinating and instructing joint bodily activities (Keevallik, 2015, 2018; Wiggins, 2002; Wiggins & Keevallik, 2021) and in organizing interaction, such as managing turn delays (Hofstetter, 2020) and initiating repair (Dingemanse, Torreira, & Enfield, 2013). Even beyond vocalizations, which originate from the vocal tract, humans can produce sounds that support coordination such as rhythmic clapping (Kerrison, 2018) or honking (Laurier et al., 2020). Current design of robot sound typically does not draw on these insights, missing a source of inspiration that could be particularly informative for designing sound that

can support interaction. I strongly build on interactional studies that demonstrate how sound is used as a resource for coordination in human interaction in exploring how sound can be designed to facilitate human-robot coordination.

1.3 Research Questions

Working at the intersection of EMCA and HRI, specifically targeting robot sound, this work aims to address three research questions:

1. How do people make sense of robot sound in everyday interaction?
2. Can robot sound play a role in the accomplishment of sociality?
3. How can EMCA reconceptualize what "interaction" in HRI is or might be?

In the first research question, I am interested in how people make sense of robot sounds in ordinary encounters. Many studies have been conducted in laboratory settings and online, asking people to specify what a sound could mean or to rate how well it conveys a certain quality. Following calls from the HRI community for more studies of robots outside the lab and "in the wild" (Jung & Hinds, 2018; Šabanović, Michalowski, & Simmons, 2006), I am interested in how robot sound gets reacted to and interpreted in homes, on the road and in ongoing collaborative work. Within the framework of this question we are encouraged to explore whether sound gets noticed at all, whether it is always interpreted in the same way, and whether the understanding displayed by participants aligns with what the designers have intended it to convey. The question enables me to explore opportunities and limitations of sound in interaction, taking a classic EMCA perspective, which is critically analyzing existing designs.

The second research question explores whether robot sound can contribute to the accomplishment of social actions, or actions-in-interaction in the EMCA sense. I take a more skeptical perspective than what prevails in HRI, which assumes *a priori* that sound is communicative. I am therefore asking whether (and how) robot sound contributes to the accomplishment of social actions, hoping to widen the conceptualization of robot sound beyond a means to convey emotions. While being rooted in an EMCA understanding of interaction, the question is also motivated by a human-centered design perspective that treats the performance of social actions as a form of interface that makes transparent how a robot works and how it can be used (Norman, 2013).

With the third question I want to explore further how EMCA can transform current HRI research and design practice, focusing on the *interaction* in human-robot interaction and moving towards a hybridisation of EMCA and HRI. I am particularly interested in going beyond evaluation of successful designs, and rethinking how to embed an ethnomethodologic understanding and conversation analytic methods in the design of specific robot behavior throughout a design process. The question encourages experimentation with how practices of working with video can be drawn upon in design. At the same time, I am interested in how EMCA could be complemented by design practices that can facilitate interdisciplinary conversations.

1.4 What is a Robot?

Different people disagree quite heavily on what should be considered a robot, and different languages may vary in their use of the term[4]. In this section I want to highlight that like for any interactive technology or interface, there are always different human roles implied (Norman, 2013): *users* (the "human" in human-robot interaction) and *designers*. I will briefly trace how popular culture may shape users' understanding of robots, and how technical definitions shape the view of the designers, in a loose sense those who build robots. Finally, I will focus on the interaction, highlighting what makes robots an interesting and somewhat special object of study from an EMCA perspective.

Robots are popular movie characters, which often feature human-like robots, such as Maria in *Metropolis* (1927), Sonny in *I, Robot* (2004) or Ava in *Ex Machina* (2014). Even more machine-like robots have gained popularity, including R2D2 from the *StarWars* franchise and Wall-E and Eve in the movie *Wall-E* (2008). In fact, even the word robot is not a technical term but originates from the drama *R.U.R. Rossumovi Univerzální Robot* (Rossum's Universal Robots, RUR) by Czech author Karel Čapek (1920/2004), who uses the word *robota*, which means "forced labor". The robots in this play are what is nowadays referred to as *humanoid* robots, robots that have bodies similar to human ones: a head, arms, legs and a trunk. Čapek's robots are workers with human-inspired but optimized bodies, "mechanically more perfect than we are, they have an astounding intellectual capacity, but they have no soul" (Čapek, 1920/2004, Prologue, p. 9), so that they can be produced faster and more efficiently. While not being known as "robots" until 1920, mechanical creatures, so-called *automata*, have a long history and have fascinated humankind for centuries. Automata appear in Greek mythology and were popular among royals during Renaissance times (see e.g. Bedini, 1964). The first physical robots were analogue, running on clock-work and were driven by cogs. They have also inspired literature. German writer E. T. A. Hoffmann (1816/2003) describes female automaton Olimpia, who can only say *ach, ach*, "oh oh" or "well well" in English, but nevertheless makes the main character in his book *Der Sandmann* (The Sandman) fall in love with her. Portrayals of robots in popculture are often gloomy, and use robots as a mirror for asking what it means to be human. Such fictitious robots may influence people's expectations about real robots (Bruckenberger et al., 2013), although these may change as robots increasingly enter everyday life. When robot vacuum cleaners or lawn mowers enter people's homes, users also draw on their practical experiences with other household appliances such as washing machines, toasters and kitchen aids (Forlizzi & DiSalvo, 2006). Through voice assistants and chatbots, people increasingly gain experience in using natural language for interacting with machines, which may further inform their expectations of what a robot can and cannot do. While the lay understanding of what a robot is may be quite broad already, it is somewhat in contrast to how the term robot is used in a technical sense.

Roboticists have jointly formulated a technical standard (ISO, 2021, p. 3.1), which defines the term robot as "programmed actuated mechanism with a degree

[4]In Swedish, the word *robot* is for instance commonly used to refer to chatbots, but also means "missile".

of autonomy (3.2) to perform locomotion, manipulation or positioning". According to this definition, a robot is characterized by its material presence and possession of actuators, implying that it can be programmed to move in some way. According to the standard, a robot also needs sensors that give it a window into what is happening around it. They enable the robot to perform certain tasks without human intervention, based on its "current state and sensing" (ISO, 2021, p. 3.2). Finally, the standard highlights that a robot includes a control system for "monitoring and controlling of the behaviour of a robot (3.1) and its interaction and communication with other objects and humans in the environment" (ISO, 2021, p. 3.4). Human–robot interaction is then described as "information and action exchanges between human and robot (3.1) to perform a task by means of a user interface" (ISO, 2021, p. 3.15). The definition highlights that in order to work together, humans and robots need to communicate in some way. Yet, it seems to simplify interaction in an input-output model, which contrasts with the back-and-forth adjustments that I try to capture in the term "coordination". The standard has changed even during the time of my PhD, and as HRI researcher Kerstin Dautenhahn (2014, p. 38.2) puts it, the concept of a robot is a "moving target". While robotics researchers typically agree that a robot should have some autonomy in achieving specific goals, as well as sensing and acting capabilities (c.f. Bekey, 2005; Matarić, 2007) the specific definitions may vary. Some distinguish between robots that are built for social and purely technical purposes (see e.g. Fong, Nourbakhsh, & Dautenhahn, 2003; Henschel, Laban, & Cross, 2021). Others challenge a definition of robots based on technical specifications, and instead suggest a characterization "based on the relation humans have with the robotic technology in question" (Mollen, van der Putten, & Darling, 2023, 8:4). They highlight that different persons may have different relationships with the same robot, and focus on a distinction between machines that are treated as subjects versus objects. This potential alterity relation with robots as an "other" is precisely what makes robots interesting from an interactional perspective.

From an EMCA stance, a robot becomes social when humans treat it as a participant in an interaction, responding to its (autonomous) actions and/or treating the actions as responses to their own actions (see also Pelikan, 2024). The potential to participate and to contribute actions to interaction distinguishes robots from objects that primarily extend human action (like a computer mouse) and technology that serves as a medium for interaction between humans (like a telephone or videoconferencing platform). When robots participate in interaction, their agency may be seen as generally distributed: The interaction designer is the *author* of the behavior that the robot *animates* (c.f. Goffman, 1981). The EMCA perspective highlights that as participation frameworks are dynamically changing, the status of a robot is never fixed but may constantly alternate between that of an autonomous agent (and active participant) and a material object, depending how participants treat the robot at a specific moment in time (Alač, 2016; Krummheuer, 2015; Pelikan, Broth, & Keevallik, 2022).

Since robots are built by humans, they can be designed with more or less capabilities to participate. Some regard robots as an interface that should facilitate efficient and socially acceptable use (c.f. Dautenhahn, 2014). One way to achieve this is by mimicking closely the ways in which humans interact (see e.g. Mutlu et

al., 2012; Yamazaki et al., 2008). Others focus on designing experiences and may see robots as artwork[5] that should please (or provoke). Aiming to design expressive robots, one may take inspiration from animation (see e.g. Hoffman & Ju, 2014; Takayama, Dooley, & Ju, 2011). Taking a human-centered design perspective that is robust for everyday use and enables the machine to work smoothly even when things go "wrong" or differently than expected (c.f. Norman, 2013, p. 8-9), both aspects will be relevant. I am particularly interested in how to incorporate insights about the organization of human interaction into robot design, without necessarily aiming to copy human behavior.

1.5 Contribution

As the previous sections have outlined, my work is located at the intersection of EMCA, HCI and HRI. Depending on the perspective, this dissertation makes different contributions.

I contribute to HRI and HCI

(a) detailed out-of-lab studies of robots in-the-world, with a specific focus on the interpretation of sound in real-time,

(b) a design practice and framework for prototyping HRI based on an EMCA understanding of social order, focused on the design of sound.

I contribute to EMCA

(a) descriptions of how robots are embedded into the social organisation by members, specifically in the context of the home and in traffic,

(b) members' discovery of how to "be" with categories of things with a dynamic participation status,

(c) work towards EMCA "hybridisation", a synthesis of EMCA and robot interaction design.

1.6 Outline

In the following chapters I will first introduce related research on sound in HRI to situate my studies within this interdisciplinary body of work (chapter 2). Subsequently, I will take an EMCA stance, describing how it informs my understanding of interaction and action, and highlighting how EMCA connects with topics in interaction design and HRI (chapter 3). Next, I will describe my methods, centering around video as data and design material, and I will introduce the video corpora that my studies build on (chapter 4). This will be followed by a summary of the five articles that contribute to this compilation dissertation (chapter 5). I will highlight how the articles contribute to answering each of the research questions and discuss contributions and implications (chapter 6). Finally, I will provide a concluding summary (chapter 7).

[5] There may be parallels between robots and other works of art, with related implications for the relationship between author and the artwork.

2 Robot Sound

Sound can always be viewed from two perspectives, the perspective of the listener and the sound producer. The literature on robot sound has dealt with both aspects. Until a decade ago, robot sound research predominantly focused on how robots can process sounds in their surroundings. In HRI research, sound has since gained increased attention with a workshop on the topic happening at the flagship conference (Pelikan et al., 2021) and a special issue on Sound in HRI forthcoming in the ACM Transactions in Human-Robot Interaction[1]. Researchers started to become interested in sounds generated by robots, with more publications emerging on the topic during recent years (Zhang & Fitter, 2023). My work falls into this latter category, sounds that robots produce, and I delimit my literature review to this subdomain.

This chapter will give an outline of the types of sounds that robots can emit, and describe what these sounds are currently used for. Subsequently, the chapter will introduce methods for design and evaluation of robot sound. I will close by formulating an opening for EMCA research to contribute.

2.1 How Do Robots Sound?

Sounds are a natural part of a robot's material presence, even when they are not deliberately designed, and accompany any physical robot movement. The sounds that result from the hardware of a machine are categorized as *consequential* sounds and stand in contrast to deliberately designed *intentional* sounds (Langeveld et al., 2013). In video demonstrations of robots both for sales pitches and research, these whirring and humming sounds are often removed, creating the illusion that robots

[1] Guest edited by Robinson, Watanbe, Pelikan, Damiano, Bown, and Velonaki.

would be silent. Consequential sounds are typically not very pleasant to listen to, and study participants asked to rate different servo motor sounds viewed such sounds as generally more negatively than no sound (Moore et al., 2017; Tennent et al., 2017). Interestingly, however, people can distinguish between different servo sounds and describe certain sounds as indicating that the robot is working harder (Moore et al., 2017). Further, sound perception seems to depend on the specific context. Study participants rated a moving robotic arm (with consequential sounds) as more competent and trustworthy when the robot was interacting with a human than when it was completing a task by itself (Tennent et al., 2017). Consequential sound can be masked by overlaying it with additional designed sound, resulting in *transformative sound* (Frid, Bresin, & Alexandersson, 2018; Latupeirissa, Panariello, & Bresin, 2023; Zhang et al., 2021), a strategy which is for instance used in the Cozmo robot. Consequential and transformative sounds fall into the category of *implicit* sounds (Moore, Currano, & Sirkin, 2020; Zhang & Fitter, 2023), following HRI terminology that distinguishes implicit from explicit signals to encourage consideration of naturally available (and often more subtle and implicit) resources (Ju, 2015). Movement and motor sounds should then be considered first before adding a new interface such as a graphical display that explicitly instructs how to use a robot (Moore et al., 2019).

For sounds that are designed with the intention to communicate, the terminology is less clearly defined. A range of sounds have been explored, with work rooted in different disciplinary traditions, ranging from musical sonification to psychology and linguistics. Yilmazyildiz et al. (2016) glossed these sounds as *semantic-free utterances*, an unfortunate term since these are certainly not without meaning. Attempts to map this area more systematically (Yilmazyildiz et al., 2016; Zhang & Fitter, 2023) and to describe more general design practices and guidelines specifically for HRI sound design (Robinson, Bown, & Velonaki, 2022) are only recently forthcoming.

Work on sonification, typically drawing on music theory and sonic interaction design, explores for instance how information about the robot's state can be conveyed by accompanying gestures and movements with specifically designed musical or natural sounds (Frid & Bresin, 2022; Latupeirissa, Panariello, & Bresin, 2020; Robinson, Velonaki, & Bown, 2021). Such sounds are usually designed as a unit with the robot's movement, communicating for instance where a robot is going to move next (Johannsen, 2001) or creating emotional gestures (Latupeirissa, Panariello, & Bresin, 2020; Savery, Rose, & Weinberg, 2019). These studies point out the value of more carefully designed sounds over pure motor sounds, highlighting that musical theories can inform pleasant designs that may better suit the particular character and physical shape of a robot. Some of this work also takes inspiration from prior work in sound in human-computer interaction, which explored the use of everyday sounds for making interaction with computers intuitively understandable (Gaver, 1986, 1993), such as the sound of crumpling paper that makes transparent that files are being deleted or a knocking sound for incoming messages.

Sounds recently described as *vocables* by Zhang and Fitter (2023) resemble or imitate human and animal vocalizations and their design often draws on a linguistic perspective. Yilmazyildiz et al. (2016) distinguish further between gibberish speech and paralinguistic utterances, with *gibberish speech* referring to concatenations of

syllables used by characters in children's movies such as *The Minions*, which may come close to developing a new language for robots. The category of *paralinguistic utterances* captures human-like elements of speech such as "affect burts", laughter, and prosody. The latter is considered as separate from speech, since from a computational perspective, prosody (how something is said) is generated separately from the lexical content (what is being said). A lack of prosody is one aspect that makes computer voices sound "robotic". Depending on the use context, users may prefer robots to speak with different voices (Torre, Latupeirissa, & McGinn, 2020). Studies that explored the benefits of *backchanneling* may be counted into this category (Jung et al., 2013; Moubayed et al., 2009; Park, Gelsomini, Lee, Zhu, et al., 2017). Backchannels are human-inspired sounds that display continued listening in robots. This seems to be the only line of research that has explored explicitly *when, at what time points* sound should be produced in an ongoing interaction, for instance by statistically modeling backchanneling opportunities (Park, Gelsomini, Lee, Zhu, et al., 2017). Conversational fillers such as "umm" have also been suggested to bridge longer robot silences, thereby identifying specific moments in which robot sound could become interactionally relevant (Ohshima et al., 2015; Shiwa et al., 2008). In the work on backchannels and affect bursts, vocalizations are often combined with facial expressions (Park, Gelsomini, Lee, & Breazeal, 2017; Schadenberg, Heylen, & Evers, 2018).

Besides vocalizations, electronic beep sounds are another resource available to robots, and humans have been demonstrated to treat such sounds as meaningful. Fischer, Jensen, and Bodenhagen (2014) show for instance that people can distinguish between beeps that were designed with rising and falling intonation inspired by human prosody, and especially women seemed to prefer to be approached by a robot that plays beeps with rising intonation rather than beeps with falling intonation or no sound. Read and Belpaeme (2012, 2014) describe beep sounds as *non-linguistic utterances* and explore how they can convey emotion. This strand of work highlights that robots may not necessarily need to sound like humans but may use more "robotic" sounds. Such sounds also have a history in human-computer interaction, which has explored *earcons*, electronic sounds that users would learn over time (Blattner, Papp III, & Glinert, 1994; Blattner, Sumikawa, & Greenberg, 1989), such as sounds for specific error messages.

In general the boundaries between the categories are often fluid, and consumer robots such as Cozmo combine human- and animal-inspired syllables with motor sounds and more robot-like beeps. In my work, vocables and non-linguistic utterances are perhaps most salient, but consequential sounds are not only ubiquitous but also occasionally become interactionally relevant.

2.2 What Do Robot Sounds Mean?

Three main strands of work can be recognized that concern the function or types of meanings that robot sounds are designed to convey: emotion, intentions or the quality of the robot as a product. The first two are particularly relevant to my dissertation.

Research on psychological topics has taken particular interest in investigating whether robot sounds can reliably communicate a set of emotions (Chan, Zhang,

& Fitter, 2021; Frid & Bresin, 2022; Kappas et al., 2015; Latupeirissa, Panariello, & Bresin, 2020; Read & Belpaeme, 2016; Zhang et al., 2021). Read and Belpaeme (2016) demonstrate for instance that study participants were able to categorize non-linguistic beeps played by a Nao robot into representing a set of basic emotions. Frid and Bresin (2022) show that sonifications blended with consequential sounds from a Nao robot could convey joy or frustration. Others focus on exploring the intertwinement of different modalities and how they contribute to different emotion displays, finding that sadness is particularly easy to convey through sound (Löffler, Schmidt, & Tscharn, 2018; Song & Yamada, 2017). This line of work tends to take the view that a specific emotion can be designed into a robot sound animation. Chan, Zhang, and Fitter (2021) experimentally validated Cozmo's emotion displays, demonstrating that they largely match the valence and energy intended by the designers. While study participants may agree to a high degree, there usually is a small number of people who may perceive a different emotion, which highlights that a meaning cannot be ultimately fixed into a sound. While Read and Belpaeme (2014) demonstrated that the context in which a sound is played may influence how it is interpreted, many studies play sound animations in isolation.

A different strand of research explores how to communicate specific intentions through sound, mostly focusing on how vehicles can communicate that they are moving into a certain direction (Cha et al., 2018; Johannsen, 2001; Moore, Currano, & Sirkin, 2020). Typically, visual tools are used for this purpose, but some studies consider sounds in combination with visualizations to communicate what an autonomous vehicle is perceiving or about to do next (Lagström & Lundgren, 2015; Mahadevan, Somanath, & Sharlin, 2018). Moore et al. (2019) suggested communicating a car's intent by drawing on existing car sounds rather than adding completely novel designs and Moore, Currano, and Sirkin (2020) played synthetic motor sounds from a speaker taped to a hybrid car during a naturalistic field study at a crossing. The driver was hidden so that the car appeared to be driving autonomously and pedestrians reported afterwards that they understood the car's intent to yield more clearly with the synthetic motor sounds than without them [2]. In-depth studies on how sound animations can contribute to making transparent what a robot is going to do next have so far been lacking.

2.3 Methods for Design and Evaluation of Robot Sound

Current HRI sound design seems to lack a formalized design approach. Techniques developed for sonic interaction design (Ekman & Rinott, 2010; Rocchesso et al., 2015; Rocchesso, Serafin, & Rinott, 2013) such as *vocal sketching* are primarily used in work on robot sonification (Panariello et al., 2019). Researchers with extensive training in music design musical sounds for robots (Robinson, Velonaki, & Bown, 2022). The design and timing of backchannels and affect bursts typically aims to imitate humans (Park, Gelsomini, Lee, Zhu, et al., 2017; Schadenberg, Heylen, & Evers, 2018). In the majority of cases, the specific robot sound animations are designed in an *ad hoc* manner, with actionable design guidelines largely missing. Only very recent work has started to formulate candidate design principles (Robinson,

[2] Synthetic sounds are obligatory on electric and hybrid cars in the European Union since 2019 as warnings that a vehicle is approaching (European Commission [EC], 2019).

Bown, & Velonaki, 2022) based on interviews with professional audio designers. These are more generally applicable.

Concerning the evaluation of sound, the large majority of studies are quantitative and test people's subjective rating of sounds based on questionnaires (Zhang & Fitter, 2023). These often involve comparisons between different sound designs or sound versus no sound. Generally, there is evidence that suggests that robot sound may be more ambiguous than lexical utterances. Researchers point out that robot sounds do not work equally well in all settings (Fischer et al., 2014) and for all robot types (Read & Belpaeme, 2010). Read and Belpaeme (2014) further highlight that the interpretation of robot sounds strongly depends on the context of the situation. In their lab study, participants interpreted the same sound differently when it was played immediately after the robot was hit or kissed. This finding points to the importance of studying non-lexical sounds in the specific sequential contexts they occur in. However, the notion of context is very narrow in the existing studies, which are typically conducted in laboratory or semi-laboratory settings (see e.g. Fischer, Jensen, & Bodenhagen, 2014; Read & Belpaeme, 2016; Zhang & Fitter, 2023), through simulations (Johannsen, 2001) or by overlaying sounds on video recordings of robots (Latupeirissa, Panariello, & Bresin, 2020; Moore et al., 2017; Read & Belpaeme, 2014; Robinson, Velonaki, & Bown, 2021). Few researchers have explored sound outside controlled settings, including a naturalistic experiment on the road (Moore, Currano, & Sirkin, 2020) and a study in a museum (Latupeirissa, Panariello, & Bresin, 2023). Besides these studies, sound is rarely explored in the environment where a robot should later be deployed.

2.4 Opportunities for Respecifying the Study of Robot Sound

Current work on robot sound typically uses quantitative evaluation methods, and design often targets one single category of sound, inscribing specific meanings or functions (c.f. Zhang & Fitter, 2023). While this has led to interesting insights, it leaves several questions open: How do people interpret robot sound outside the lab? How do different types of sound play out when combined into sequences? How do robot sound animations contribute to the organization of social interaction?

EMCA provides an approach for the evaluation of sound that differs from the current mainstream in HRI. Instead of asking people to describe various sounds along some dimensions provided by the researcher, EMCA observes participants' manifest understanding of sounds, which they display in ongoing interaction. This approach is particularly suited for studying the complexity of out-in-the-world interactions. Furthermore, EMCA asks different questions, which encourage a focus on how robot sound contributes to interaction.

First, EMCA can contribute to a respecification of the question "How does a robot sound?" which seems to invite descriptions in terms of pitch and timbre as well as quantifications of perceptual qualities. Instead, EMCA encourages questions like "How do people make sense of robot sound in interaction?" which encourages exploring how a sound is interpreted in different ways by different people in different situations. My analysis will thus focus on how participants display their understanding of sound in real time. If participants treat a motor sound as a rel-

evant contribution to their interaction, it is likewise relevant for the analysis. If participants seem to ignore the sound, it will not be treated as relevant for the analysis. Rather than categorizing sound types from the perspective of the researcher, the EMCA approach invites a participatory stance in the analysis of sound.

Similarly, EMCA invites a reconsideration of the question "What does a sound mean?". As the literature review showed, HRI research tends to focus on validating potential meanings. EMCA replaces the question with "What does a sound do?" inviting an examination of how sound animations contribute to the accomplishment of actions in ongoing interaction. EMCA work demonstrated previously how machine sounds triggered by humans contribute to the accomplishment of actions when people use horns to coordinate movement in traffic (Laurier et al., 2020) or summon others to pick up the phone (Schegloff, 1968). As Garfinkel and Wieder (1992, p. 197) point out, such "sound is recognized and identified as a sounded doing", in other words, as relevant contribution and performance of an action in an ongoing interaction. An analytical departure from actions can help exploring how robots could use sound to make transparent what they are currently doing, what they have done, and what they will do next.

While EMCA does not immediately offer design solutions, it can help to shift the focus from specific sound characteristics to interaction. The jargon in HRI studies may suggest that sound can be designed as a separate channel. From an EMCA perspective, sound animations occur as holistic *multimodal Gestalts* (Mondada, 2019), in which sound is intertwined with movement, gesture, facial expressions, lights, and colors. Putting a particular focus on aural elements can be a helpful exercise for becoming aware of this intertwinement, which may easily be missed when focusing on aspects in the dominant visual domain. Therefore, a focus on sound as an understudied modality seems a suitable way for starting to rethink more generally what human-robot interaction is or might be.

3 Interaction, Action and Design

The chapter introduces the most important theoretical concepts for this dissertation. Since my work focuses on the *interaction* in human-robot interaction, I will start by describing my understanding of interaction. I will then zoom in on actions (in interaction) before introducing literature on design, particularly highlighting meeting points between EMCA and design of interactive technology.

3.1 Interaction

In the following sections I will introduce principles in the organization of interaction that are most relevant for understanding (challenges in) interaction with robots: co-construction, turn-taking and sequence organization. I will start by explaining how the concepts have been introduced for human interaction before speaking to what extent they apply in human-robot interaction.

3.1.1 Co-Constructing Understanding

Contemporary studies of human sociality and communication agree that human social interaction is inherently collaborative (Levinson & Enfield, 2006). This collaborative nature has been described as "joint" (Clark, 1996), stressing that humans need to work together to build a shared understanding. Others have highlighted that interaction is not a combination of separate monologues by different speakers, but that it is inherently "dialogical" (Linell, 2009). Interaction is not only a division of labor with one person doing one part and another person doing the other. In contrast, many approaches, including EMCA, underline that meaning is co-constructed.

Interaction is then not the addition of single actions, one after another, but instead an intertwinement of the actions that different participants are contributing. As Schegloff (1995, p. 201-202, emphasis in original) puts it, "*actions by other than the speaker* are relevant to understanding a speaker's construction of discourse". What a turn means is not determined by the speaker alone, but it is only fully established in the following turn, when the recipient has displayed what they understood (and this understanding can get ratified in a third turn by the original speaker). This view of interaction challenges the idea that robots should convey fixed meanings, as for instance present in approaches that aim to design robot sound that conveys a predetermined emotion (Chan, Zhang, & Fitter, 2021; Read & Belpaeme, 2016).

The process of co-construction has been described in different ways. Psycholinguists Clark and Brennan (1991) refer to the process of building a shared understanding as "grounding", in which participants try to reach a mutual belief that the listener has understood what the speaker meant well enough for their current purposes. This view is already present in some HRI work (Jung, 2017; Kontogiorgos & Pelikan, 2020)[1]. Clark and Brennan explain that participants do not always need to understand each other in every single detail. A degree of understanding that is sufficient for accomplishing their joint action and for reaching their common goal is enough. As Clark and Brennan focus on modeling interaction, they have mostly been concerned with distinguishing clear phases in this process. In the tables they use to describe the grounding process, one may easily miss the reflexive element that is inherent in co-construction. Charles Goodwin (2018) perhaps captures co-constructive element in interaction best when arguing that human action is "co-operative", reflexively building on the actions of others. Even when we are not jointly working on a task but having a fight, we are "creating new action by operating on, and transforming, the detailed organization of [another participant's] utterance" (Goodwin, 2018, p. 6). Goodwin (2018, p. 7) further adds that "participants not only attend to, but actively participate in, the detailed organization of each other's action as it unfolds through time". This reflexive engagement and operating on each other's actions is central to how interaction is understood in this dissertation.

Adopting this view to machines raises interesting questions and challenges, since it highlights that the machine's contributions cannot be completely "pre-planned" (Suchman, 1987), but only develop their specific situated meaning in interaction with others. When we gain new insights after talking to others, this is not only based on the feedback that we receive on our own view, but in talking with and for a specific person, our own view is simultaneously transformed (Linke, 2009). Some scholars stress alterity, the role of "the other" in creating meaning. This will not be a central theme in the present work but taken seriously, questions arise as to whether a machine can take the role of 'the other", touching deeply also on questions of human singularity. My focus is more practical, asking not whether a machine can ever be equivalent to a human interaction partner, but rather how machine actions can be designed so that they are more easily understood by humans, making machine actions "co-operable". How should a machine react to human action in order to

[1] In my view, CA and grounding theory can be productively combined as long as a clear distinction between both approaches is made in the analysis (see e.g. Pelikan, 2018)). Some differences remain, most notably how mental states are dealt with.

contribute to interaction? To support and enhance sociality by the machines that we build, we need to understand the basic principles of human interaction. This will also help us to understand how and why humans treat interactive objects, specifically robots, in certain ways.

3.1.2 Turn-Taking and Progressivity

One of the key mechanism that drives interaction is turn-taking. In joint activities, parties in an interaction cannot act all at once, but need to do things one at a time, so that their contributions to the joint project are recognizable. Initially described for conversation, turn-taking organizes interaction so that people do not all speak at the same time. Participants work hard to avoid overlap as well as too long silences between turns, so that it is still recognizable that they are responding to the previous turn (Sacks, Schegloff, & Jefferson, 1974). Sacks, Schegloff, and Jefferson (1974) famously distinguish between turn-constructional units (TCU) and transition-relevance places (TRPs). At each TRP, participants in the interaction have the opportunity to change who is speaking, or the current speaker can continue their talk, further expanding their turn and arriving at a next possible place for transition to another speaker. Turn transition usually does not happen at random moments but is accomplished through grammatical, pragmatic and prosodic structure (Ford & Thompson, 1996; Sacks, Schegloff, & Jefferson, 1974). Contemporary work has provided further nuance to the organization of turn-taking, demonstrating that not only talk but even bodily-vocal behavior may be treated as a turn (Keevallik, 2013) and bodily conduct is crucial for turn-transition (Mondada, 2007). Turn-taking rules may even apply to purely embodied activities, such as when dance couples negotiate who is next to perform during an improvisation session (Keevallik & Ekström, 2019) or when skateboarders take turns at riding a pool (Ivarsson & Greiffenhagen, 2015). This highlights that one does not necessarily need to speak in order to contribute a turn to an ongoing activity.

With the advent of computers that can talk and move, the question arises whether even non-human partners can engage in turn-taking, contributing to joint activities. Work on dialogue systems (see e.g. Bohus & Horvitz, 2011) and robots has extensively explored turn-taking, studying how robot gaze (Mutlu et al., 2012), and multimodal signals such as breath, gaze and filled pauses (Skantze, Johansson, & Beskow, 2015) can facilitate human-robot turn-taking. Even in robots, turn-taking is not limited to speech, but can be purely embodied, for instance when humans and robots jointly make music by engaging in drumming (Kose-Bagci et al., 2009) or doing jazz improvisation (Hoffman & Weinberg, 2010). However, while robots and voice agents may succeed at contributing turns to interaction, EMCA work has repeatedly pointed out that robots and voice assistants fail to time their turns in a way that can keep up with human interaction (Pelikan & Broth, 2016; Porcheron et al., 2018). Robots and voice agents also often struggle with progressivity (Fischer et al., 2019; Stommel, de Rijk, & Boumans, 2022), keeping the interaction moving forwards (Stivers & Robinson, 2006). Their limited hearing and understanding capabilities can require several repetitions and rephrasings before the interaction can move on (Fischer et al., 2019; Pelikan & Broth, 2016; Stommel, de Rijk, & Boumans, 2022). Robots are not listening for TRPs in the same sense as humans do but are

often only receptive to input during pre-programmed time slots. Thus, robots do not orient to the emergence of possible TRPs, dynamically reacting and moving interaction forward. Instead, robots produce responses to human input following their programmed sequence. Despite these limitations, robots may momentarily contribute turns that move interaction forward (Pelikan, Broth, & Keevallik, 2022). A crucial concern for analyzing human-robot interaction is then to determine whether robot turns are treated as contributions, which lead to progression in a joint activity or whether they are ignored. This is particularly important for sound animations. Since sounds are more ambiguous than words, there may be more variability in whether they are treated as interactional turns or talked over as background noise.

3.1.3 Sequential Organization

Another important interactional mechanism, which forms the basis for co-construction of action is the sequential organization of action (Schegloff, 2007): participants display their understanding of the previous action in their response (or non-response to it). This aspect is also reflected in Clark and Brennan's (1991) grounding model, in which listeners demonstrate what they took the speaker's utterance to mean. For conversation analysis, participants' understanding is a central focus of analysis: participants demonstrate what they recognize a previous action to be in their next turn, through the precise way in which they respond to that action.

Schegloff and Sacks (1973, p. 76) coined the question "why that now?" to emphasize that participants (and analysts) constantly need to assess what a certain behavior can be recognized to be doing in the specific context at the specific time. Thus, the focus is not on information transfer but on what an action accomplishes here and now (Schegloff, 1995). Different behaviors (e.g. saying "hello", waving, or playing a beep sound) can occasionally accomplish the same action, a greeting. Examples of actions are greeting, asking, answering, offering and requesting. There is no exhaustive list of actions, so with the help of sequential analysis we can even describe actions that we have not yet "discovered" or ones we do not yet have names for (Schegloff, 2007). Certain actions make specific responses relevant (Schegloff, 2007), meaning that a specific next action is expected after specific previous actions. In the case of the greeting, a return greeting is expected (and its absence will be noticed, resulting in people making inferences such as "the other didn't see me"). Response relevance can be understood as a normative issue, rather than a mathematical-logical one. There is no rule, which states that *if* there is a greeting, *then* there will always be a return greeting. Rather, it is normatively expected. Its absence is accountable (c.f. section 3.2.3), if it is not produced there will be an element of surprise, and an explanation may become relevant.

When studying robots, sequential analysis offers an alternative to assessing what is happening from a technical perspective (did the sensor detect what a human just did?) or inferring intentions (did the human intend to do X?). EMCA analysis can provide a new perspective by targeting people's manifest behavior in which they display their understanding of actions. By looking at how participants respond to what a robot does, the analyst can gain insight into what they recognize the robot to be doing. In their reactions, humans display their sense-making of the robot's behavior, for instance as a response to a question that they just asked. For the

investigation of robot sound, sequential analysis provides not only an opportunity to study how sound is interpreted in real time. More importantly, it also offers an alternative to asking people what an isolated sound could mean, focusing instead on what action they recognize the robot to be doing, involving the sound.

3.1.4 Human-Robot Interaction

In the previous sections, I have described key principles in the organization of human interaction. Considering the limitations that robots have in keeping up with this organization, it is clear that there will be differences in how humans interact with each other and how they interact with robots. Prior work has highlighted that robots do not have the same capabilities as humans and may require human assistance (Fischer, Jensen, & Bodenhagen, 2014; Pitsch, 2016). Robots can gain the status of an agent, but this status is transient, and robots may also just get treated as material objects (Alač, 2016; Pelikan, Broth, & Keevallik, 2022). Momentarily, people may respond to robots in similar ways as they would to other humans. In the next instant, the robot may turn into a plastic object. When analyzing human-robot interaction, it is thus essential to be particularly sensitive to how humans scaffold and adapt to robots in different ways. Rather than starting from the robot's perspective, asking "What does the robot perceive?", or a human perspective, asking "What does the robot's behavior mean?", a focus on interaction then encourages exploring how humans interact with robots. Building on this focus on interaction, I will scrutinize human-robot interaction in the home, in groupwork, and on public roads.

3.2 Action

Interaction involves the participation of at least two parties, who contribute to an ongoing collaborative effort. Actions in this perspective do not appear as in a vacuum, but always occur as action-in-interaction, as part of and contribution to ongoing interactional sequences. This section presents interactional perspectives on action as a multimodal, situated, and accountable phenomenon. Similar to the previous section, I start by discussing how human behavior gets recognized as action, and then outline implications for the understanding of robot actions.

3.2.1 Multimodal Gestalts

The body plays a crucial role in social interaction (Streeck, Goodwin, & LeBaron, 2011), an insight which has been stressed in the work on proxemics (Hall, 1966), spatial formations (Kendon, 1990) and ethnomethodologic studies of walking (Ryave & Schenkein, 1974). Viewing language as action, early conversation analytic and interactional linguistic work focused on talk (partly also due to the technologies available at the time, when most studies were conducted with audio tape recordings). Charles Goodwin (1979) was one of the first to demonstrate that gaze plays a crucial role in how turns evolve, stimulating a stronger interest in the interplay of talk and bodily conduct. The "embodied turn" (Nevile, 2015) in EMCA describes the development towards an increased analytic interest in the body. More recent research showed how emotion (Peräkylä & Sorjonen, 2012) as well as senses like touch (Cekaite &

Mondada, 2021) and taste (Mondada, 2019) are achieved in interaction. Similarly, a specific analytic focus on non-lexical vocalizations yields insights on how voice and body are intertwined (Keevallik & Ogden, 2020; Reber, 2012) and highlights that this interplay also features in the forming of grammatical utterances (Doehler, Keevallik, & Li, 2022; Keevallik, 2013).

In multimodal conversation analysis actions are studied as holistic, multimodal Gestalts (Mondada, 2019) that may involve language, gesture, gaze, body postures, movement and embodied manipulations of objects (see e.g. Broth & Keevallik, 2020; Streeck, Goodwin, & LeBaron, 2011). Taking a multimodal view, EMCA has demonstrated among others how people skillfully coordinate movement in different settings, ranging from dance (Albert, 2015; Broth & Keevallik, 2014) to traffic (Deppermann et al., 2018; Haddington & Rauniomaa, 2014). Particular practices, such as interactional closings, have also been demonstrated to heavily rely on embodied action and are negotiated through changes in body posture (Broth & Mondada, 2013; Tuncer, 2015). Manipulation of material objects has been explored in a broad range of settings ranging from surgery (Mondada, 2014) to video calls (Licoppe et al., 2017). EMCA work has further studied the engagement with material objects and technology in complex workplace settings (Heath, Knoblauch, & Luff, 2000; Heath & Luff, 2000; Suchman & Trigg, 1992).

Working with robots, which tend to have different sensors for each modality, makes it tempting to think of isolated communication channels. EMCA work highlights in contrast that body, voice, and facial expressions cannot be separated. They come as a full package, a multimodal Gestalt, and the whole body is involved in accomplishing actions. This is particularly important for my perspective on sound. Since robots may generate output through entirely different pipelines, sounds, movements, and text-to-speech output may not necessarily be perfectly synced. Robotic prosody is for instance not treated as inherent to an utterance, but added on top of lexical phrases that are modified to be uttered with different prosodic contours (Fischer, Jensen, & Bodenhagen, 2014; Yilmazyildiz et al., 2016).

3.2.2 Situated Action

Actions do not take place in a vacuum but occur in specific local contexts. As Linell (2009) highlights, situated meaning emerges as a combined product of meaning potentials and aspects of context; or vice versa, words and movements do not have a fixed meaning, but are only meaning potentials before they get placed into a specific context, where they then gain situated meaning. While many other disciplines agree that context matters, EMCA work takes a particular view on the notion of context. According to this perspective, context is not a static "bucket" but constantly unfolding. Actions are both shaped by the specific context and at the same time renew this context, which can then be considered a product of the participants' actions (Heritage, 1984). Humans design their actions to fit these specific contexts, building on what has just happened, their mutual history of previous encounters, as well as culture. In interaction, members produce and display their commonsense knowledge of everyday activities (Garfinkel & Sacks, 1970).

Lucy Suchman famously introduced this dynamic notion of context to HCI, challenging mainstream views on artificial intelligence at the time (for discussions

on the impact of her work see Heath & Luff, 2000; Sharrock & Button, 2003). Stating that actions are "situated", Suchman (1987) highlights that human actions are not indifferently following a pre-determined plan. Instead they evolve based on previous interactions and the specific situation. Suchman illustrates her argument through the example of navigation at sea: Micronesian sailors reach their destination without maps and compasses, by constantly observing and adjusting to the changing environment. They are navigating not according to a predetermined sequence of steps but rather adjusting to how the situation unfolds. Suchman argued that instead of focusing on interaction as a sequence of fixed steps, HCI needed to focus on how planned steps get practically adjusted when they are executed. Suchman (2007) later clarified that she is not arguing against plans, but rather that these need to be adjusted due to the local and situated dynamically changing situations.

While the concept of situatedness is tightly coupled with EMCA work, it can be viewed as slightly broader (which perhaps is also reflected in Suchman's gradual move away from ethnomethodology and towards STS and feminist studies). Situatedness captures the "thickness" (Geertz, 1973) of the context, in which associations that influence actions in the here and now may be evoked. As Stayton (2020) highlights with respect to the design of autonomous cars, a focus on situated actions acknowledges the local interpretation of designed moves: If a vehicle follows the law and strictly abides to a speed limit, this will naturally lead to different situated interpretations, such as that there is a conservative or non-local driver in the car.

Suchman's work remains relevant and can teach important lessons for HRI (see also Šabanović, 2010). A mismatch between how the designers planned an action trajectory and the situated emergence of events may lead to interpretations of robot behavior that are not in line with what the designers intended. Suchman (1987) has illustrated this vividly by transcribing for each moment in an interaction with a Xerox copying machine the *design rationale*, *machine effects available to the user* as well as *user actions available to the machine* and *user actions not available to the machine*. Clearly distinguishing between the perspectives of the designer, the user and the machine can help to uncover how misunderstandings and trouble emerges in situ. While some work in HRI has actively sought to design for situated interaction (Šabanović, Reeder, & Kechavarzi, 2014), the term "situated" is also used in a more loose sense to describe the differences between interaction in the lab and in the wild (Andrist et al., 2017).

3.2.3 Accountability

Accountability is a distinctive concept for how EMCA regards action and closely ties to the idea of reflexivity and meaning being co-constructed in interaction. EMCA highlights that a prerequisite for interacting with others is that others recognize one's behavior as intelligible action. What one is doing needs to contribute in some meaningful way to an ongoing interactional project. Following the ethnomethodological tradition, there is thus a normative component to action and interaction. Actions are accountable, that is intelligible or explainable (ten Have, 2002a). The term was originally introduced by Garfinkel (1967, p. vii), who highlighted that

> "ethnomethodological studies analyze everyday activities as members' methods for making those same activities visibly-rational-and-

> reportable-for-all-practical-purposes, i.e., "accountable," as organizations of commonplace everyday activities. The reflexivity of that phenomenon is a singular feature of practical actions, of practical circumstances, of common sense knowledge of social structures, and of practical sociological reasoning."

Put differently, the question is not what exactly someone is saying or in what way they are moving their arm, but rather what others are making of this behavior, which intelligible action they treat this behavior to accomplish (Garfinkel & Sacks, 1970). Dourish and Button (1998) use the example of someone saying "Hello" to highlight that it is not the word as such that makes it an action, for instance a greeting. Rather, it is the work that the word does in a particular interaction, where the specific circumstances (the situated context), make it recognizable as a greeting. A person that randomly goes around saying "hello" at the end of a party may no longer get treated as a competent member of society, possibly raising questions about their sanity or soberness. Yet, this is exactly the way in which robots sometimes behave, as they typically lack this commonsense understanding of when it is appropriate and normatively expected to perform certain actions and not others.

Ten Have (2002a, p. 8) explains the concept of accountability highlighting that "actors are supposed to design their actions in such a way that their sense is clear right away or at least explicable on demand". Commonly people succeed in designing their actions to be immediately intelligible. If one is not behaving in line with the expectations in a particular setting, others may explicitly ask for an "account", an explanation of why one is behaving in this particular way (Sacks, 1992). The design of intelligible actions is not limited to verbal activities but includes all behavior. Even movement in traffic is accountable and people use indicators, movement and positioning in a lane to help others project what they are about to do (Broth, Cromdal, & Levin, 2018). Non-lexical vocalizations are a particularly interesting case regarding accountability, since it may sometimes be hard to tell whether they are coincidental products of bodily strain or deliberate contributions. Vocalizations are therefore particularly suitable for navigating delicate matters such as recruiting help in an implicit way (Keevallik, 2018).

The concept of accountability highlights that it is not enough to use the right words (for instance a robot saying hello), but one also needs to be able to competently use them in interaction, deploying resources like language and movements in such a way that they appear as relevant and meaningful for others. Prior work has highlighted that even non-humans, including animals and AI can be "held to be sequentially accountable members" (Due, 2021, p. 141), stressing that people may not only expect other humans but also robots to act intelligibly. Discussions about trust and failure in HRI reflect the challenges that come with accountability (see e.g. Salem et al., 2015; Tolmeijer et al., 2020). While robots and conversational agents can succeed at displaying understanding of human actions and contributing recognizable actions for short stretches of interaction, they typically fail to do so at some point during an interaction. At these moments, questions arise about whether the conversational partner can be trusted, which may have larger implications for the interaction (for an ethnomethodologic perspective on trust see Ivarsson & Lindwall, 2023).

3.2.4 Robot Action

EMCA takes the stance that language, body, situated context and sequential timing contribute to the formation of actions. Together, they form "conformations designed to be, and to be recognizable by recipients as, particular actions" (Schegloff, 2007, p. xiv). The specific shape of the body influences the range of possible behavior for performing actions. Modifying human bodies, for instance by attaching an artificial tail to one's hips (Svanaes & Solheim, 2016), may afford new ways of interaction. Similarly, decisions about a robot's embodiment influence how the robot can participate in interaction. HRI researchers have started to challenge traditional ideas of robots as humanoid metal boxes. Soft, hand-crafted robot bodies (Suguitan & Hoffman, 2019), robothings (Zaga, 2021) and robotic objects (Anderson-Bashan et al., 2018; Tennent, Shen, & Jung, 2019) are just some examples of alternative robot shapes. These provide different opportunities for interaction than humanoid bodies. HRI has particularly explored movement as an expressive interaction modality for robots (Anderson-Bashan et al., 2018; Hoffman & Ju, 2014; Sirkin et al., 2015), demonstrating that robots do not necessarily need to copy human ways of moving to contribute actions.

The literature on robot sound highlights the tight connection to robot movement, which often results in consequential sound (Frid & Bresin, 2022; Moore et al., 2017; Robinson, Velonaki, & Bown, 2021; Tennent et al., 2017). Many HRI studies on movement sonification build on the idea that meaning can be "set" by the design team, and users only need to "correctly" decode it. Designed behaviors are often quantitatively validated to ensure that they get interpreted "correctly", in line with what the designers intended. This may seem at odds with the EMCA notion of holistic multimodal Gestalts that humans make sense of in situated interaction. However, it might also be seen as a way to deal with meaning potentials as a practical problem. Designers validate that robot sound gets interpreted in line with the intended meaning at least to a satisfying percentage. Reducing the range of typical interpretations in this way may be a way to limit the potential meanings.

The EMCA perspective highlights that it is not the producer of some behavior alone, who determines that they are now contributing an action. Others make sense of the specific behavior, and may recognize it as a relevant action in its situated and sequential context. From this perspective, the exact same behavior can be intelligible for some (such as the robot's designers or expert users) but not others (such as the novice users). A robot's behavior can be made sense of in different ways depending on (among others) the specific institutional and cultural context. From an EMCA perspective, robots can perform actions when their behavior gets recognized as meaningful by the participants in an interaction (Pelikan, Broth, & Keevallik, 2022; Suchman, 2007). Any behavior that a robot displays (such as a move, an utterance or just staying idle) can be interpreted as performing an action. Robots may therefore be held accountable, and are expected to behave in ways that are intelligible or explainable. Contemporary discussions of building trustworthy robots have acknowledged this problem, typically grounded in the psychological concept of trust. EMCA work can contribute insights on what situated explanations may look like, and offer alternative ways to think about why robots need to be transparent about what they will do next.

3.3 Designing for Action-in-Interaction

Humans carefully design their turns for specific recipients (Goodwin, 1979; Sacks, Schegloff, & Jefferson, 1974)[2]. This includes robots, which often get treated as participants with limited capabilities (Fischer, 2016; Pelikan & Broth, 2016). In this chapter I will focus not so much on how humans design their turns to adapt to robots, which is naturally achieved by doing recipient design, but instead on how robots can be designed to be intelligible for humans.

I will first describe intersections between EMCA and human-computer interaction and workplace studies, paying particular attention to efforts in embedding EMCA in the design process. I will then introduce EMCA work on human-robot interaction before discussing how this work can speak to efforts to design robots for social interaction.

3.3.1 EMCA and HCI

To understand how EMCA has contributed to and shaped studies of human interaction with computers, it is helpful to consider the history of HCI. As an interdisciplinary field, HCI has seen several paradigms and methods emerge throughout the decades, and EMCA has been more present during some periods than others. Three main phases seem to emerge, referred to as "waves" (Bødker, 2006) or "paradigms" (Harrison, Tatar, & Sengers, 2007). In the first phase, the focus was on human factors and ergonomics, as reflected in the title of the premier HCI conference "CHI Conference on Human Factors in Computing Systems". Initially, studies centered around single users in the lab, and influence came mostly from artificial intelligence and cognitive science, which applied the metaphor of a computer to humans.

As computers moved from laboratories to workplaces, new methods and approaches were necessary to study how they were used in complex organizational contexts, "centers of coordination" (Suchman, 1993, p. 114) such as air traffic control centers and cockpits (Goodwin & Goodwin, 1996), subway control centers (Heath & Luff, 1992) as well as Navy ships (Hutchins, 1995). Bødker (2006) characterizes this period as the second wave and Harrison, Tatar, and Sengers (2007) as the third paradigm of HCI, stressing that interaction is regarded as situated in this stream of research. Methods such as ethnography and interaction analysis become gain popularity in this paradigm, as they allow to include context and provide thick descriptions (Geertz, 1973). During this time, workplace studies emerged (Heath, Knoblauch, & Luff, 2000) and EMCA (Luff & Heath, 2000; Suchman, 1987), distributed cognition (Hollan, Hutchins, & Kirsh, 2000) and activity theory (Bødker, 1989; Kaptelinin & Nardi, 2006) gained popularity. As Bannon (1995) puts it, a shift in focus "from human factors to human actors" occurred and users' own perspectives were taken more seriously. In Scandinavia, worker unions called for stronger involvement of users in the design process leading to the development of participatory methods, which can be seen to be paralleled by user-centered design approaches emerging in the USA (Asaro, 2000; Greenbaum & Kyng, 1991; Zim-

[2]This is known as recipient design in EMCA, and has been called "audience design" in other disciplines (see e.g. Bell, 1984; Clark & Murphy, 1982).

merman & Forlizzi, 2014). Today, workplace studies are more typically part of the Computer-Supported Cooperative Work (CSCW) community.

Gradually the collaboration technologies developed during this time entered homes and public areas, which Bødker (2006) characterizes as a third wave of HCI, involving a move away from rationality towards emotion, suggesting that designers now need to embrace "people's whole lives" (Bødker, 2006, p. 6). Until today, researchers seek to overcome the Cartesian divide between body and mind, and advocate for involving the whole *soma* - body *and* mind in the design process (Höök, 2018). While EMCA perhaps was most popular during second wave HCI, it seems to have contributed to the establishment of a more nuanced and phenomenologically-rooted perspective than the original mind-as-a-computer paradigm.

Since the turn of the century EMCA work has mostly explored technologies that mediate human interaction (Arminen, Licoppe, & Spagnolli, 2016), including collaboration in text-based interfaces (Due & Toft, 2021; Musk, 2021), interaction through social media (Reeves & Brown, 2016) and video-mediated interaction (Broth, Laurier, & Mondada, 2014; Gan, Greiffenhagen, & Reeves, 2020; Licoppe et al., 2017; Luff et al., 2003), as well as mobile technologies such as phones (Arminen & Weilenmann, 2009; Brown, McGregor, & Laurier, 2013) and telepresence robots (Boudouraki et al., 2021; Due, 2021; Jakonen & Jauni, 2021; Nielsen, 2020). Studies on voice user interfaces, virtual agents and robots may manifest an emerging new stream of work that focuses on direct, spoken interaction with the computer (Krummheuer, 2010; Porcheron et al., 2018; Tuncer et al., 2023; Yamazaki et al., 2009).

3.3.2 Intertwining EMCA and Interaction Design

Before taking a closer look at how EMCA has studied robots, it is helpful to reflect on the perspective that EMCA has taken on human-computer interaction. Grounded in an interest in how people use technology, EMCA work has typically fallen in the empirical studies track at HCI conferences, showing the skillful ways in which people interact with each other and how this interaction is facilitated, or hindered by technology, demonstrating novel practices that emerge in using this technology. Typically, discussions take a critical-evaluative stance, pointing out problems in the design of the technology, rather than actively shaping technology or participating in the re-design of existing technology (Reeves, 2022).

This is not necessarily because EMCA researchers do not want to engage in design work more actively. Garfinkel (1986) has actively advocated so called *hybrid studies*, in which ethnomethodologists gain *unique adequacy* - knowledge in the domain that they are studying, which enables them to contribute to the field that they are studying. Hybrid studies were less successful in disciplines such as mathematics, where practitioners did not see the value in ethnomethodologic descriptions of their own work practices (c.f. Greiffenhagen & Sharrock, 2019). HCI is a different case, since "the primary audience of that HCI study are not the practitioners in that setting, but instead system designers who are developing technologies for that setting" (Greiffenhagen & Sharrock, 2019, p. 276). As the continued presence of EMCA work at HCI conferences demonstrates, interdisciplinary communities in-

volving designers, engineers and computer scientists seem to value learning about how technologies are actually used.

Despite seeking to inform design, it appears that entering a dialogue and close collaboration with system and interaction designers historically has not been easy. In the middle of what Bødker (2006) called the second wave of HCI, Button and Dourish (1996) articulate this difficulty, arguing that:

> "Ethnomethodologists interested in technology and design have thus so far, and after much effort, managed to make the first step from the study of the use of technology to the critique of technology. The next step however is yet to be taken and will be more problematic which is to move from critique to design."

To enable such move from critique towards design, Button and Dourish proposed a hybrid discipline that would combine design and ethnomethodology into a "technomethodology", that is designing HCI based on an ethnomethodologic understanding of interaction. Interestingly, while Button and Dourish (1996, p. 21-22) stress that mutual learning will take place when combining ethnomethodology and design in this way, they only demonstrate how system design can learn from ethnomethodology. A methodological reflection of what ethnomethodology will learn from system design is missing. Until today there have been rather cautious attempts at intertwining EMCA and design in this way. Ethnomethodologists and computer scientists have collaborated in the development of platforms for mediated interaction (Aoki et al., 2002; Aoki et al., 2003; Fraser et al., 1999; Hindmarsh et al., 2000; Woodruff & Aoki, 2004; Woodruff et al., 2002), and more recently more examples of EMCA informing the design of voice interfaces are emerging (McMillan et al., 2019; Moore & Arar, 2019).

While many have recognized that the practices described by ethnomethodology and conversation analysis can be useful for design, there has been debate how exactly they can be applied. The metaphor of human-computer interaction as "conversation" was popular even before voice interfaces were invented, and the prevalence of technical metaphors such as "machinery" (Sacks, 1984) and "rules" (Sacks, Schegloff, & Jefferson, 1974) in early conversation analytic work seems to have made a direct translation particularly appealing (see e.g. Frohlich & Luff, 1990; Norman & Thomas, 1991; Norman & Thomas, 1990). Others have been more hesitant and challenged the idea of conversation between humans and machines (Button, 1990). Button argues that what Sacks, Schegloff, and Jefferson (1974) call "rules" may better be coined as "practices" (Button, 1990) to stress that the observed patterns in human interaction are not easily implementable rules but much more complicated (see also Clayman & Maynard, 1995, p. 17 for a discussion of how Sacks' use of mechanistic terms may have fostered this misunderstanding). Today, voice user interfaces have reinforced the idea of conversing with computers, and contemporary EMCA work has again challenged this notion (Reeves, Porcheron, & Fischer, 2018).

A general struggle in combining ethnomethodology and design has been the differences between specific and abstract (Button & Dourish, 1996). This becomes evident in an early argument on how interaction between users and visual interfaces could be productively informed by conversation analysis (Norman & Thomas, 1990, p. 55, emphasis in original):

> "Guidelines generated on the basis of Conversation Analysis offer the premise of advice *specific* enough to provide the designer with leverage into the design of particular interactional sequences, yet *general* enough to be applicable to current and emerging technologies."

Since EMCA scholars are usually not trained as designers, they cannot make this translation themselves but rely on the goodwill of others to build designs based on their suggestions. This puts the EMCA expert in a service position, where they can only hope that their offers will be taken up by designers (see also Reeves, 2022).

As I highlighted in the previous chapter, what is described as "system design" by Button and Dourish has changed quite considerably since the 1990s and is today more commonly known as interaction design (see e.g. (Löwgren, 2014)), which does not always have to build a finished product but may also be more research focused (Zimmerman & Forlizzi, 2014; Zimmerman, Forlizzi, & Evenson, 2007). As the interdisciplinary field has evolved, it may be easier to seek a dialogue between EMCA and design research than it was in the 1990s. When applying EMCA to HRI, I am building on this prior work that has not only studied how humans interact with technology, but also explored how it can be embedded in its design.

3.3.3 EMCA's Place in HRI

HRI as a field is only starting to mature, with the *18th ACM/IEEE International Conference on Human-Robot Interaction* taking place in 2023. Paralleling the early days of HCI, robots were only studied in the lab due to technical constraints, with some researchers calling for studies in situated interaction outside of laboratories (Dautenhahn, 2018; Jung & Hinds, 2018; Šabanović, 2010). Recent work has demonstrated that the field remains dominated by computer scientists, engineers and psychologists (Winkle et al., 2022), who call for more formalized experimental lab studies to ensure replicability (Hoffman & Zhao, 2020; Kress-Gazit et al., 2021). Others criticize that this makes the field too narrow, and propose alternative approaches, including EMCA (Fischer, 2021). Recent work problematizes the treatment of study participants as interchangeable subjects that are removed from context (Lee et al., 2022) as well as the dyadic one-robot-one-user paradigm (Hornecker et al., 2022; Schneiders et al., 2022) that is still prevalent in HRI work. HRI aims to be an interdisciplinary field and the community seems open to recent attempts to broaden design research in HRI including also more critical and participatory practices (Lupetti, Zaga, & Cila, 2021; Serholt, Ljungblad, & Bhroin, 2022; Weiss & Spiel, 2022). The calls for more critical and situated work highlight that EMCA work can make contributions that will be valuable for the HRI community. At the same time, recent discussions about more rigorous methodology may highlight a need for thorough, mature approaches to studying human-robot interaction. EMCA can contribute such an empirically grounded and mature approach, which goes beyond more general qualitative insights based on thematic analysis. Particularly in its focus on studying deviant cases, EMCA provides a way to analyze "outliers" or robot failures in a systematic way.

While not always addressed at the HRI community, a growing body of EMCA work has investigated how people interact with robots in a variety of settings, including museums (Arend, Sunnen, & Caire, 2017; Pitsch et al., 2009), shops (Velkovska,

2019), pre-schools (Alač, Movellan, & Tanaka, 2011; Ferm et al., 2015) and healthcare (Krummheuer, Rehm, & Rodil, 2020; Robins et al., 2004). Specific attention has been given to openings (Gehle et al., 2017; Pitsch et al., 2009) and closings (Licoppe & Rollet, 2020) of encounters between humans and robots. Some of this work has sought the dialogue with roboticists, exploring how robots could be used to support specific users (Due, 2022; Krummheuer, Rehm, & Rodil, 2020; Rehm et al., 2016), how to establish a focused encounter with a robot (Gehle et al., 2017) and identifying what aspects of interaction a robot needs to perceive (Tuncer, Gillet, & Leite, 2022). A small body of work has attempted to bring EMCA interaction principles into the design of robot behavior, focusing on the case of robots as museum guides (Gehle et al., 2017; Kuzuoka et al., 2008; Pitsch et al., 2009; Yamazaki et al., 2009). However, as Pitsch (2016) points out, situatedness and contingency may not be completely addressable by a robot, highlighting that there may be limits to designing robots as stand-ins for humans. Pitsch argues that instead robots and humans should be seen as one interactional system, and Alač, Movellan, and Tanaka (2011) highlight how roboticists scaffold a robot to be social in an experimental setting. Recent HRI work has also acknowledged that robots rarely are completely autonomous, and often get assisted by "robot wranglers", who scaffold robots in real world settings (Takayama, 2022).

While EMCA has made a contribution to HCI by critiquing existing approaches, it can take a more active role in HRI: To avoid having to redo work that has been done, EMCA approaches to HRI may benefit from going hand-in-hand with HCI. Rather than treating HRI as utterly distinct from HCI and introducing concepts like situated interaction from scratch, it may be beneficial to build on how they have been established for the interaction with computers, and to instead explore how they apply to human-robot interaction, how they may need to be tweaked, and what elements in the organization of human interaction that have not been relevant for computers may be relevant for robots. In this process, EMCA robot studies can learn from discussions and frustrating experiences in HCI (Reeves, 2022), and avoid misleading promises such that turn-taking "rules" (Sacks, Schegloff, & Jefferson, 1974) could be easily implemented in robots. Instead EMCA approaches to HRI may engage more productively with the question how robots can be designed to deal with the contingencies of the social world, that have been already demonstrated in HCI and CSCW.

4 Methods and Data

Video data plays a central role in this work. Taking an ethnomethodology and conversation analysis approach, I drew on video recorded interactions as empirical data. Exploring intersections with interaction design, I used the same video data as design material. I will first introduce the background to these different approaches and describe practical considerations. I will then introduce the three different video corpora that I worked with, featuring three different robots. I recorded a Cozmo palm-sized toy robot in family homes and autonomous shuttle buses on public roads. I also built on videorecorded groupwork sessions with the Vero robot, a researcher-controlled interactive video prototype. Finally, I will discuss my own background and training, highlighting how it enabled me to carry out this work.

4.1 Video as Data

Taking an EMCA approach, video recordings of naturally occurring interaction take a central role. I followed conversation analytic procedures for data analysis (Broth & Keevallik, 2020; Heath, Hindmarsh, & Luff, 2010; Hutchby & Wooffitt, 2008; Sidnell & Stivers, 2012). I reviewed all video recordings and fieldnotes, noting down observations. While I did not bring any specific hypotheses to the data, I paid particular attention to non-lexical aspects of interaction. I worked with ELAN (Wittenburg et al., 2006) and DOTE (McIlvenny et al., 2022) to annotate general observations and to make first transcript drafts. The systematic marking of all robot sounds and human reactions in my first recordings in the Cozmo corpus helped me to get a better grasp of the sounds that occurred in the data and whether humans generally treat them as relevant for the interaction. Working with this corpus helped

me to develop a sense of how robot sound features in interaction, which has then informed the way in which I looked at the other corpora. In the bus corpus I paid particular attention to the interplay of sound and movement. In the Vero corpus, I was particularly interested in comparing human vocalizations to the behavior of the robot. Generally, I developed an interest in moments when a smooth flow interaction was disrupted as these can turn into opportunities for (re-)design. In the following sections, I will provide more details on the steps that I followed. While mapping the data, I started to build collections of similar clips. While doing so, I carefully watched and transcribed selected extracts. I applied the next-turn proof procedure to deepen my understanding of the transcribed extracts. To develop ideas and to validate observations, I discussed video clips with other researchers at data sessions throughout the analytic process.

4.1.1 Building Collections

As similar cases gradually emerged when going through a corpus, I grouped these clips into collections, typically by clipping video snippets and saving them in separate folders for later analysis. Collecting similar instances is particularly easy when working with recordings of programmed robots, since their turns can be repeated infinitely. This provides an exceptionally regular ground for building collections, as I could easily identify video segments in which an identical sound is played in a programmed sequence in different situated contexts. For the Cozmo robot, I collected all recordings of a face-learning sequence that the robot can perform. This gave me the opportunity to scrutinize how people react to that particular programmed behavior. For the buses, I collected video snippets in which the bus was braking in between stops - often an indicator of failed coordination. This data is more complex and exhaustive searches for all cases were more difficult. Skimming through the data with the help of my fieldnotes, I could build collections that reflected the range of observed behavior. At later stages of the project, I mapped the moments in which we tested different sound designs. As the data also features longer stretches in which no sound was played, focusing on stretches in which people were around the bus was helpful. Again, I built collections in a slightly opportunistic manner, aiming to reflect a range of different reactions (from no visible reaction to gazing at the bus or swerving). For the Vero corpus, the recordings are comparatively short, and I could watch all clips in full length several times. The creation of collections was strongly design-driven, and I opportunistically selected clips for design sessions focusing on moments in which Vero was relatively silent. Since Vero moved differently when the person controlling the robot was speaking, moments of Vero being silent could also be identified easily when skimming through the data.

4.1.2 Transcription

An important conversation analytic tool is a detailed transcription of interaction, in which talk (Jefferson, 2004), movement, gaze and gesture (Mondada, 2019) as well as engagements with material objects (see e.g. Licoppe et al., 2017) are noted down. Transcripts enable documenting and pinpointing the sequential unfolding of interaction and serve as a shared resource for re-inspection and re-analysis (Laurier, 2014). Writing down multimodal behavior helps to visualize matters that would

otherwise evade attention. A transcript illustrates and captures phenomena of interest but at the same time supports the comparison of different interaction snippets and stimulates discussion about what can be seen in a particular video. Transcripts are never final and fixed. Transcription is analysis (Roberts, 2012), meaning that transcripts do not represent an objective truth, but are necessarily shaped by the subjective perspective of the researcher (see e.g. Ochs, 1979).

While conventions are well-established for transcribing verbal and embodied behavior, conventions for the transcription of non-lexical sound, specifically machine sound are lacking. While Schegloff (1968, p. 1080) analysed machine sounds as summons that need to be answered, he typically refrains from transcribing them and instead starts his transcripts with someone answering the phone[1]. To learn more about how robot sound is treated in interaction I found it crucial to include machine sound in my transcripts. Specifically for analyzing in detail how sound affects the temporal organization of actions, I produced more thorough transcriptions of how a robot sounds in a local context. Taking inspiration from Jefferson's (2004) transcription system for verbal utterances, I drew on the letters of the alphabet to capture what I heard. I noticed early on that despite dealing with non-lexical sounds, I had to pick a language for transcription, since individual letters are pronounced differently in different languages, and onomatopoetic renderings may differ quite substantially. A car horn may be transcribed as "beep" in English (Laurier et al., 2020), but a German may capture the same sound as a "möp" or "tuut". Since my publications are in English, I chose to transcribe robot sound in English for practical purposes. The spectrum of sounds that robots can generate is much broader than human vocalizations, and standard orthography is limited in capturing how a robot actually sounds. In some way, this poses similar challenges as transcribing bird sounds. As ten Have (2002b, p. 26) points out, "the language of humans is of limited use in providing a recognizable image of calls and songs produced by birds". For more complex sounds, I also looked at spectrograms, which show the spectrum of frequencies of a selected sound. Such visualizations can help to demonstrate the interplay of beeps and motor sounds (Article I) and to highlight differences between sounds (Articles I and II).

Another issue with transcribing robots is that because they are programmed, they can infinitely repeat the same behavior (see also Pelikan, 2024). While the specific sequence still depends on context, such as whether Cozmo can detect a face in a given amount of time, the analyst can identify the underlying script after looking at a number of recordings of the same behavior. I have chosen to make this script explicit (see Article II), when it enabled me to recognize moments in which the robot deviated from its typical sequence. Providing a detailed multimodal transcription of the animation in focus (Article I, Article III) allowed me to later refer to a shorter or simplified version.

4.1.3 Next-Turn Proof Procedure

To formulate conversation analytic observations, the next-turn proof procedure (Sacks, Schegloff, & Jefferson, 1974) is an essential guide for the analysis. Seeking

[1]For an exception see Schegloff (2007, p. 131- 132) where participants discuss whose phone is ringing.

answers to the question "why that now" (Schegloff & Sacks, 1973, p. 76) analysts can inspect how a turn relates to a previous turn and study how it is responded to. Put differently, in their next turn, humans display their understanding of the preceding turn, which is then also available for the analyst. Working with robots, the application of the next-turn proof procedure can be challenging, because robot behavior may not always (recognizably) build on previous human actions. In human interaction, speakers typically follow the turn-taking systematics described by Sacks, Schegloff, and Jefferson (1974), minimizing moments at which speakers are talking at the same time. In human interaction "overlaps in particular are bound up with the management of turn-taking and the observable achievement of mutual understanding in talk-in-interaction" (Hutchby & Wooffitt, 2008, p. 74). Participants do seem to apply human turn-taking practices when interacting with a robot. For some of Cozmo's sounds, people abandon their turns when overlapping with the robot. However, robots are not taking turns in the same way as humans do. This results in many overlaps, with the robot producing sounds while participants are speaking. In these cases in particular it may be unclear for the participants and the analyst whether the robot's sound should be treated as a response to what people have been doing or as a somewhat "random" move. When robots move or sound at relevant moments in the interaction, their behavior tends to get recognized as a contribution to the interaction, and can be treated as an action. However, there may also be aspects of what a robot does that are not treated as relevant by the participants. The challenge for the analyst is to carefully inspect how people react to the robot, refraining from stepping into a positivist view of regarding anything that the robot does as an action. Staying with the participants' understanding means being open to the fact that a robot's agency status is not fixed but changes dynamically throughout an interaction (Pelikan, Broth, & Keevallik, 2022). Related to this, achieving sufficient mutual understanding to progress to a next relevant action in an extended sequence is a fundamental challenge in human–robot interaction. Robots can initiate sequences by providing a first turn that humans build on in their response (second turn). However, robots typically struggle with *third turns* (Schegloff, 2007), responses to a human response. While there are often strong expectations for what constitutes an appropriate second turn (for instance an answer to a question), the third position is more flexible and can range from a sequence-closing "okay" to more complex expansions of the second turn. Interestingly, third turns are often missing in the way robots are designed, perhaps because robot design tends to build on an input/output logic where a response to the output is not considered.

Determining how a robot "interpreted" a human turn can a practical problem for both the participants in the interaction and the analyst, and questions about the inner workings of the technology arise easily. The functioning of robots' sensors (what they can perceive) and actuators (what range of behavior is possible) is essential for how interaction evolves. Since I focused on the perspective of the human participants interacting with the robot, I have not studied log files of the interactions. However, a basic familiarity with the underlying technology has been crucial when disseminating recommendations, ensuring that the findings could contribute to design. Similar to the families who interacted with Cozmo and the safety drivers on the buses, I successively gained practical experience in how the robots work and the problems that typically occur through my fieldwork and video analysis.

4.1.4 Data Sessions

Data sessions have been a particularly important part of my work. At a data session, researchers meet to discuss and jointly analyze data, usually a video clip accompanied by a transcript brought by one of the attendees. After watching the video snippet several times to get familiar with the data, attendees start formulating their understanding of (or questions about) what is happening, which may be challenged by others. As ten Have (2007, p. 140, emphasis in original) highlights, "What is most important in these discussions is that the participants are, on the one hand, *free* to bring in anything they like, but, on the other hand, *required* to ground their observations in the data at hand". This is where EMCA analysis is very noticeably different from other forms of video analysis. It forces the analysts to focus on what can be observed, rather than offering guesses about what someone may have intended or sorting behavior into pre-defined categories. As Jordan and Henderson (1995, p. 44) point out: "Collaborative viewing is particularly powerful for neutralizing preconceived notions on the part of researchers and discourages the tendency to see in the interaction what one is conditioned to see or even wants to see". Through repeated watching of the video and reading of the transcript and joint discussion, the group eventually starts to converge on formulating a shared understanding of what people are doing in the video recording. In this sense, data sessions are inherently dialogical (Linell, 2009). They facilitate building scientific knowledge through discussion with others. Data sessions were therefore an important step in validating my observations and I presented data from all the articles in this dissertation at data sessions. During my PhD studies, I chaired two data session series (AI in Interaction, online with Philippe Sormani and the Robot's in Groups data sessions, at Cornell University with Malte Jung). While getting feedback on my own data has crucially informed my analysis, even engaging with other's data has been important in forming a better understanding of how people typically interact with robots and what types of problems may persist with robots, including those that I have not specifically worked with.

Since the discussion format is particularly useful for teaching and learning conversation analysis (Heath, Hindmarsh, & Luff, 2010; ten Have, 2007) data sessions made it easy to involve people with different backgrounds. This turned out as particularly important for the collaboration on Article IV. The data sessions formed a bridge between EMCA and interaction design by gathering the interdisciplinary research team around a specific video clip that we then also used for design. In the following sections I will describe how we turned the video data into a design material.

4.2 Video as Design Material

Video is not only central to conversation analytic work, but there is also a tradition of using video for the design of human-computer interaction (see e.g. Harrison et al., 1989; Mackay, Ratzer, & Janecek, 2000; Suchman & Trigg, 1992; Vertelney, 1989; Ylirisku & Buur, 2007). Video can be used at different stages of design and is specifically beneficial for illustrating how a not-yet-developed system may look like (Henderson, 1989). In the late 1980s, replaying and editing physical video tapes was

a cumbersome process. Early studies focused on recording intermediate prototypes and discussions between designers (Minneman, 1991; Suchman & Trigg, 1992; Tang, 1989). In HRI, video-recorded prototypes remain a popular tool for studying how potential users perceive a robot or how they interpret what a robot does (Syrdal, Otero, & Dautenhahn, 2008). Video is a cost-effective way to give more interactional context than a picture could convey (c.f. Randall & Šabanović, 2023). Article IV also draws on a video-prototype of the Vero robot. Rather than only showing a pre-recorded sequence, Vero is interactive and different animations can be triggered during an interaction in real time.

Video can also be used beyond capturing scenarios or finalized designs. Interaction designers can use video to sketch out possible interactions, recording a range of ideas during a brainstorming phase (Mackay, 2002). Using the full potential of today's video editing tools, video can thereby "directly aid invention" (Henderson, 1989, p. 105). In the following sections, I will describe how I prototyped possible robot behavior without implementing it. I will introduce techniques that informed my sound design process and explain how I turned video into a design material. I will also highlight participatory elements in my design work.

4.2.1 Improvising a Robot's Behavior

My work can be best described through a Research through Design (RtD) paradigm that describes how knowledge can be gained by engaging in design activities (Zimmerman & Forlizzi, 2014; Zimmerman, Forlizzi, & Evenson, 2007). In contrast to commercial design practice, which is concerned with making products that can be sold, RtD explores what futures could and should be designed, asking "what is the right thing to design" (Luria, Zimmerman, & Forlizzi, 2019). RtD does not necessarily strive for building fully functioning prototypes but focuses on what can be learnt on the way. I specifically focused on low-fidelity designs, drawing on rapid prototyping techniques (Hoffman & Ju, 2014; Sirkin & Ju, 2014; Zamfirescu-Pereira et al., 2021).

A common way to investigate capabilities that are not yet there, is the *Wizard-of-Oz* (WoZ) paradigm, in which a researcher acts through the robot, without users knowing that the robot is controlled by a human. It was initially developed for testing natural language dialogue systems (see e.g. Dahlbäck, Jönsson, & Ahrenberg, 1993; Porcheron, Fischer, & Reeves, 2020). Even in HRI, WoZ is a popular tool for exploring how users would react to a robot's verbal and non-verbal behavior (Riek, 2012)[2]. In the context of rapid and exploratory prototyping, WoZ is typically used with the goal to quickly prototype a range of ways in which a robot could behave in different contexts (Bartneck & Jun, 2004; Zamfirescu-Pereira et al., 2021). Such exploratory work is more improvisational in nature, acting out different possible behaviors in real time (Sirkin et al., 2016; Sirkin & Ju, 2014; Sirkin et al., 2015). The Vero robot prototype draws on this improvisational application of the WoZ

[2]EMCA work has described in detail how an experiment leader and the person who covertly acts as the wizard cooperatively manage the performance (Porcheron, Fischer, & Reeves, 2020). Exclusively focusing on the interaction between the (enacted) robot and the user, one may miss how the interaction is collaboratively achieved. In my work I therefore carefully scrutinized actions as enacted by a human wizard, rather than being a true representation of what the robot will be doing in the future.

paradigm, with a researcher-wizard testing a range of different ways in which Vero could participate in a groupwork activity. In my work with autonomous buses I also adapted the WoZ paradigm for testing sounds on the road.

4.2.2 Sound Design

Working with sound design requires both careful listening and techniques for creating sound. Paying attention to everyday sounds (Gaver, 1993; Gaver & Norman, 1988) is an important element of sound design, and fits well with analytic practices of listening carefully in conversation analysis. A crucial difference is a shifted focus away from human voices towards sounds coming from the (moving) body when walking (see e.g. Westerkamp, 1974) or the environment, including machine sounds (Bijsterveld & Krebs, 2013). Particularly when exploring sound for the autonomous buses, I took inspiration from *soundwalking* (Baharin, Rintel, & Viller, 2013; Westerkamp, 1974), excursions during which one carefully listens to the environment. During my rides on the buses, I paid careful attention to the sounds of the bus. I also walked along the route, listening to environmental sounds along the way. During my design process, I tested sound ideas during these walks to explore how they blended into the existing soundscape and observing whether people would react to them.

In exploring how robots can make use of sound in interaction, I had to come up with my own sound designs. Following a rapid prototyping approach (Bartneck & Jun, 2004; Hoffman & Ju, 2014; Sirkin & Ju, 2014; Zamfirescu-Pereira et al., 2021) and focusing on low-fidelity, my goal was not to create aesthetically complex designs. I was rather interested in exploring how sound can be heard as a "sounded doing" (Garfinkel & Wieder, 1992, p. 197) by the robot, accomplishing actions in interaction. I therefore drew on sound design methods that lend themselves for quick iteration. In particular, I drew on *vocal sketching* (Ekman & Rinott, 2010; Rocchesso, Serafin, & Rinott, 2013), using one's voice to prototype different sounds. This method allows to immediately start experimenting with different sounds, and the designer can move at the same time, enacting how the robot should move. Vocal sketching is not always immediately comfortable for everyone (Rocchesso, Serafin, & Rinott, 2013) and previous work in HRI invited a professional mime actor to perform vocal sketching (Panariello et al., 2019). In Article IV, I describe how we came up with a specific warm-up exercise before engaging in vocal sketching as a group.

Besides sketching with my own voice, I used sound samples from online databases and created simple jingles in Audacity[3], an open source audio software. In sonic interaction design, samples are used for *sonic overlays* (Rocchesso, Serafin, & Rinott, 2013), in which the sound track of a video recording is modified by adding the designed sounds. In HRI, this technique is applied to video recordings of robots to evaluate how potential users rate different ways in which a robot could sound (Robinson, Velonaki, & Bown, 2021; Tennent et al., 2017). I took inspiration from such overlays but combined them with vocal sketching, repeatedly vocalizing over a looping video. I thus performed voice-overs on recordings of the robot in interaction, exploring alternative ways of how the robot could have sounded. Brief video snippets

[3]https://www.audacityteam.org

from my collections were an excellent basis for this. Transitioning from the data sessions in which video is looped and rewatched, to design, we performed a number of iterations of possible alternative ways for the robot to sound. The video recordings then turn into a canvas for sketching possible robot sounds and how they could be placed in specific interaction sequences (see Article IV for a detailed description of the methodology). Video is a resource that can bridge EMCA and interaction design, since it allows transitioning smoothly between analysis and design. I explored using my video recordings not only as data for analysis of how humans do things with robots, but also as a design material for prototyping robot sounds and possible interactions.

Combining sound design with the WoZ paradigm was particularly useful, giving me the opportunity to test audio on the autonomous buses while observing how people would react. The WoZ paradigm typically requires some interface, which enables the researcher to control the robot. For prototyping sound, I found that a phone app that could play sounds through a Bluetooth speaker taped to the front of the bus was enough. I experimented with smartphone apps that enabled me to play different sound files on a button press, such as a beep button and a hand bell app. Since the Bluetooth speaker already introduced a small lag, it was particularly important to find ways to trigger the sounds as fast as possible. The hand bell app did not even require looking at the phone and was preferred by the safety driver, who also acted as a wizard.

4.2.3 Participatory Design

The EMCA focus on participants' understanding lends itself well to a participatory approach (see also Greenbaum & Kyng, 1991). In HRI, participatory design (PD) aims to go beyond asking users what they want, towards actively involving them as co-designers (see e.g. Lee et al., 2017; Stegner, Senft, & Mutlu, 2023). Ideally, the different skills and understandings that lay and professional designers bring to the table allow mutual learning, but they can also make it difficult to collaborate on equal footing (c.f. Axelsson et al., 2022; Lee et al., 2017; Weiss & Spiel, 2022). Typically, PD processes focus on conceptual work, exploring how a robot could be designed to support a specific user group in a particular setting (Azenkot, Feng, & Cakmak, 2016; Björling & Rose, 2019; Krummheuer, Rehm, & Rodil, 2020; Lee et al., 2017; Weiss & Spiel, 2022).

In my study on autonomous buses, I worked in a setting where the robot is already deployed. During my explorations of how the buses could be modified to improve their coordination abilities, I involved participants that professionally work in the setting as safety drivers. I therefore drew on slightly different methods, than the commonly deployed formal interviews, focus groups and workshops (see e.g. Axelsson et al., 2022; Azenkot, Feng, & Cakmak, 2016; Lee et al., 2017). A close collaboration with the safety drivers grew out of my fieldwork, during which I continuously visited the safety drivers at their workplace, on the autonomous buses. I could only take rides if the buses were running and if the safety drivers agreed to have me and my equipment on board. Since I did not have access to the buses without the safety drivers, power dynamics were different than in typical PD studies, in which the researchers are in control of the robot. While the safety drivers were

always welcoming, it was clear that passengers would always take priority, and as soon as someone stepped on the bus, I had to pause any unstructured interviewing or discussion about design choices.

Throughout the project, the safety drivers became more familiar with typical problems that the buses would face and through my video observations, I learned along with them. I discussed my observations informally with the safety drivers during the rides and presented my findings formally at meetings where representatives of the bus operators were present. My observations overlap with challenges identified by the bus project team (Anund et al., 2022). I therefore started to explore sounds aiming to address the identified coordination problems. I contributed the focus on sound based on my expertise, while the safety drivers drew on their professional expertise when instructing me how sound can be used on public transport vehicles. I worked particularly closely with one safety driver who was eager to do the sound tests together. As he made nuanced observations of what happened on the road, I considered this an especially good basis for learning from each other, as we could share and discuss our observations. He always wanted to listen to the sounds first before letting me deploy them on the road, and I dropped some of the sounds because he strongly opposed them. I learnt early on that I could not play sounds at random moments, but playing honking sounds from public transport vehicle is only okay when an obstacle or a dangerous situation is ahead. At times, the safety driver would instruct me to play a sound, as he observed emerging problems on the road. When I asked the safety driver early on whether he would like to play sounds, he refused. However, after several sound testing sessions, as we had mutually learnt about each other's work, he volunteered to play sounds himself. This yielded more nuanced results than my own tests, in which I had focused on emulating the ways in which a machine could trigger sound. The safety driver clearly draw on his professional experience in coordinating movement on the road when he triggered the sounds. I presented my analysis of how the sounds were reacted to both to the safety drivers and their manager, which stimulated discussion on how the findings could be deployed on the bus.

4.3 The Video Corpora

I am particularly interested in exploring sound in non-anthropomorphic robots. To explore how my findings generalize from one robot to another, I identified two robots of different shapes and with different purposes: The Cozmo robot is a small, palm-sized robot designed for playful interaction in the home. Autonomous shuttle buses are large robots that hold space for up to 15 adults, designed for last-mile transportation on public roads. Exploring sound design, I worked with the Vero virtual team assistant, which is a non-commercial research prototype developed by the Robots in Groups Lab at Cornell University. Rather than having one fixed shape, Vero is a concept that explores robots in teamwork settings. While the physical Vero prototype consists of a floating ball-shaped microphone, I focused on a video prototype of Vero as an animated sound-wave on a screen.

The three different robots which feature in this work have been video recorded in three different settings. In the following sections I will describe these settings, the participants who interacted with the robots and the robots themselves. I will also

describe how I worked with the different corpora and provide considerations that are specific to each robot. Table 4.1 provides an overview of the three corpora.

Robot (Manufacturer)	**Robot type**	**Setting**	**Participants**	**Country (Language)**	**Year**
Cozmo (Anki)	Palm-sized toy robot with animated display face and forklift arms	Home	Families, adult pairs	Sweden (Swedish), Germany (German)	2019-2020
Autonomous shuttle buses (Navya and EasyMile)	Autonomous vehicles for public transport	Public roads	Safety drivers, pedestrians, cyclists, car drivers	Sweden (Swedish)	2020-2022
Vero (Robots in Groups Lab, Cornell University)	Robot prototype for teamwork controlled by a researcher	Zoom calls	University students	USA (English)	2021

Table 4.1: Overview of the video corpora featuring in this work.

4.3.1 Corpus I: Cozmo in Family Homes

Since I was interested in how people make sense of robot sound in everyday interaction, I decided to study robots in the most ordinary and intimate context: people's homes. I collected video data of participants playing with a Cozmo toy robot in family homes in two countries, Germany and Sweden. In both countries I visited participants in their homes and accompanied them during their initial encounter with the robot. In the German study, I asked participants to interact with the robot for as long as they wanted, which turned out to be between 10-30 minutes. I was in control of the robot's app but mostly let the robot roam freely, a mode in which the robot takes randomized initiatives. In the Swedish study, families played with the robot for 8-13 days. They controlled the robot through their own smartphones and were free to explore all its functions. I recorded participants during introduction and retrieval of the robot and they recorded themselves with a simple camcorder on a tripod when interacting with the robot during the days in between. During my drop off and pick up visits, I recorded with two video cameras from two different angles, one focusing on the robot and one on participants. The families were instructed to record not only the robot but also their faces and bodies and were asked not to stand behind the camera but to join the everyday family activities. In all cases, I explained the aim of the study and obtained written consent before cameras and the robot were turned on.

The Cozmo Robot

Anki's Cozmo robot (see Figure 4.1) is a palm-sized robot inspired by Pixar's Wall-E and Eve, which is marketed as a toy for children in the age of 8-14. Lukas a teenage boy who has owned a Cozmo robot for a year, presents Cozmo as follows:

> "I describe Cozmo as a tiny tiny little robot that can express itself in different ways [gestures around his own face] that can go forward and backwards, turn and do things with the arms. So kind of like a small friend, it can talk but if you say something to it, it will not respond." (February 2020, translated from Swedish).

Cozmo's designer puts it in slightly more technical words:

> "Cozmo has two treads, has this lid that moves up and down, has an LED screen with a front-facing camera, which a lot of people confuse for a mouth. Also there is just that little speaker on top of his head. So his head is a little bit mobile, [he has] his lift and it can drive around. And that's really the core of what Cozmo is." (Ben Gabaldon, lead sound designer for Cozmo robot, Sound in HRI workshop 12-03-2020)

As pointed out by Lukas, Cozmo does not have speech recognition and mainly interacts through beeps, movements and animated eyes. Cozmo recognizes human faces and its toy cubes, and uses different sensors to track its body orientation. As pointed out by Lukas and Ben, Cozmo drives on cambelt wheels and has a forklift that can move up and down. Among many other sounds and animations, Cozmo displays emotions, which according to the sound designer are fitted to specific contexts, such as a particular game or activity (e.g. learning a user's name and face). The Anki robots Cozmo and Vector were carefully designed to exploit several hundred different sounds in their animations, including beeps, human and animal inspired sounds but also sounds that would cover up or enhance noises from moving parts. The robot is controlled through a smartphone app with different modes such as letting the robot roam freely, teaching it a person's name and face, playing games or creating simple programs.



Figure 4.1: Cozmo robot with toy cubes and its smartphone app.

Participants

A total of 35 people (20 adults, 15 children) figure in the corpus of video recordings. The German study included four pairs of German adults. The Swedish study involved six Swedish families with at least one child in the age of 8-14. The entire family was encouraged to play with the robot and the collected data often involve more than one participant interacting with the robot, including siblings and parents. Participants were recruited by convenience sampling through personal contacts and local social media groups. I decided to include participants with different native languages and from diverging age groups to explore cross-cultural patterns.

Ethics

Cozmo is a commercial robot that was not produced for research purposes but for the consumer market. Therefore, it was not only me, the researcher, who was collecting data, but the company that produced the robot and potential third-party contractors were continuously collecting data, too. Robot manufacturers are private actors who have "no obligation to disclose data, or keep data secret, unless this follows from special legislation covering their activities" (Vetenskapsrådet, 2017, p. 75). As a researcher, I can thus promise participants no more privacy than the robot manufacturer allows, without even being able to use the data that is collected by the robot for my own research (unless I collaborate with the robot manufacturer). Nevertheless, the researcher needs to take responsibility in investigating potential threats to participants' privacy through the robot (High-Level Expert Group on Artificial Intelligence [AI HLEG], 2018). This involves reading the company's privacy statements to understand how they may use the data, and thinking of potential privacy threats. One also needs to carefully select participants, limiting the number of participants to what is needed to make the research valid. During the consent procedure, I made sure that participants understood the potential threats to privacy as well as their rights to contact the company to have their data deleted (Association for Computing Machinery [ACM], 2018). This involved thoroughly informing participants about potential privacy risks, especially since users tend to consent to data collection by technology without knowing what this implies (AI HLEG, 2018). When using commercial robots for research, both the researcher and the robot manufacturer need to protect the research participants' privacy. Accordingly, we need both "trustworthy AI" (AI HLEG, 2018) technology and trustworthy researchers, who do their best in minimizing potential threats to the participants' privacy and provide subjects with sufficient information so that they can make an informed choice.

As collecting video recordings in family homes can be particularly sensitive, I sought participants' consent at several steps during the study. I first explained in the call for volunteers that the study entails videorecordings. When parents contacted me to learn more about how to participate, I sent them a copy of the consent form so that they could decide in advance whether they still wanted to participate. When leaving Cozmo in the Swedish homes, I scheduled my drop off visits so that all participants would be present. Before switching the robot and camera equipment on, I always explained the study in detail and asked participants to sign consent and video usage forms (see appendices A and B). I also presented an adapted version of

the consent procedure for the children, and ensured that they were comfortable with being recorded. When instructing participants' to record themselves, I particularly highlighted the importance of ongoing consent, making sure that everyone agreed to being recorded whenever the camera was switched on. If children who I had not spoken to featured in the data, I immediately deleted those recordings. In two cases adult family members who had not been present during the initial consent procedure played with Cozmo in front of the camera, and I sought their written consent before analyzing the data.

4.3.2 Corpus II: Autonomous Shuttle Buses on Public Roads

To compare the palm-size Cozmo robot with a different type of robot, I chose to study autonomous shuttle buses that can be stepped onto. The autonomous buses are tested at Linköping University on campus Valla and in the Vallastaden neighborhood as part of a larger public transportation project, called Ride the Future[4]. The goal is to explore how autonomous buses could complement existing transport solutions in an urban environment, specifically by offering last-mile transport.

Autonomous Shuttle Buses

Buses from two French manufacturers feature in the data. One bus is built by a company named Navya and two by a company called EasyMile. The buses are electric and navigate based on data from LIDAR and GPS sensors as well as cameras. When someone is getting too close, the buses play a distinctive beep and start braking. The sound is reminiscent of the sound that old trams use, a sharp "ding" that may be repeated. The same sound is also used when the bus is slowing down to pass an obstacle (such as a wrongly parked bike) and when it is about to stop at a bus stop.

Participants

The main participants in this study were the safety drivers who are compulsory on the buses and how take over control in case the buses get stuck. Over the course of the project, I worked with six safety drivers on the three different buses. The safety drivers are professional drivers employed by the local public transport provider. They do not exclusively work on the autonomous buses but also drive regular buses and trams and have alternating schedules.

Ethics

I first established an informal contact with the safety drivers to find out whether they would be interested in participating in the project. When they agreed to meet me more formally at the garage where the shuttle buses are parked, I explained the project in more detail. The drivers signed informed consent and video usage forms (see Appendix C). I particularly focused on making sure that the drivers

[4] www.ridethefuture.se



Figure 4.2: The autonomous buses during their inauguration on campus. Left: EasyMile EZ10, right: Navya DL4.

understood that I was not reporting back to their boss and explaining the different ways in which video could be used. Most of the safety drivers had not come in contact with researchers before starting to work on the university campus and were not familiar with consent procedures, which made it especially important to enter the setting in a gradual manner, and to explain that they could always opt out. One of the safety drivers refused to sign any papers but said I was welcome on the bus. While speaking with this safety driver has informed my perspective, I refrained from performing any video recordings. With all safety drivers I practiced ongoing consent. I sent them a text message asking whether it was okay to meet them on the bus on each particular occasion (usually a day in advance), and asked whether they still agreed to be videotaped before starting any recordings. During early stages of the project, I met the safety drivers at the garage, where we would jointly set up camera equipment, speaker and recording signs before leaving onto the open road. Since the safety drivers were concerned about the passengers' safety, minimizing the risk of flying equipment was a top priority. As the safety drivers gained more experience in navigating the buses, they would typically pick me up somewhere on the road.

In addition to the safety drivers, hundreds of pedestrians, cyclists and cars figure in the video recordings. Recording for research purposes is legal in public areas in Sweden, but it was still important to ensure that this data was collected ethically. During all recordings, I made clear that I was recording. When videorecording from the road, me and the students who helped me put on yellow vests with signs that said that we were researchers. When recording from the bus, I put on signs, with a large camera and a note that the recordings are for research purposes in Swedish and English. Since it was not possible to obtain active consent from people on the road during the rides, I am blurring their faces in my renderings of the data. I deliberately recorded with GoPros, which capture wide-angle views of the area but usually have rather grainy video quality, not sufficient for recognizing individual faces from the recordings.

In a few cases, passengers hopped on the bus during the recordings. As soon as the doors opened, the safety driver always welcomed potential passengers, and I then immediately informed them that I was video recording the ride, offering to switch off the cameras. When they were showing signs of hesitation, I switched off the cameras that record the inside of the bus and erased any material that featured them. If they agreed I handed them a printed paper with my contact information in case they would change their mind afterwards. If passengers stayed on the bus longer, I explained in more detail what the study was about.

4.3.3 Corpus III: Vero as Teamwork Support

I did not collect the data for the Vero corpus but I am building on this corpus in my sound design work. In contrast to the other two robots, which are commercially available, Vero is a research prototype developed in the Robots in Groups lab at Cornell University.

The Vero Robot Prototype

Vero is a prototype for exploring whether and how robots could be teammates, following a RtD paradigm. Vero is controlled by a researcher, following the WoZ paradigm, which facilitates exploration of how the robot could interact. Vero exists in different forms, which have in common that they demonstrate a non-anthropomorphic robot for team support. The physical prototype of Vero (see Figure 4.3) consists of a floating ball with a microphone. The video prototype is slightly different, but also consists of a non-anthropomorphic shape (see Figure 4.4, (Hohenstein et al., 2022; Schecter et al., 2023)). I worked with the latter, since a corpus of video recordings with this prototype had already been collected and was readily available for use. Data collection was approved by Cornell's ethics board (Institutional Review Board). I was not involved in collection of the data but have been added to the ethics protocol (Cornell-IRB0010288, 2104010288) and have permission to work with the data.



Figure 4.3: Physical prototype of the Vero robot. Picture: Yoyo Tsung-Yu Hou

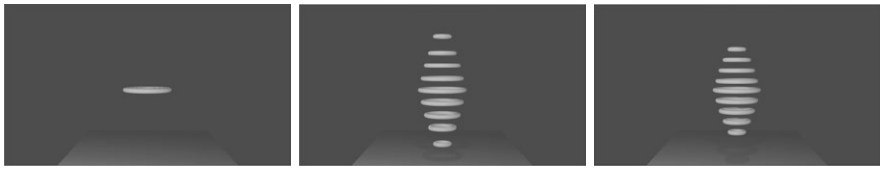


Figure 4.4: Video prototype of the Vero robot. The floating animated figure moves vertically, and vibrates slightly whenever the WoZ operator is speaking.

Participants

The participants in this study were students recruited through the mailing lists at Cornell University. They have given consent for the videos to be publicly released. The corpus consists of 7 recorded sessions with up to 4 participants each.

The Experiment Tasks

During the twenty-minute sessions, the study participants engage in a brainstorming and a discussion task with Vero. During the brainstorming task, participants come up with creative uses of a paperclip. In the discussion task, participants rank 13 possible causes of poverty according to how accurate they perceive them to be. At the beginning of the interaction, Vero introduces itself, but it is left up to participants to decide and discover how Vero can be used during the tasks. While some participants extensively rely on Vero's suggestions in the brainstorming task, others largely ignore Vero's presence. In the discussion task, the researchers controlling Vero were instructed to be more or less active: In some recordings, Vero challenges others' arguments. In other cases, Vero remains relatively idle during the discussion and is turned to when the group comes to a conclusion. In general, the researchers acting as wizards were following a loose script for the introductions and when refusing to provide information (e.g. "Sorry, this is not in my database"), but acted relatively freely and spontaneously during other parts of the interaction, testing different ways of responding to requests from the participants.

Ethics

Participants agreed to participate in the experiment and to be recorded during their interaction with Vero. Participants were made to believe that Vero was acting autonomously, by presenting a video that introduced Vero as drawing on advanced speech recognition and generation. While this deception could be seen as ethically problematic, does not cause any immediate harm. After the experiment, participants were debriefed and told that Vero was controlled by a human. The majority of people had fallen for the deception, and displayed surprise when learning that Vero was in fact a researcher.

As highlighted by Porcheron and colleagues (2020), wizards and experimenters are collaboratively performing the robot's behavior, engaging in hidden interactional work. This corpus is no exception, and while one may pretend that Vero is acting like a robot from a design perspective, I have been careful to observe how Vero is

brought into existence by the researchers when analyzing the data. Some of the wizards were part of data sessions, and rather than pretending that they were all acting out the same Vero's, we discussed how their vocal characteristics and accents as well as personal style of enacting Vero influenced the interactions.

4.4 A Note on My Competence in the Field of Study

My work spans different disciplines, bringing together EMCA and design methods. While EMCA forms the starting point of this dissertation, I would not have been able to carry out my work without extensive training in computer science, robotics, natural language processing and human-centered design. In this section I will briefly describe my background and training to highlight how it equips me with competence in the practices that I am studying, what ethnomethodologists refer to as "unique adequacy" (Garfinkel & Wieder, 1992, p. 182)[5]. While analysts can "use common sense knowledge of the society" (Garfinkel, 1967, p. 31) when studying everyday conversations, gaining first hand experience and members' knowledge is particularly relevant when studying institutional settings (c.f. Crabtree, 2000; ten Have, 2002a). More generally, reflecting on one's own involvement and competence in the studied practice is particularly relevant when the researcher is participating in the studied activity themselves (see e.g. Hofstetter, 2021; Pehkonen, Rauniomaa, & Siitonen, 2021). In the case of system design research (including HRI), the analyst may benefit from competence both in the practices of the studied communities (the users) and of the design research community (c.f. Greiffenhagen & Sharrock, 2019). In the previous sections, I have touched on my involvement with the users and my collaboration with the professional safety drivers. Here I would like to reflect on my competence as a roboticist, since this is key for being able to read and understand HRI literature (weak unique adequacy), which forms the basis for being able to meaningfully contribute to that field (strong unique adequacy) (c.f. Garfinkel & Wieder, 1992).

I have been working with robots in a variety of ways for more than 15 years, beginning already as a teenager when my parents gave me an electronics kit that was designed to build a robot through small experiments with LED lights and resistances. At the end of high school, I was one of the first students in Munich to take a written final exam in computer science. I studied Cognitive Science with specialisations in computer science and artificial intelligence on the one hand, and cognitive neuropsychology on the other. During my time as exchange student in Sweden, I encountered multimodal interaction analysis (Broth & Keevallik, 2020), taught by Mathias Broth. I ended up writing my bachelor thesis on a conversation analytic study of human-robot interaction, which formed the basis for a CHI-paper (Pelikan & Broth, 2016). After an internship with Karola Pitsch working on analysis of interactions with a Nao robot, I enrolled in an interdisciplinary master program in Interaction Technology. My focus was on design and robots, and in a course project, I even built a robot prototype. As part of the program, I spent two years in the Netherlands and six months at Cornell University. Once again, I ended up doing a conversation analytic study for my thesis work, this time looking at how

[5]The following reflection may bear resemblance with positionality statements in anthropology and design with a focus on competence.

surgical robots influenced teamwork in the operating room. While encountering EMCA relatively early during my studies, I first acquired a Bachelor and a Master of Science before starting graduate level training in the graduate school of Language and Culture.

I consider myself to have weak unique adequacy in the perspective of the robot, particularly in the way in which robots "see" the world and how they generate language. My studies included training both in what is now termed *good old-fashioned AI*, focused on logical reasoning, and in machine learning, particularly in computer vision and natural language processing. While I would require further training before being able to publish a scientific paper or to work as a practitioner in these areas, I gained a good theoretical understanding of the state of the art and the practical challenges through my coursework. This has been invaluable in understanding why Cozmo sometimes fails to recognize people and why the buses sometimes brake abruptly even if there is no obstacle for a human eye. My background helps me to be aware of the limits of speech generation and what "understanding" can mean for a machine.

I would argue that I do have strong unique adequacy in certain areas of computer science. I have worked several years as a freelance programmer, and while I would not say that this is my strongest skill today, I certainly would be able to continue a career in this direction. Particularly concerning HCI and HRI, I consider myself as part of the community. I was selected for the Pioneers workshop at the 2023 HRI conference, which according to the website "brings together a cohort of the world's top student researchers (...) to present and discuss their work with distinguished student peers and senior scholars in the field"[6]. This manifests that it would be wrong to refer to roboticists as "them", when I am in fact beginning to actively shape the field. In this sense, I take a different approach than many of the scholars who paved the way towards a hybrid form of EMCA and systems design, many of whom come from humanities and social science backgrounds and were much more dependent on collaborators (c.f. Crabtree, 2004; Sharrock & Button, 2003). While certainly lacking skills to build full-fledged aesthetic prototypes, my competence goes beyond user studies and reaches into interaction design.

As part of my dissertation work I also had to engage with methods and tools for sound design. While I would never consider myself a professional sound designer, I had to learn some practical skills in how to generate sound. During the course of the project, I seem to have developed a form of professional hearing, finding myself indulging in the nuances of electronic music in ways that I never could before.

In this chapter, I described both the EMCA and design methods that I used, delineating them by their different approaches to working with video, as data or as design material. I then introduced the three different video corpora, the robots and humans that they feature. Finally, I reflected on my background as a robot designer, or in other words, my unique adequacy in the methods of HRI. In the next chapter, I will summarize the articles that constitute this compilation dissertation.

[6]See https://hripioneers.org

5 Summary of the Articles

The articles in this dissertation provide different answers to the research questions. This chapter summarizes each article. Articles I and II draw on the recordings of Cozmo in family homes, Articles III and V are based on the autonomous shuttle buses, and Article IV focuses on the teamwork prototype Vero. Articles I, II and III document in detail how humans make sense of robot sound and how sound Gestalts can contribute to the accomplishment of sociality. Further, Articles II and III identify moments when sound is missing. Article IV and Article V explore how robot sound could be designed for interaction, based on EMCA insights.

5.1 Article I: "Are You Sad, Cozmo?" How Humans Make Sense of a Home Robot's Emotion Displays

Article I analyzes how displays of robot emotions are interpreted in interactional sequences in family homes. Focusing on Cozmo, the sounds are always accompanied by movement as well as animated eyes on the robot's small display. Since human non-lexical vocalizations typically are multimodal displays, we analyzed Cozmo's sounds as multimodal Gestalts (Mondada, 2019), looking at the entire animation.

Drawing on my data from German (4 adult pairs) and Swedish (4 families[1]) homes, the article explores how humans make sense of Cozmo's professionally designed emotion displays in everyday interaction. The article focuses on two particular emotion displays, one animation that is designed to display happiness, and one to display sadness. According to the sound designer, the happy animation should

[1] Data collection was still ongoing when the article was published, I later received recordings from two more families that are not included in this study.

express that Cozmo is "happy to meet you" at the end of the face recognition sequence. The sad animation occurs in the context of a fist bump activity, and is triggered when Cozmo cannot detect a human fist pressed against its forklift arms. We present six excerpts from the video-recorded data, three featuring the happy animation (2 German pairs, 1 Swedish family) and another three showing the sad animation (2 Swedish families, 1 German pair). Taking a multimodal conversation analytic perspective, we show how participants make sense of the robot's emotion displays in the specific contexts that they occur in.

The analysis shows that the display of happiness is treated as unproblematic and moves the interaction forward. Participants minimally acknowledge it by smiling and laughing (Excerpt 1), by petting the robot (Excerpt 2) or translating it as "oh yes" (Excerpt 3). They then move on in the interaction, either by switching the topic (Excerpts 1-2) or by further building on the animation (Excerpt 3). In contrast, the display of sadness gets verbally formulated as "are you sad? (Excerpt 4, Excerpt 5) and "no" (Excerpt 4, Excerpt 6). It makes participants reconsider their previous actions, offering to redo the fist bump in the cases where participants had understood that this was the action that the robot tried to accomplish (Excerpt 4, Excerpt 5). This article shows that emotion displays through sound and visible behavior are not just a feature that could make robots more relatable or natural, but that they can take powerful interactional functions.

However, we also demonstrate that these animations do not have a fixed meaning but are interpreted in situ, depending on the specific sequential context that they occur in. While the designers may envision a particular context, their plan is not always in line with what users actually do around the robot (Suchman, 2007). In Excerpt 3 we show for instance how the participants interpret the happy animation as Cozmo liking the local beer, which they offered to the robot just before it happened to play the animation. In Excerpt 6, we illustrate how the sad animation is made sense of as Cozmo not wanting to play with its toy cubes when participants do not recognize the lift forklift arms as a request for a fist bump.

While the notion of situated meaning making (Linell, 2009) is well-established in research on human social interaction, HRI work continues to focus on validating that study participants will interpret a specific sound design in the same way. In human-human interaction, non-lexical vocalizations are used to navigate complex social matters through their ambiguity by design (Keevallik & Ogden, 2020). We make a similar observation for robot sounds and argue in the article that this can be drawn on as a feature, rather than a bug (Dingemanse, 2020). While the sounds were still interpreted in line with the designed positive or negative valence, they could mean different things depending on the sequential context. The article argues that this makes them particularly suitable for "quick and vague"[2] communication. Displays of happiness can be used to quickly communicate that everything is going according to plan, while displays of sadness may show that humans need to reconsider their actions, for instance by providing a different input for the robot.

Prior work has been focusing on validating that a sound animation designed as happy would indeed get interpreted as happy, and a sad one as sad. Our article shows

[2]This is to highlight that there may be a tradeoff between quick (but vague) and slow (but specific). The phrasing takes inspiration from Daniel Kahneman (2011) who makes a similar distinction for cognitive processes.

that while it is important to design meaning potentials as sufficiently narrow (e.g. positive versus negative), it should not become an end in itself. Robot sounds (as part of multimodal Gestalts) may be interpreted more broadly than intended when deployed in interaction. They can be used to highlight quickly where an interaction is headed, and be used on budget systems (like Cozmo) as a low-cost alternative to expensive speech generation.

5.2 Article II: Managing Delays in Human-Robot Interaction

Article II analyzes how non-lexical sounds can accomplish a certain practice, in this case the management of what we call "delay", drawing on literature in human-computer interaction (Miller, 1968; Shneiderman, 1984). Delay occurs when a computer has received a command and is working on it, but the response is not yet forthcoming. In computers, graphical solutions such as progress bars are commonly used to indicate delay (see e.g. Ghafurian, Reitter, & Ritter, 2020; Gutwin et al., 2004). In our article, delay is introduced when one party momentarily withdraws from a joint activity to engage in an individual activity that needs to be completed before the joint activity can proceed. The article analyzes how a Cozmo robot has been designed to manage delay using sound, movement and its screen-eyes, and compares this with the embodied practices that humans draw on to manage their temporal disengagement from a joint activity. The comparison sheds light on the interactional function of non-lexical sounds and embodied conduct in sequences of interaction.

The study is based on my recordings of Cozmo in family homes (4 German adult pairs, 6 Swedish families), as well as on two corpora featuring human-human interaction, which have been previously analyzed (Hofstetter, 2016, 2020). The human-human corpora show how people temporarily withdraw from a joint activity, to look up information while talking to a caller on the phone (Extract 1) and while planning their moves in a board game (Extract 2). The robot corpus features several run-throughs of a face-learning sequence, in which Cozmo is scanning a face, indicates that the face was scanned successfully and then utters the name of the person whose name was learnt. Cozmo plays a "rrr" sound and displays squinted eyes during the first delay, and remains silent and idle during the shorter second delay.

Our analysis demonstrates that delay management is not an individual process, but involves the waiting party, who supports the delay. Typically, this is done by disengaging and closely monitoring for indicators that the joint activity can proceed. We first describe how humans commonly manage their withdrawal from a joint activity through non-lexical vocalizations such as singing "uh deuh deuh" on the phone (Extract 1) or by embodying that their action is on hold, for instance by hovering their hand over a game board (Extract 2a). We also show that humans support the waiting party, typically by disengaging and giving them space to perform the individual action, such as by starting a side conversation (Extract 2b). During the disengagement they closely monitor the delayer for indicators that the joint activity can proceed. We then compare this to Cozmo's delay animation, which consists of squinted eyes and a sound that resembles turning cogs. Participants similarly treat

this as putting the activity on hold. They disengage from the robot to support the delay, for instance by talking about the robot between themselves (Extract 4), gazing away (Extract 5) and moving their bodies (Extract 6). While most people stayed put so that the robot could scan their face (see Table 1 in the article for details), Extracts 5 and 6 provide examples of how the robot's delay management may interfere with the face scanning. Participants instruct each other to "sit still" (Extract 6) and "don't move" (Extract 5) to prevent this. This highlights the importance of aligning the delay management practice to suit the current activity. The article shows how sound animations can legitimize a robot's delay.

At the end of the first delay, Cozmo plays a "ding" sound, followed by a sound that resembles a hesitating "uh". Participants treat the "ding" as projecting the end of the delay (Extracts 3a, Extract 4) and "uh" draws the waiting party's gaze towards Cozmo (Extract 7, see also Extract 2b). However, the "uh" is followed by silence and the robot stays idle, leading to an unmanaged delay. This results in participants frowning (Extract 7) or raising their eyebrows (Extract 8), displaying that they anticipate trouble, and even abandoning the joint activity (Extract 3b). This shows the importance of not only managing the time during a delay but designing for a coordinated return to the joint activity. The article highlights that there is "no time out" (Garfinkel, 1996, p. 11) from interaction, and even brief moments of delay are accountable, making relevant a situated explanation - for instance in the form of a sound animation.

Contributing to the study of human-computer interaction more broadly and of human-robot interaction specifically, the article highlights the importance of delay management, providing situated explanations for what is happening with the machine. Even if a robot is not designed to perform delay management, humans will use its body to make sense of the delay – suspecting trouble if it does not give off signs of processing a response. While today's computers are designed to provide accounts through graphical solutions, we argue that robots can use their material presence to manage delay. This includes moving their body, but also producing sounds related to their physical presence such as the turning of cogs. This enables humans to support the robot by "doing waiting". The article also demonstrates that for successful coordination, the delay exit needs to be designed in an explainable way. In the presented extracts, humans return to the joint activity in a gradual manner. Vocalizations like "uh" draw gaze back to the party who produced them and can be suitable for coordinating a gradual exit from the delay.

5.3 Article III: Why Autonomous Driving is So Hard: The Social Dimension of Traffic

Article III looks at a case of autonomous shuttle buses, where the few existing sounds do not seem to support coordination between humans and the robotic buses. The paper provides examples in which robot sound does not support coordination, and serves as a pre-study for the sound design work described in Article V.

I draw on video recordings of two autonomous shuttle buses on campus Valla at Linköping University, recorded at different times during one year. The buses drive on invisible "tracks", much like trams, and are programmed to stop whenever they detect obstacles in front of them. The recordings reveal how the buses brake

repeatedly in between their designated stops, at moments that would not appear as dangerous or relevant for braking from the perspective of a human driver. This occurs particularly often when cyclists and other road users get close to the bus, for instance during overtaking maneuvers. The article demonstrates that the buses fail to provide situated explanations for their limited ability to coordinate with humans.

Human drivers constantly coordinate their actions on the road. In an overtaking maneuver, the party that is being overtaken also participates, typically in a cooperative way by letting the other pass, not speeding more, lifting the foot off the gas pedal or moving towards the side of the lane (Deppermann et al., 2018). While the buses have visual displays instructing people in Swedish to "keep the distance" and play warning "ding" sounds when braking, these seem to be ignored by overtaking cyclists (see Figure 1 in the article). The sounds are played when the overtaking cyclist is already in front of the bus, too late to instruct or warn them. To the contrary, the "ding" sounds can lead to dangerous situations as other cyclists around the bus may get startled, pull away from the bus and into oncoming traffic (Figure 1 in the paper). The analysis further shows that by pulling in closely in front of the shuttle buses, cyclists demonstrate their expectation that the buses would participate in the coordination of the overtaking maneuver similar to human drivers (Figure 2 in the paper). However, the buses continue with fixed speed and only react to surrounding road users once they get within the range of their sensors.

The article contributes to HRI research by demonstrating the benefit of studying autonomous shuttle buses in live traffic throughout entire rides, not just at particular landmarks such as intersections, as has been commonly done (see e.g. Moore, Currano, & Sirkin, 2020). This can help to better understand the most challenging problems on the road, before diving into design work. In the current case overtaking and passing the slow-moving shuttles is a major issue. More importantly, the paper highlights that the absence of sound (before people get too close) can be problematic. Cyclists next to the bus do seem to react to the sounds – however they are not the addressees, and those who caused the brake are already far away ahead of the bus. The paper argues that sound in the form of a bell-inspired "ding" sound could potentially instruct movement. In the existing design on the buses however, it is played too late to meaningfully instruct humans around the bus.

5.4 Article IV: Minimal Sound for Maximal Interaction: Sequential Timing of Expressive Robot Behavior

Article IV explores how robot sound can be designed from a perspective that is grounded in an EMCA understanding of interaction and sequentiality. It presents an empirical design case study inspired by prior work in HCI (Gaver et al., 2010) and describes the sound design process for the Vero teamwork robot. The paper raises the question *when*, at what moments, robot sound could contribute in a relevant way and discusses design considerations for achieving this. The article does not present a working end product. It focuses on describing challenges that arise when designing robot sound and the methods we developed for addressing them. The article argues that it is important to prototype sound in specific sequences from the very beginning, rather than designing standalone sounds that are later placed into situated contexts.

The case study builds on video recordings of study participants working on two teamworking tasks with the Wizard-of-Oz controlled video-prototype version of Vero. The article walks through six questions that arose during the design process. For each question, we present design solutions and reflects on insights gained during process.

First, we looked through the video recordings in a data session format. it became evident that Vero was silent for long stretches of talk by others. While humans would nod, breathe or react to what others said with non-lexical vocalizations, Vero seemed absent for long stretches of time, randomly asserting itself verbally later during the interactions.

Second, we asked ourselves how Vero could display minimal participation. Our discussions very quickly ended up being about what specific sounds should be added to the wizard interface. The article highlights the challenges we faced when searching sound samples that were suitable for Vero in sound databases. Searching for non-lexical vocalizations like "uh" and "mmh" resulted in a lot of results that were not relevant for our purposes, some of them evoking sexual connotations or simply being very clearly not "naturally occurring" but clearly enacted. Exploring machine-inspired sounds, we discussed in detail the associations that these would evoke (c.f. Gaver, 1986). While these considerations focused on how the sounds would fit Vero's character, they led us away from the question how the robot could minimally indicate its presence, agreement, or disagreement through sound. We then took a step back and attempted to use our own voices to produce sounds.

Third, we discovered that using one's voice to prototype sounds for Vero often felt strange, especially when others were around. While vocal sketching is a common sound design technique (Ekman & Rinott, 2010; Rocchesso, Serafin, & Rinott, 2013), there seemed to be reservations among the members of the team who had never engaged in sound design. The paper reflects on these barriers and draws a connection to the long nurtured view of non-lexical vocalizations as not being professional or even "primitive" (Wundt, 1900) and bodily experience being removed from scientific practice (Höök, 2018). We developed a warm-up drawing on the sound ball improvisation game. In this exercise a sound gets "thrown" like an imaginary ball from person to person, with the "catcher" repeating the utterance of the "thrower" and then coming up with a new sound. While the goal of this type of exercise is usually to open up the design space and come up with as many sounds as possible, we slightly changed the rules an only allowed the use of one specific syllable, such as "ma". This helped exploring the possibilities of communicating with a relatively fixed sound through varying duration, pitch and repetition. The article highlights the importance of finding a way to comfortably sketch with our voices, which then enabled a focus on how Vero should sound at specific moments of interaction.

Fourth, the article reflects upon how to design for situated and sequential contexts, which is very much grounded in an EMCA stance. The EMCA approach provides ready tools for dealing with specific interactions: we turned to the video recordings of concrete interactions with the Vero prototype. The article explains how the video recordings were not used as data for analysis but instead as prototyping material. Looping short snippets of video in which Vero had been quiet, the team vocalized during the looping video, acting as Vero. The article argues that the voice-over technique is especially interesting because designers can productively

work with overlap (Sacks, Schegloff, & Jefferson, 1974) when sketching with the voice. When several people vocalize at the same time, they simultaneously display to each other that they have found a moment that they agree on is relevant to design for.

Fifth, the article deals with the challenge of moving towards more general designs while still remaining grounded in specific situated cases. Working in a culturally diverse team, we could not agree on how exactly human-like vocalizations should sound and kept producing versions with different pitch curves when vocalizing suggestions. We addressed this by moving away from our own voices as a design tool. We were interested in limiting the variability of the sounds, and achieved this process of estrangement by using different objects to create sound, such as a MIDI keyboard or even just paperclips to mark moments in the looping videos. The article reflects on how painters and visual designers often make a pencil sketch before adding color. Similarly, our tools enabled us to focus on sketching out when sound should be added rather than its specific realization.

Sixth, the article reflects on the tools that are currently available for preserving the designs for further iteration, specifically for HRI designers who typically are not professional sound designers. We argue that it is important to record the iterations to be able to review and compare them. Recording a superimposed video stream of the design team on the original looping clip was one way to preserve the discussions in the team along with the prototyped sounds. However, this did not provide a possibility to later move sounds in the recording. While the free audio editor Audacity allows moving sound snippets within a larger recording, it cannot include video. Video editors have this flexibility and may support some stages of the process. The paper highlights that there are no readily available tools that can support the design of sound specifically for human-robot interaction.

The article demonstrates how designs can be analyzed through conversation analytic transcription and analysis. The timing of the sounds turned out not be not random, but designers seem to produce sounds at TRPs (Sacks, Schegloff, & Jefferson, 1974). This may not be very surprising for EMCA scholars, but the article validates and demonstrates that the voice-over method is a way to mark TRPs in an intuitive way, without any prior academic knowledge of turn-taking. This is an important finding for HRI, validating the design method. Designers can use sound ball and voice-overs to engage with their intuitive understandings as members of human-human interaction. These findings also show how EMCA observations can be made explicit and engaged with in a creative way through design methods.

The paper contributes to HRI by pointing out aspects of sound design that have so far been neglected, and by walking through the concrete steps in the design process. The article also contributes to EMCA by providing a detailed account of how the EMCA stance can be practically combined with interaction design. Further, we discuss how the different techniques can contribute to a larger design framework that combines EMCA and interaction design (see Article V).

5.5 Article V: Designing Robot Sound-in-Interaction: The Case of Autonomous Public Transport Shuttle Buses

Article V presents a novel method for designing robot sound, which intertwines EMCA and interaction design. The article applies some of the sound design methods developed in Article IV to the case of autonomous buses. Exploring sound design in live traffic, the article highlights the importance of designing robot sound in interactional sequences. It builds on the ethnomethodological observation that traffic is social, and is inspired by how drivers use the horn to communicate in regular cars (Laurier et al., 2020). The article contributes to the development of technomethodology (Button & Dourish, 1996; Crabtree, 2004), providing an example of how EMCA can be tightly embedded into the design process and formulating a design framework that intertwines EMCA and interaction design.

The article reports on a RtD study on the autonomous shuttle buses that were introduced in Article III, exploring how the shuttles could use sound to communicate with cyclists and pedestrians. The study draws on video recordings of the shuttle buses in their original design (9 hours). It also builds on video recorded sound design sessions on one of the buses (8 hours). During a period of two and a half years, I also collected notes on ethnographic observations and conducted interviews with the safety drivers, the project leader at the local transport provider, and a sales director at one of the bus manufacturers.

The sound design approach can be described through three major design phases, which we call "describe","intervene", and "reflect", as in Article IV. The design work iterated between these phases, going from description to intervention and back, reflecting on interventions and using this to inform new interventions.

The process started with *describe*, which entailed getting to know the setting through observation and formulating a design problem. Building on Article III, the current article also highlights how rides on the buses are characterized by many unnecessary stops, which slow them down. The initial phase was supported by video recordings and observations on the road, and the analysis and transcription of selected clips of such brakes. This preparatory work was important for identifying moments when sound could become relevant. Working with video recordings and ethnographic observations helped to ground the design in real-world problems rather than addressing an imaginary scenario.

The *intervene* phase entailed the active exploration of different sound designs. I developed a sound design technique using voice-overs, producing sounds in real time on top of looping video snippets, which is inspired by vocal sketching (Ekman & Rinott, 2010). Some of the voice-overs built on the designers' physical voice, by uttering vocalizations while the video was playing. Later, I triggered pre-recorded sound samples while the videos were looping and we implemente sounds that could be manipulated in duration by button presses or shaking a digital "bell". The voice-overs were important to explore how a sound would fit the size of the shuttle buses and the soundscape on the road. It helped to rule out horns used in heavy vehicles such as buses or trucks. Following the goal to ground the designs in the real world as much as possible, I tested the sound designs on the road early on. With the help of a Bluetooth speaker taped to the outside of the bus, I could trigger sounds

during rides on the road, whenever cyclists and pedestrians were about to get close to the bus. By acting out a capability that the buses did not yet have, I followed the Wizard-of-Oz (Dahlbäck, Jönsson, & Ahrenberg, 1993; Porcheron, Fischer, & Reeves, 2020) paradigm, which enabled me to test the designs on the road early on, without having to implement a system that would trigger them automatically. I acted as the wizard in three sound tests, while at the same time discussing the sounds with the safety driver. Following a participatory design approach (Asaro, 2000; Bødker & Iversen, 2002), we switched roles during the fourth iteration, with the safety driver acting as the wizard and me observing and reflecting on the sound designs.

Finally, in the *reflect* phase, I selected clips of specific sound designs from the video recorded rides and analyzed them following EMCA practice. I grouped similar instances into collections and transcribed several clips. Following the RtD approach, this analysis was opportunistic in nature, focusing on examples that could stimulate ideas and inform the next iteration.

The article presents three transcribed extracts to illustrate what we learned from the process. First, humming sounds, including the buses' natural engine sounds (Extract 1) were not enough to explain where the bus was going. While synthetic motor sounds may work for autonomous vehicles, they seem not sufficient for explaining how humans should coordinate with the bus. We also demonstrate how safety drivers can anticipate trouble long before the bus comes to a halt, highlighting that the unnecessary braking could be prevented if a human was in charge of the driving. Second, a jingle with the melody of the song "the wheels of the bus go round and round" (Extract 2) seemed useful in settings where many people were present and it would not have made sense to address one individual. Interestingly, the jingle seemed to invite interaction with the bus, with people waving (Extract 2), dancing and approaching the bus. Since the communal bus project has been struggling with low passenger numbers, this might be a way to encourage people to step on the bus. The friendly melody that evokes association with ice cream trucks in Swedish neighborhoods may work as a situated explanation of how to interact with it. Finally, the article highlights how repetition of short sounds like honks and beeps can instruct cyclists and pedestrians (Extract 3) to move away from the bus. Repeating a sound can be understood as indicating that the first response was not sufficient, effectively working as a repair initiator.

The paper makes a contribution to HRI by demonstrating how EMCA can be embedded in the a design process for human-robot interaction. It also contributes to robot sound design by presenting a method for designing what we call *sound-in-interaction*, focused on what sound should achieve in the interaction.

6 Discussion

Sound is necessarily part of a robot's material presence. It accompanies movements and can be designed as a resource for interaction. Taking an EMCA perspective and drawing on a RtD approach, this work has started to shed light on the question how robot sound can support coordination with humans in joint activities. In the following sections, I will highlight how this work provides answers to the questions of this dissertation. I will point out directions for future work and formulate implications.

6.1 Making Sense of Robot Sound

I investigated how people make sense of robot sound in everyday interaction in family homes, during collaborative team sessions and on public roads. My research confirms that people may treat robot sounds as relevant contributions to interaction. I highlight that sound occurs not in isolation but as part of multimodal Gestalts, in which movement and for instance facial expressions contribute to how a sound animation is perceived. I described how sounds played from an approaching autonomous bus are responded to by keeping its way clear - by slowing down, swerving or moving out of the way (Article V). Especially when repeated or playing continuously, such sounds are interpreted as meaningful. I also showed how sound animations that are designed to display emotions get recognized in family homes (Article I). I demonstrated that these emotion displays may be interpreted very differently than intended by the designers. In a context in which a German man offers beer to a Cozmo robot, the "happy to meet you" animation gets interpreted as "full agreement" by Cozmo to the question whether the robot likes beer (Article I). While the happy and sad animations were recognized as affiliative or disaffiliative, their

meaning potentials prove to be much broader in everyday interaction than intended in the original design. Prior work in HRI focused on validating specific emotions. My work gives more nuance and situates these studies as part of a larger endeavor. I suggest that the goal should not be to program a fixed meaning into a sound, but to narrow meaning potentials in such a way that they support progressivity, moving interaction forward. For emotions, it may be more relevant to work with positive and negative valence (Russell & Barrett, 1999; Zhang et al., 2021) than with very specific emotion labels. I showed how participants treated a sound animation that evokes positive interpretations as a "go-ahead" signal (Article I). A sound that was designed to display sadness was interpreted as highlighting a problem on the robot's end, implying that the problem needs to be addressed by the human before the joint activity can move on.

Second, I demonstrated that sound does not automatically get treated as relevant and meaningful but may also be ignored. The sounds that autonomous buses played during an emergency brake were not visibly reacted to by the pedestrians and cyclists who caused the bus to brake. Instead of turning around and apologizing, they simply kept moving. I showed that the sounds are initiated after those who triggered the brake have already passed the bus and therefore occur outside a recognizable context (Article III). In the Cozmo studies (Articles I and II), I focused on the sounds that do get treated as meaningful. However, also with Cozmo, there may be sounds that get ignored and talked over. While robots can participate in interaction, humans can also treat them as mere objects (see e.g. Pelikan, Broth, & Keevallik, 2022). Keeping in mind that humans will not automatically consider sounds as meaningful is an important (and humbling) lesson for design. Whether a sound makes a relevant and recognizable contribution depends crucially on the sequential context that it is played in. If a sound animation is not contributing in a recognizable way to the ongoing interaction, humans ignore the robot and talk over whatever it does. Robots that talk are often abandoned and left behind when their contributions are not recognizable (see e.g. Tuncer, Gillet, & Leite, 2022). As sound Gestalts are more ambiguous, they can be flexibly ignored in one moment, or attended to in another. Using sound animations in robot design may therefore make robots easier to integrate in human activities, as they do not constantly break the interactional sequence through entirely unfitting turns.

Third, my work showed that the absence of sound, silence, is not neutral but can be problematic. When Cozmo suddenly turns idle after managing a delay through sound, participants respond with frowns, anticipating trouble (Article II). Similarly, on an autonomous bus, the absence of sounds at interactionally relevant moments makes coordination difficult (Articles III and V). An AI team agent that has an entirely quiet presence may be perceived as awkward and not part of the discussion (Article IV). I highlighted that sound and silence may be important to consider together. EMCA speaks of moments in which highly expected responses, second pair parts to adjacency pairs, are noticeably absent when not actually produced (Sacks, 1992). This similarly holds for machines in which prolonged silence may be treated as an indicator of failure (Miller, 1968; Suchman, 1987). I demonstrated how sound can fill such silence and is often interpreted as a reaction, even if not always fully sufficient, that indicates that more actions are forthcoming or at least that the robot is not broken (Article II). My work provides empirical evidence that HRI

should not strive for entirely silent robots, but instead for designing robot sounds that can contribute to interactions in meaningful ways (c.f. Cha et al., 2018; Moore et al., 2017).

To summarize, I demonstrate that sound gets interpreted as meaningful when it can be recognized as constituting sequentially relevant actions. At the same time, it may also be ignored, specifically when it does not occur in a context in which participants expect a response from the robot, or when it appears as a random "outburst". Robot sound does not convey fixed meanings or information, but gets interpreted according to the situated and sequential context. This potential for robot sound to be understood in different ways is not a bug. It is a feature that is present in all utterances, in particular in human non-lexical vocalizations (Dingemanse, 2020; Keevallik & Ogden, 2020). Such vocalizations are less accountable than lexical speech and can therefore be particularly useful for navigating delicate matters (see e.g. Ogden, 2020) or for conveying a strained body (Hofstetter, Keevallik, & Löfgren, 2021; Keevallik, 2023). For robots, this means that sound is a subtle tool, which is particularly suitable for facilitating interaction in intuitive and implicit ways (Ju, 2015). My work also highlights that sound can be noticeably absent, which may even be treated as projecting trouble, highlighting that transitions between sound and no-sound should be carefully designed.

6.2 The Role of Sound in the Accomplishment of Sociality

While the previous section addressed sound very generally, I also provided specific examples of how sound contributes to the accomplishment of social actions. I analyzed both sound animations designed by a professional design team and explored what could be learnt through engaging in sound design myself.

First, my study of robot sounds designed by a professional audio designer on Cozmo revealed that sound animations designed to display emotions can be used as a go-ahead signal (display of happiness) and an "interactional rewind button" (display of sadness) that can initiate repair (Article I). Previous work on backchannels (Park, Gelsomini, Lee, & Breazeal, 2017) in robots has demonstrated that sound can make robots preferred interlocutors in storytelling activities. I demonstrated that sounds are not just a means to make robots more expressive, leading to more "natural" or "human-like" interactions. Instead, robot sound animations can make interactional contributions. Even non-humanoid robots can use such "quick and vague" indicators to make transparent where the interaction is going or whether the robot has received input that it can process. A frown combined with "huh?" in human interaction quickly highlights that something is off and talk needs to be repeated (Dingemanse, Torreira, & Enfield, 2013). Similarly, a display of sadness highlighted that something had just gone wrong, and thereby initiated repair. In contrast, participants interpreted the happiness display as an indicator that the interaction could continue.

Second, my work has shown how robot sound can contribute to the interactional management of delay, when the robot needs to disengage from a joint activity (Article II). As I highlighted before, and as Cozmo's design team demonstrably recognized in their design, simply turning silent while the robot is processing would

be problematic. Instead the robot needs to indicate its upcoming disengagement from the joint activity, manage the time during which it is occupied, acknowledging that there still is an ongoing joint activity, and then return to this joint activity in an organized manner. Filling the silence and playing sounds inspired by turning cogs while the robot was displaying squinted eyes seemed to accomplish "doing processing": it enabled humans to support the robot's delay by disengaging while monitoring its progress. This demonstrates that even mechanic and motor sound can contribute to the organization of social action. A subsequent "ding" sound that evokes associations with toaster ovens or microwaves was treated as evidence that the robot had completed its task. My work provides further evidence that machine sounds can be used productively for the accomplishment of interaction. Such consequential sounds do not only provide insights on what the robot will do next (Moore, Currano, & Sirkin, 2020), but also help managing multiactivity (in an interactional sense) or delay (in interaction design terms). The "rrr" cog-like sounds do not resemble human thinking displays and are entirely robotic, evoking associations with other technology such as old hard-disk PCs or old vending machines, sounds which have been culturally learnt. Similarly, participants may feel familiar with the "ding" through everyday interactions with other machines such as microwaves (c.f. Gaver, 1993). Cozmo also played a range of different "uh" sounds, which drew attention to the robot at the end of the face learning sequence, projecting a return to the joint activity in progress. Even though some of the sounds were more robotic than human-like (see also section 4.1.2), they achieved a similar interactional function as "uh" in human-human interaction. "Uh" is often used as a floor holding device, showing that talk is forthcoming. My work highlights that robots may not need to produce identical copies of human vocalizations. Similarity with sound-function relations that members of a society are already familiar with may help recognizability, making it easier to understand what the robot is trying to accomplish.

Third, I explored how sound can be designed for interaction more systematically. Many of Cozmo's sounds are treated as meaningful contributions by the participants, demonstrating the skill of Cozmo's design team. However, the sound designer also admitted that they went through a lot of trial and error and did not follow a specific procedure. Both Articles I and II highlight elements of the sound design that are partly confusing for participants, indicating that the designs could be further improved by a stronger focus on how the sounds could constitute actions-in-interaction. I therefore explored methods for designing robot sound in and for interaction, starting from identifying what actions the robot needs to accomplish. Working with the Vero teamwork robot prototype (Article IV), focus was on exploring how the robot could participate in the team discussions through minimal contributions. Rather than aiming to play sounds at random intervals to make Vero more expressive, focus was on identifying moments in interactional sequences where contributions from the robot would be expected. The work demonstrates that robot responses are especially needed at transition-relevance places (see section 3.1.2), moments when speaker change becomes relevant in human interaction. Designing sound for the autonomous buses, I further explored how sound can instruct humans where to move (Article V). Prior work has shown that consequential motor sounds can indicate where an autonomous car is going (Moore, Currano, & Sirkin, 2020). Their study starts from a robot-centric view, exploring how the robot can

make its internal state recognizable to allow for an ascription of intention by humans. I expanded this perspective, shifting the focus towards the coordination work that these displays could accomplish. My goal is to identify relevant actions that the robot needs to contribute in order to coordinate with others on the road. Since the autonomous shuttle buses cannot divert from their pre-programmed route, it is important to make this aspect in which they deviate from other wheel-based vehicles recognizable, by providing situated explanations that enable humans to support the buses by adjusting their own movement. Working very closely with the professional safety drivers on the shuttles, it became evident early on that continuously producing sounds risks rendering the sounds meaningless and playing sounds at random moments may even be accountable as harassment (e.g. evoking associations of men honking at women in skirts). Instead, the autonomous buses should only play sound when there is a need for coordination, i.e. when they become interactionally relevant. Jingles can announce the buses' presence when approaching a larger crowd, while repeated bell sounds can instruct pedestrians to gradually move out of the way of the public transport autonomous bus.

To summarize, I demonstrated that robot sound can contribute to the accomplishment of sociality. I provided several examples of how participants treat sound animations as constituting actions. Prolonged machine sounds put joint activities on hold, while human-inspired "uh" sounds attract attention back to the robot. Individual "ding" sounds indicate that a robot finished its task. Upbeat sounds combined with smiley eyes and quick movements of Cozmo's forklift move the interaction forward when played after accomplishing a task. In contrast, participants treat slow-tempo and falling-pitch sounds as repair initiators, indicating failure to finish the task and the need to repeat the prior actions. Jingles and repeated bell or horn sounds on an approaching vehicle invite coordination, displaying that humans need to adjust and give way to the bus.

6.3 Focusing on the Interaction in Human-Robot Interaction

Through the detailed analysis and design of sound, I explored the understanding of interaction in HRI more generally. Taking an EMCA perspective encourages a focus on accountability (Dourish & Button, 1998; Garfinkel, 1967) and sequentiality (Heritage, 1984; Schegloff, 2007). Behavior is not interactionally meaningful by default, but needs to be recognizable to members of a social group as a relevant action in the specific context. Actions are not standing by themselves, but occur as actions-in-interaction. They are understood by others as responses to what has just happened before, and shape opportunities and expectations for what actions should occur next. My empirical studies demonstrate how human-robot interaction can be scrutinized on a moment-by-moment level through video recordings, and what type of insights such analysis can provide. Building on EMCA work towards hybrid studies in HCI (Button & Dourish, 1996; Crabtree, 2004; Woodruff et al., 2002), I formulated a framework that tightly intertwines EMCA and interaction design (Articles IV and V). Interaction design facilitates the exploration of alternative interaction trajectories and stimulates reflection on how a robot could be designed differently. My design framework (see Figure 6.1) deeply embeds a focus on in-

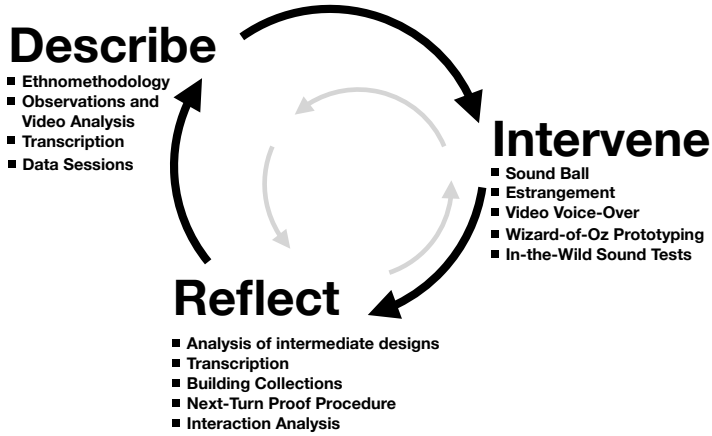


Figure 6.1: Framework for designing robot sound in interaction.

teraction in the sound design process through an iterative cycle with three main phases, *Describe*, *Intervene* and *Reflect*. While describe and reflect are grounded in the EMCA approach, the intervene phase draws stronger on interaction design. Building on the insights gained in a previous phase and iterating back and forth between the different phases, the framework facilitates learning and designing through practices and techniques that originate in both approaches.

First, drawing on the video recordings that are central to the EMCA approach, I developed a video voice-over technique. Looping a snippet of an interesting or problematic interaction, and producing a voice-over for how the robot could have sounded (by using one's own voice or tools such as a MIDI-keyboard) has proven a helpful way to explore not only how a particular sound would fit into a specific sequential context, but also to work on elements such as duration and latency of a sound (Articles IV and V). I thereby adjusted and expanded the sound design technique of vocal sketching (Ekman & Rinott, 2010; Panariello et al., 2019; Rocchesso et al., 2015) to make it suitable and compatible with the EMCA approach. Working on video clips, which may be shown in a data session before starting to explore the design space, the transitions between the describe an intervene phases become fluid. The same material which can work as data in a classic EMCA approach can turn into design material from which prototypes are emerging. This work is thereby extending earlier attempts at making video a more central aspect of interaction design work (Mackay & Fayard, 1999; Mackay, Ratzer, & Janecek, 2000; Suchman & Trigg, 1992; Ylirisku & Buur, 2007) to the design of robots.

Second, this work explored the role of transcripts in design. The bottom-up inductive approach of dealing with data in EMCA is compatible with RtD, which takes an opportunistic stance. During the active design work, the goal can never be to transcribe all collected data, but rather to familiarize oneself with a setting through unmotivated looking, coming to the data without a pre-conceived hypothesis. Transcription of specific cases may be particularly informative when trying to understand and identify design problems in a specific setting (see Article III and IV). It is also a very important tool for taking a step back and reflecting on design

when the design process is converging (Articles IV and V). Transcripts and video recordings manifest intermediate-level knowledge (Höök & Löwgren, 2012; Lupetti, Zaga, & Cila, 2021). They are grounded in the specific, demonstrating a design in situated interaction, and at the same time, since transcription is analysis (see e.g. Roberts, 2012), they encourage reflection and facilitate moving towards generalization of designs. In Article V, we demonstrated how transcripts can be used to showcase prototypes and to reflect on their interactional implications. Such analysis can provide insights on why and how a design worked in specific contexts, providing an alternative to the prevalent quantitative approaches in HRI for evaluating whether a design was effective.

Third, I demonstrate how human-robot interaction cannot only be studied but also be designed for in the wild (Jung & Hinds, 2018; Šabanović, 2010). Reflecting the EMCA interest in the ordinary and mundane, I have taken WoZ prototyping out of the lab and onto the road (Article V). While prototyping in the world can be challenging, it yields important insights. Sounds that appeared as too loud and intrusive inside a building were for instance barely noticeable outside where they were supposed to be deployed. Moving out of the safe space of a design studio also made it easier to get feedback from ordinary members of society. It broadened the perspective beyond the immediate users (passengers of the autonomous shuttle buses) towards anyone who was present on the road at different times of the day. Since EMCA works with the displayed understandings as opposed to interviews, it can include even perspectives that may be otherwise inaccessible or easily overlooked. This can help to introduce a participatory element to the design process. While mainstream HRI-PD requires co-designers to attend interviews and workshops (Björling & Rose, 2019; Lee et al., 2017; Ostrowski, Breazeal, & Park, 2021; Winkle, Senft, & Lemaignan, 2021), an EMCA informed perspective can include even those who may not immediately volunteer their opinions. Since the safety drivers were constantly present on the bus, they became unanticipated but important stakeholders in the process. Due to the nature of their job as service providers, they were initially timid when asked about their opinions on sound design, but they could demonstrate their expertise while working on the bus. Their inclusion as co-designers naturally emerged from existing ethnographic practices in EMCA. Intertwining the EMCA perspective with design can thereby offer new perspectives to participatory approaches in HRI, asking more critically who the participants are or should be.

Finally, working in the phenomenological tradition that EMCA stands in has provided new perspectives on sound design for robots. Since non-lexical vocalizations are often strongly tied to bodily activities and may have a negative connotation as primitive or unprofessional (see e.g. Dingemanse, 2020; Wundt, 1900), it can be difficult to produce them in a professional setting, even when one's workplace is a design lab. The EMCA perspective highlights that anything, not just words and lexical items can be meaningful contributions to interaction. Playing the "sound ball" improvisation game (Article IV) as a warm up helped to overcome difficulties in getting started with vocal sketching in our design team. Limiting the contributions to a single syllable sensitized designers to the communicative potential of sound. Since the exercise involves the whole body, it supports a practical engage-

ment with the interplay of prosody and gesture, and the exploration of sound as part of multimodal Gestalts.

To summarize, EMCA can promote a shift in HRI from a focus on expressivity towards interaction. To stimulate a focus on the interaction throughout the design process, I developed a framework that intertwines EMCA and interaction design (Articles IV and V). Turning video data into a design material is a particularly suitable way of allowing EMCA to inform interaction design. The EMCA focus on how interaction is practically accomplished can inform design in real-time, through voice-overs and improvisation following the Wizard-of-Oz paradigm. In turn, videorecorded prototyping sessions can feed back into EMCA analysis. Transcribing videos helps design teams to reflect upon their designs and contributes intermediate-level knowledge, which makes it easier to identify how specific elements of a design contribute to ongoing interactions. The EMCA focus on everyday interaction can inform design work that is performed in the actual settings where the design should be used and promotes participatory approaches. Finally, the phenomenological and multimodal focus of EMCA can help designers to become more sensitive to how the whole body is involved in interaction.

6.4 Future Work

This work opens up several directions for further research. First, I focused on sounds that get treated as relevant by humans. While Article III and Article V provide examples of sounds that do not seem to be interpreted as meaningful by humans, I did not analyze them in further detail. While EMCA may be limited in its ways to study aspects of interaction that do not get further picked up on by the participants in the interaction, it may be interesting to explore more systematically whether certain sounds tend to be picked up on more than others. There may for instance be an orientation towards whether sounds seem to be produced deliberately or not. In human interaction, bodily sounds like sniffs (Hoey, 2020) and grunts (Hofstetter, Keevallik, & Löfgren, 2021; Keevallik, 2023) may be less accountable than vocalizations that are clearly designed for a specific audience, such as vocalizations that instruct movements in dance classes (see e.g. Keevallik, 2021) or that convey sensory experiences (Keevallik et al., 2023; Wiggins, 2002). It would be interesting to further investigate whether hardware sounds in robots (including synthetic motor sounds) are also less accountable and perhaps also missed more easily than explicitly designed beeps or even human-inspired utterances.

Second, while this work has explored sound in a small range of settings with three different kinds of robots, more work needs to explore how this can scale to other systems. My work has started to map a range of different ways in which robot sound plays a role in action constitution, but this is certainly only scratching the surface of what can be done. Vocalizations are particularly interesting for humanoid robots and dialogue systems with human-like voices, where non-lexical vocalizations are only starting to emerge as a topic of discussion (Dingemanse & Liesenfeld, 2022). Article II on delay management gives some pointers on how non-lexical vocalizations such as “uh” could be used in these systems. Further exploring non-lexical sound as a means to transition into speaking (for instance through equivalents of inbreaths) seems like a promising area for further research. Similarly, exploring further how

autonomous vehicles and other mobile robots can draw on sound to coordinate movement around humans is a promising avenue, as Article V has started to tap into.

Third, this work has explored how EMCA can contribute to more situated designs by developing methods that will naturally nudge the design process into a certain direction, even when the design team has no formal experience with EMCA literature. I see this work as taking a first step towards this direction, with much more work being possible in the area. Sound is a minimal aspect and has allowed me to ignore the complicated world of a specific language. Expanding these methods beyond sound would be interesting, and exploring further tools for enabling designers to reflect on their knowledge as members will likely yield interesting insights. I am particularly interested in how video can be used to document interactions and as design material (Pelikan, Porfirio, & Winkle, 2023). Working further towards other ways of integrating interaction design and EMCA work is fruitful for both sides: EMCA researchers gain tools that they can deploy in collaborative projects with engineers and computer scientists even if they are not trained as designers. Designers who draw on EMCA informed design methods can ground their work more explicitly in reflections on their own knowledge as competent members in interaction.

6.5 Implications for Theories on Interaction, Language, Mind and Body

Being part of the project "Vocal Practices for Coordinating Human Action" (VR 2016-00827, PI: Leelo Keevallik) that investigates the role of sound in interaction, my research has theoretical implications for the interplay of body and mind, and for what is considered language.

First, I contribute to overcoming the mind-body dualism in scientific work. The division of mind and body as separate entities in philosophy has been challenged by phenomenology (see in particular Merleau-Ponty, 1945; Schuetz, 1945). Ethnomethodology is grounded in phenomenology (Coulter, 1979), and multimodal conversation analysis that builds on those traditions has recently taken an embodied turn (Nevile, 2015), consolidating a focus on the interplay between bodily and verbal resources in interaction. A reconciliation of mind and body has also happened in cognitive science, where more recent theories on embodied and distributed cognition have argued for thinking that goes beyond the brain of one person (Hollan, Hutchins, & Kirsh, 2000; Kirsh, 2010). More recently, enactivism is further challenging existing models of cognition, studying the intertwinement of language and body (Jaegher, Peräkylä, & Stevanovic, 2016; Paolo, Cuffari, & Jaegher, 2018). While these fields have moved beyond the Cartesian understanding of the mind-body relationship, computer science remains dominated by a dualistic view of mind and body. HCI research has started to challenge this (Dourish, 2001), developing design methods that include the whole body (Höök, 2018). In robotics, mind and body remain neatly separated: (Electrical) engineers build the moving robot, the hardware, while computer scientists work on the software, the inherent logic. This is only being challenged recently, through an influence of HCI work in HRI (see e.g. Serholt, Ljungblad, & Bhroin, 2022; Weiss & Spiel, 2022; Winkle et al., 2022). In exploring multimodal sound Gestalts as a (minimal) resource for communication,

this dissertation contributes to more embodied approaches to design and HRI in a very concrete way. The evidence I presented challenges the notion that robots need to communicate through lexical utterances or visual displays, demonstrating that a robot's body can be designed to give off information about its status (Article II). I have also developed robot design methods that help engaging with norms of how voice and body may be used in professional work (Article IV).

Second, my studies contribute to a reconceptualization of interaction. While traditionally only lexical items have been regarded as being communicative, with non-lexical items being dismissed as primitive, or accidental mistakes (c.f. Chomsky, 1965; Wundt, 1900), this work has set out to explore how relatively recent fundamental research into how humans use non-lexical vocalizations (Dingemanse, Torreira, & Enfield, 2013; Keevallik & Ogden, 2020; Wiggins, 2002) could be translated to robots. The fields of HRI and even HCI are still grounded in a more traditional focus on lexical interaction, which is only recently getting challenged (Dingemanse & Liesenfeld, 2022). Having demonstrated that humans treat robot sounds as meaningful (when they occur at sequentially relevant positions, Article I, II, V), I have contributed a piece to this still largely underexplored puzzle. Robots may not need to use lexical instructions in a specific language, but may instead use sound to communicate.

Finally, my work challenges the notion that robots always need to understand and produce full lexical utterances. Studying Cozmo and the autonomous shuttle buses, I have analyzed two systems with no speech recognition components embedded that cannot process what is said around them. Demonstrating how these robots can still (temporarily) participate in interaction with humans (Pelikan, Broth, & Keevallik, 2022), my work highlights that there may be solutions for building robot interfaces that do not rely on large language models, potentially contributing not only to cheaper but also more sustainable interfaces: Sound design does not require collection, storage, and analysis of vast amounts of data in server centers (Crawford & Joler, 2018). In many cases, sounds also do not require translation[1]. Sound may therefore be a more inclusive interaction resource - if it is well designed.

6.6 Implications for Practitioners

This work also has practical implications for designing interaction with robots. These implications may be relevant both for professional designers and anyone who practically ends up being involved in robot development, such as students working on robotics projects and EMCA scholars collaborating with roboticists.

I highlighted how robot sound design could take inspiration from the role that non-lexical vocalizations play in human interaction. Being semantically underspecified (Keevallik & Ogden, 2020; Reber, 2012), non-lexical vocalizations are more flexible than words and can gain different local meanings depending on the context that they are uttered in (Dingemanse, 2020). Similarly, robot sound can be especially appealing in settings where no particular language should be used and where it may even be beneficial to express multiple things through one type of sound -

[1] An adaptation to the specific cultural context may be necessary but still less complex than translating between different languages.

think of the creative use of car horns (Laurier et al., 2020). Sounds are less accountable than verbal utterances, which makes them a more implicit (Ju, 2015) design resource. Their inherent ambiguity also makes them particularly well suited for supporting the dynamic status of a robot, which may alternate between participating in interaction and being treated as an object. Therefore, sound as an interactive resource is especially interesting for non-humanoid robots and can be a cheap solution where expensive language models are not an option - but it needs to be designed carefully to support interaction.

When designing sound for interaction, aesthetics is only one aspect that needs to be addressed. A key question designers should ask themselves at the very beginning of their sound design process is what a sound animation should accomplish in a specific interaction. I provided examples of how sound contributes to the accomplishment of actions-in-interaction. For instance, happy sound animations at the end of a successful task can be interpreted as a "go ahead". In contrast, sad sound animations after a failed collaborative action get treated as "go back to the previous action" (Article I). Mechanic cog-like sounds played during a joint activity may get treated as "current activity is on hold" and "ding" sounds as "ready to resume joint activity" (Article II). Repetition of sounds by a bus while approaching a person gets interpreted as "keep moving out of the way" (Article V). These examples highlight that aesthetics is only one aspect that needs to be addressed in a robot sound design process. Exploring early on how sound contributes to the organization of interaction is equally important.

Ensuring a focus on interaction requires careful design and testing. In this dissertation, I have presented several methods for sound design in interaction. First, I developed a warm up for starting collaborative sound design sessions. I found that vocal sketching (Panariello et al., 2019; Rocchesso et al., 2015) can be challenging even for people who have experience with design because of social norms related to how the voice should be used in professional settings, where we have been socialized into avoiding "um" during speeches and "mmh" or "yuk" when tasting food for instance. Creating a welcoming atmosphere is therefore an essential prerequisite for collaborative design work. Playing sound ball with a single syllable (see Articles IV for details) can open the design team's eyes (or rather ears) to how sound can accomplish interactional goals. Second, using video-recorded interactions as design material in video voice-overs (see Article IV and V) is an excellent way to become aware of one's own knowledge as a member and much better than making assumptions about interaction. As Enfield (2013) highlights, it is easy to miss aspects of interaction that are very common and to imagine interactions that generally occur rarely. Grounding design in concrete cases can help to avoid this pitfall. Data of several people performing voice-overs on video recordings could potentially also be used as training data for machine learning models. This could help to identify appropriate moments for robots to produce a sound (or to speak), effectively identifying transition-relevance places (Sacks, Schegloff, & Jefferson, 1974), moments where speaker change is possible and it would be normatively expectable for a robot to take a turn.

This work also tapped into questions about what sounds are appropriate for different robots and at different moments. While several designs may accomplish similar social actions, sounds with specific features may be generally better suited

to accomplish certain actions. For robots in close interaction with humans in a relatively quiet environment such as a home, consequential sounds originating from gears and servos may not necessarily be bad (if they are not too loud). Just like the rustling of fans and harddrives in old PCs, they can make transparent what is happening inside a robot, for instance, how burdensome a task or movement is at a particular moment. In manual cars, people rely heavily on their hearing to identify when to switch gears (Bijsterveld & Krebs, 2013; Cromdal et al., 2020). In line with prior work that has argued for the benefits of implicit sound (Moore, Currano, & Sirkin, 2020), my dissertation highlights that consequential sound, if it is not too loud nor poses a health risk, can be an important tool for supporting interaction with robots. In contrast, when autonomous public transport vehicles move contrary to common expectations, such implicit (Ju, 2015; Moore, Currano, & Sirkin, 2020) sounds may not be enough to communicate that surrounding traffic participants should move away from the vehicle. In some settings, sounds that evoke clear associations and expectations may be needed. Such sounds could be inspired by human vocalizations or other household technology, including "ding"s and "beep"s. HCI research has previously taken inspiration from everyday sounds (Gaver, 1986, 1993) grounded in experiences with the material world, which can also inspire HRI design.

Finally, I have explored the intersection of EMCA and interaction design, contributing to hybrid studies and developing what has been labeled technomethodology (Button & Dourish, 1996) for interaction with robots. Intertwining both approaches is generally feasible, especially when staying grounded in the phenomenological tradition that EMCA work stands in. While EMCA cannot cover all aspects of the design process and new methods need to be developed (see Articles IV and V), I have highlighted how the materials that EMCA works with, video recordings and transcripts, can be embedded into a design process not only as informative data or for evaluating a design but also as a design material. They facilitate a stronger grounding of the design in situated interaction (Suchman, 2007), in which users may interact with robots differently than expected by the design team. They also enable designing for actual interaction in real-world contexts rather than imagined scenarios. EMCA further lends itself to participatory design processes, especially in settings where robots are already deployed (see Article V). Rather than focusing on researchers' theories, EMCA puts the lived experience and understanding of the members in the setting at the center. This focus on participants' understanding makes working on eye level with lay designers easier (see also Hornecker et al., 2022; Krummheuer, Rehm, & Rodil, 2020).

7 Conclusion

This dissertation investigated sound in human-robot interaction. Taking an EMCA approach, I explored how sound can support the coordination of humans and robots. I provided answers to the three research questions set out at the beginning through five studies with three different robots.

First, I asked how humans make sense of robot sound in everyday interaction. I illustrated how humans attribute situated meaning to robot sound, which may or may not align with what the designers intended. I also showed examples of sound that gets ignored when it is not intelligible as a relevant contribution to an ongoing interaction, and identified moments in which sound is noticeably absent, opening opportunities for design. I argued that the inherent ambiguity of sound animations can be a resource for design.

Second, I asked whether robot sound can play a role in the accomplishment of sociality. My work challenges the prevalent HRI paradigm that aims to inscribe a specific meaning (usually an emotion or intention) into designed robot sounds. I show how robot sound animations get treated as relevant actions that have consequences for the future trajectory of the interaction. Sounds designed to display emotions gave participants a quick and vague indication of whether the interaction with Cozmo was still on track. A combination of machine sounds and human-inspired "uh"s supported the withdrawal of the robot from a joint activity when its response was not yet ready. In my study of autonomous buses, I demonstrated that the repetition of sounds could support coordination, by instructing humans to move.

Third, I asked how EMCA can reconceptualize how interaction is conceived in HRI. I explored how EMCA and interaction design can be practically combined and created a design framework that tightly intertwines both approaches. Within this

framework, I developed practices that combine EMCA practices of closely working with video with sonic interaction design. Specifically, I introduced a video voice-over method that is grounded in vocal sketching and uses video as a design material. Engaging in video voice-overs can be particularly beneficial for interdisciplinary teams that want to bridge EMCA and interaction design. I also provided an example of a long-term case study with autonomous buses that explores sound design during rides on public roads in a participatory manner.

I demonstrated different ways in which EMCA can contribute to HRI, ranging from empirical studies to the development of novel design methods that can deal with the contingencies of real-world settings. My work contributes field studies in the home and on the road to HRI, demonstrating how humans interact with robots "in the wild". It also contributes to EMCA through detailed descriptions of how robots are embedded into the social organization of interaction. My work can be seen as a contribution to technomethodology, a hybrid discipline that combines ethnomethodology and design. Learning from the history of EMCA work in HCI, it is unlikely that large numbers of HRI researchers will undergo years of training in EMCA analysis. Instead, I am interested in how EMCA can inform the development of methods for robot design that are easily accessible to both researchers and practitioners. In future work, I aim to explore further how EMCA perspectives on accountable, situated, and sequential action may be embedded into tools for designing human-robot interaction.

Bibliography

Alač, M. (2016). Social robots: Things or agents? *AI & SOCIETY*, *31*(4), 519–535. https://doi.org/10.1007/s00146-015-0631-6

Alač, M., Movellan, J., & Tanaka, F. (2011). When a robot is social: Spatial arrangements and multimodal semiotic engagement in the practice of social robotics. *Social Studies of Science*, *41*(6), 893–926. https://doi.org/10.1177/0306312711420565

Albert, S. (2015). Rhythmical coordination of performers and audience in partner dance: Delineating improvised and choreographed interaction. *Etnografia e Ricerca Qualitativa*, (3), 399–428. https://doi.org/10.3240/81723

Anderson-Bashan, L., Megidish, B., Erel, H., Wald, I., Hoffman, G., Zuckerman, O., & Grishko, A. (2018). The greeting machine: An abstract robotic object for opening encounters. *2018 27th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 595–602. https://doi.org/10.1109/ROMAN.2018.8525516

Andrist, S., Bohus, D., Kamar, E., & Horvitz, E. (2017). What went wrong and why? Diagnosing situated interaction failures in the wild. In A. Kheddar, E. Yoshida, S. S. Ge, K. Suzuki, J.-J. Cabibihan, F. Eyssel, & H. He (Eds.), *Social Robotics* (pp. 293–303). Springer.

Anund, A., Ludovic, R., Caroleo, B., Hardestam, H., Dahlman, A., Skogsmo, I., Nicaise, M., & Arnone, M. (2022). Lessons learned from setting up a demonstration site with autonomous shuttle operation – based on experience from three cities in Europe. *Journal of Urban Mobility*, *2*, 100021. https://doi.org/10.1016/j.urbmob.2022.100021

Aoki, P. M., Grinter, R. E., Hurst, A., Szymanski, M. H., Thornton, J. D., & Woodruff, A. (2002). Sotto voce: Exploring the interplay of conversation and mobile audio spaces. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 431–438. https://doi.org/10.1145/503376.503454

Aoki, P. M., Romaine, M., Szymanski, M. H., Thornton, J. D., Wilson, D., & Woodruff, A. (2003). The mad hatter's cocktail party: A social mobile audio space supporting multiple simultaneous conversations. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 425–432. https://doi.org/10.1145/642611.642686

Arend, B., Sunnen, P., & Caire, P. (2017). Investigating breakdowns in human robot interaction: A conversation analysis guided single case study of a human-robot communication in a museum environment. *International Journal of Mechanical, Aerospace, Industrial, Mechatronic and Manufacturing Engineering*, *11*(5). http://www.waset.org/publications/10006962

Arminen, I., Licoppe, C., & Spagnolli, A. (2016). Respecifying mediated interaction. *Research on Language and Social Interaction*, *49*(4), 290–309. https://doi.org/10.1080/08351813.2016.1234614

Arminen, I., & Weilenmann, A. (2009). Mobile presence and intimacy: Reshaping social actions in mobile contextual configuration. *Journal of Pragmatics*, *41*(10), 1905–1923. https://doi.org/10.1016/j.pragma.2008.09.016

Asaro, P. M. (2000). Transforming society by transforming technology: The science and politics of participatory design. *Accounting, Management and Information Technologies*, *10*(4), 257–290. https://doi.org/https://doi.org/10.1016/S0959-8022(00)00004-7

Association for Computing Machinery. (2018). ACM code of ethics and professional conduct. https://www.acm.org/code-of-ethics

Axelsson, M., Oliveira, R., Racca, M., & Kyrki, V. (2022). Social robot co-design canvases: A participatory design framework. *ACM Transactions on Human-Robot Interaction*, *11*(1), 1–39. https://doi.org/10.1145/3472225

Azenkot, S., Feng, C., & Cakmak, M. (2016). Enabling building service robots to guide blind people a participatory design approach. *Proceedings of the 2016 11th ACM/IEEE International Conference on Human-Robot Interaction*, 3–10. https://doi.org/10.1109/HRI.2016.7451727

Baharin, H., Rintel, S., & Viller, S. (2013). Rhythms of the domestic soundscape: Ethnomethodological soundwalks for phatic technology design. In P. Kotzé, G. Marsden, G. Lindgaard, J. Wesson, & M. Winckler (Eds.), *Human-Computer Interaction – INTERACT 2013* (pp. 463–470). Springer Berlin Heidelberg. https://doi.org/10.1007/978-3-642-40498-6_36

Bannon, L. J. (1995). From human factors to human actors: The role of psychology and human-computer interaction studies in system design. In R. M. Baecker, J. Grudin, W. A. S. Buxton, & S. Greenberg (Eds.), *Readings in Human–Computer Interaction* (pp. 205–214). Morgan Kaufmann. https://doi.org/https://doi.org/10.1016/B978-0-08-051574-8.50024-8

Bartneck, C., & Jun, H. (2004). Rapid prototyping for interactive robots. *Proceedings of the 8th Conference on Intelligent Autonomous Systems (IAS-8)*, 136–145. https://doi.org/10.6084/M9.FIGSHARE.5160775

Bedini, S. A. (1964). The role of automata in the history of technology. *Technology and Culture*, *5*(1), 24–42. https://doi.org/10.2307/3101120

Bekey, G. A. (2005). *Autonomous Robots: From Biological Inspiration to Implementation and Control*. MIT Press.

Bell, A. (1984). Language style as audience design. *Language in Society*, *13*(2), 145–204. https://doi.org/10.1017/S004740450001037X

Bijsterveld, K., & Krebs, S. (2013). Listening to the sounding objects of the past: The case of the car. In K. Franinović & S. Serafin (Eds.), *Sonic Interaction Design* (pp. 3–38). MIT Press.

Björling, E., & Rose, E. (2019). Participatory research principles in human-centered design: Engaging teens in the co-design of a social robot. *Multimodal Technologies and Interaction*, *3*(1), 8. https://doi.org/10.3390/mti3010008

Blattner, M. M., Papp III, A. L., & Glinert, E. P. (1994). Sonic enhancement of two-dimensional graphics displays. In G. Kramer (Ed.), *Auditory Display: Sonification, Audification, and Auditory Interfaces* (pp. 447–470). Addison-Wesley.

Blattner, M. M., Sumikawa, D. A., & Greenberg, R. M. (1989). Earcons and icons: Their structure and common design principles. *Human–Computer Interaction*, *4*(1), 11–44. https://doi.org/10.1207/s15327051hci0401_1

Bødker, S. (1989). A human activity approach to user interfaces. *Human-Computer Interaction*, *4*(3), 171–195.

Bødker, S. (2006). When second wave HCI meets third wave challenges. *Proceedings of the 4th Nordic Conference on Human-Computer Interaction: Changing Roles*, 1–8. https://doi.org/10.1145/1182475.1182476

Bødker, S., & Iversen, O. S. (2002). Staging a professional participatory design practice: Moving PD beyond the initial fascination of user involvement. *Proceedings of the Second Nordic Conference on Human-Computer Interaction*, 11–18. https://doi.org/10.1145/572020.572023

Bohus, D., & Horvitz, E. (2011). Multiparty turn taking in situated dialog: Study, lessons, and directions. *Proceedings of the SIGDIAL 2011 Conference*, 98–109.

Boudouraki, A., Fischer, J. E., Reeves, S., & Rintel, S. (2021). "I can't get round" Recruiting assistance in mobile robotic telepresence. *Proceedings of the ACM on Human-Computer Interaction*, *4*(CSCW3), 1–21. https://doi.org/10.1145/3432947

Broth, M., Cromdal, J., & Levin, L. (2018). Showing where you're going. instructing the accountable use of the indicator in live traffic. *International Journal of Applied Linguistics*, *28*(2), 248–264. https://doi.org/https://doi.org/10.1111/ijal.12194

Broth, M., & Keevallik, L. (2014). Getting ready to move as a couple: Accomplishing mobile formations in a dance class. *Space and Culture*, *17*(2), 107–121. https://doi.org/10.1177/1206331213508483

Broth, M., & Keevallik, L. (Eds.). (2020). *Multimodal Interaktionsanalys*. Studentlitteratur.

Broth, M., Laurier, E., & Mondada, L. (Eds.). (2014). *Studies of Video Practices: Video at Work*. Routledge.

Broth, M., & Mondada, L. (2013). Walking away: The embodied achievement of activity closings in mobile interaction. *Journal of Pragmatics*, *47*(1), 41–58. https://doi.org/https://doi.org/10.1016/j.pragma.2012.11.016

Brown, B. (2017). The social life of autonomous cars. *Computer*, *50*(2), 92–96. https://doi.org/10.1109/MC.2017.59

Brown, B., McGregor, M., & Laurier, E. (2013). Iphone in vivo: Video analysis of mobile device use. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1031–1040. https://doi.org/10.1145/2470654.2466132

Bruckenberger, U., Weiss, A., Mirnig, N., Strasser, E., Stadler, S., & Tscheligi, M. (2013). The good, the bad, the weird: Audience evaluation of a "real" robot in relation to science fiction and mass media. In G. Herrmann, M. J. Pearson, A. Lenz, P. Bremner, A. Spiers, & U. Leonards (Eds.), *Social Robotics* (pp. 301–310). Springer. https://doi.org/10.1007/978-3-319-02675-6_30

Button, G. (1990). Going up a blind alley. In P. Luff, N. Gilbert, & D. B. T. Frohlich (Eds.), *Computers and Conversation* (pp. 67–90). Elsevier. https://doi.org/10.1016/B978-0-08-050264-9.50009-9

Button, G., & Dourish, P. (1996). Technomethodology. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems (CHI '96)*, 19–26. https://doi.org/10.1145/238386.238394

Čapek, K. (2004). *R.U.R. (Rossum's Universal Robots)* (C. Novack-Jones, Trans.). Penguin Random House. (Original work published 1920)

Cekaite, A., & Mondada, L. (2021). *Touch in Social Interaction*. London: Routledge.

Cha, E., Fitter, N. T., Kim, Y., Fong, T., & Matarić, M. J. (2018). Effects of robot sound on auditory localization in human-robot collaboration. *Proceedings of the 2018 ACM/IEEE International Conference*

on Human-Robot Interaction, 434–442. https://doi.org/10.1145/3171221.3171285

Chan, L., Zhang, B. J., & Fitter, N. T. (2021). Designing and validating expressive Cozmo behaviors for accurately conveying emotions. *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*, 1037–1044. https://doi.org/10.1109/RO-MAN50785.2021.9515425

Chomsky, N. (1965). *Aspects of the Theory of Syntax* (1985th ed.). MIT Press.

Clark, H. H. (1996). *Using Language*. Cambridge University Press. https://doi.org/10.1017/CBO9780511620539

Clark, H. H., & Murphy, G. L. (1982). Audience design in meaning and reference. *Advances in Psychology*, *9*, 287–299. https://doi.org/10.1016/S0166-4115(09)60059-5

Clark, H. H. H., & Brennan, S. E. (1991). Grounding in communication. In L. B. Resnick, J. M. Levine, & S. D. Teasley (Eds.), *Perspectives on Socially Shared Cognition* (pp. 222–233). American Psychological Association.

Clayman, S. E., & Maynard, D. W. (1995). Ethnomethodology and conversation analysis (P. ten Have & G. Psathas, Eds.). *Situated Order: Studies in the Social Organization of Talk and Embodied Activities*, 1–30.

Coulter, J. (1979). *The Social Construction of Mind*. Palgrave Macmillan UK. https://doi.org/10.1007/978-1-349-09379-3

Crabtree, A. (2000). Methodological numbers concerning the practical availability of work-practice to EM & CA. In *Proceedings of 2nd Workplace Studies Conference*. Language Study Group of the British Sociological Association.

Crabtree, A. (2004). Taking technomethodology seriously: Hybrid change in the ethnomethodology–design relationship. *European Journal of Information Systems*, *13*(3), 195–209. https://doi.org/10.1057/palgrave.ejis.3000500

Crawford, K., & Joler, V. (2018). *Anatomy of an AI System: The Amazon Echo as an Anatomical Map of Human Labor, Data and Planetary Resources*. https://anatomyof.ai/

Cromdal, J., Broth, M., Björklund-Flärd, D., & Levin, L. (2020). Sensational driving: Instructing and calibrating sensory perception in early driver training. In S. Wiggins & K. Osvaldsson Cromdal (Eds.), *Discursive Psychology and Embodiment* (pp. 169–196). Springer. https://doi.org/10.1007/978-3-030-53709-8_7

Dahlbäck, N., Jönsson, A., & Ahrenberg, L. (1993). Wizard of Oz studies: Why and how. *Proceedings of the 1st International Conference on Intelligent User Interfaces*, 193–200. https://doi.org/10.1145/169891.169968

Dautenhahn, K. (2014). Human-robot interaction. In *The Encyclopedia of Human-Computer Interaction* (2nd Edition). Interaction Design Foundation.

Dautenhahn, K. (2018). Some brief thoughts on the past and future of human-robot interaction. *ACM Transactions on Human-Robot Interaction*, *7*(1). https://doi.org/10.1145/3209769

Deppermann, A., Laurier, E., Mondada, L., Broth, M., Cromdal, J., Stefani, E. D., Haddington, P., Levin, L., Nevile, M., & Rauniomaa, M. (2018). Overtaking as an interactional achievement: Video analyses of participants' practices in traffic. *Gesprächsforschung*, *19*, 1–131.

Dingemanse, M. (2020). Between sound and speech: Liminal signs in interaction. *Research on Language and Social Interaction*, *53*(1), 188–196. https://doi.org/10.1080/08351813.2020.1712967

Dingemanse, M., & Liesenfeld, A. (2022). From text to talk: Harnessing conversational corpora for humane and diversity-aware language technology. *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, 5614–5633. https://doi.org/10.18653/v1/2022.acl-long.385

Dingemanse, M., Torreira, F., & Enfield, N. J. (2013). Is "huh?" a universal word? Conversational infrastructure and the convergent evolution of linguistic items. *PLOS ONE*, *8*(11), 1–10. https://doi.org/10.1371/journal.pone.0078273

Doehler, S. P., Keevallik, L., & Li, X. (2022). Editorial: The grammar-body interface in social interaction. *Frontiers in Psychology*, *13*. https://doi.org/10.3389/fpsyg.2022.875696

Dourish, P. (2001). *Where the Action is: The Foundations of Embodied Interaction*. MIT Press.

Dourish, P., & Button, G. (1998). On "technomethodology": Foundational relationships between ethnomethodology and system design. *Human–Computer Interaction*, *13*(4), 395–432. https://doi.org/10.1207/s15327051hci1304_2

Due, B. L. (2021). Distributed perception: Co-operation between sense-able, actionable, and accountable semiotic agents. *Symbolic Interaction*, *44*(1), 134–162. https://doi.org/10.1002/symb.538

Due, B. L. (2022). Guide dog versus robot dog: Assembling visually impaired people with non-human agents and achieving assisted mobility through distributed co-constructed perception. *Mobilities*, *18*(1), 148–166. https://doi.org/10.1080/17450101.2022.2086059

Due, B. L. (2023). A walk in the park with robodog: Navigating around pedestrians using a spot robot as a "guide dog". *Space and Culture*, 1–17. https://doi.org/https://doi.org/10.1177/12063312231159215

Due, B. L., & Toft, T. L. W. (2021). Phygital highlighting: Achieving joint visual attention when physically co-editing a digital text. *Journal of Pragmatics*, *177*, 1–17. https://doi.org/10.1016/j.pragma.2021.01.034

Ekman, I., & Rinott, M. (2010). Using vocal sketching for designing sonic interactions. *Proceedings of the 8th ACM Conference on Designing Interactive Systems*, 123–131. https://doi.org/10.1145/1858171.1858195

Enfield, N. J. (2013). *Relationship Thinking: Agency, Enchrony, and Human Sociality*. Oxford University Press.

European Commission. (2019). *Electric and hybrid cars: New rules on noise emitting to protect vulnerable road users* (tech. rep.). https://single-market-economy.ec.europa.eu/select-language?destination=/node/1136

Ferm, U. M., Claesson, B. K., Ottesjö, C., & Ericsson, S. (2015). Participation and enjoyment in play with a robot between children with cerebral palsy who use AAC and their peers. *Augmentative and Alternative Communication*, *31*(2), 108–123. https://doi.org/10.3109/07434618.2015.1029141

Fernaeus, Y., Håkansson, M., Jacobsson, M., & Ljungblad, S. (2010). How do you play with a robotic toy animal? A long-term study of Pleo. *Proceedings of IDC2010: The 9th International Conference on Interaction Design and Children*, 39–48. https://doi.org/10.1145/1810543.1810549

Fischer, J. E., Reeves, S., Porcheron, M., & Sikveland, R. O. (2019). Progressivity for voice interface design. *Proceedings of the 1st International Conference on Conversational User Interfaces*. https://doi.org/10.1145/3342775.3342788

Fischer, K., Soto, B., Pantofaru, C., & Takayama, L. (2014). Initiating interactions in order to get help: Effects of social framing on people's responses to robots' requests for assistance. *The 23rd IEEE International Symposium on Robot and Human Interactive Communication*, 999–1005. https://doi.org/10.1109/ROMAN.2014.6926383

Fischer, K. (2016). *Designing Speech for a Recipient*. John Benjamins Publishing Company.

Fischer, K. (2021). Effect confirmed, patient dead: A commentary on Hoffman & Zhao's primer for conducting experiments in HRI. *ACM Transactions on Human-Robot Interaction*, *10*(1). https://doi.org/10.1145/3439714

Fischer, K., Jensen, L. C., & Bodenhagen, L. (2014). To beep or not to beep is not the whole question. In M. Beetz, B. Johnston, & M. Williams (Eds.). Springer. https://doi.org/10.1007/978-3-319-11973-1_16

Fong, T., Nourbakhsh, I., & Dautenhahn, K. (2003). A survey of socially interactive robots. *Robotics and Autonomous Systems*, *42*(3-4), 143–166.

Ford, C. E., & Thompson, S. A. (1996). Interactional units in conversation: Syntactic, intonational, and pragmatic resources for the management of turns. In E. Ochs, E. A. Schegloff, & S. A. Thompson (Eds.),

Interaction and grammar (pp. 134–184). Cambridge University Press. https://doi.org/10.1017/CBO9780511620874.003

Forlizzi, J., & DiSalvo, C. (2006). Service robots in the domestic environment: A study of the roomba vacuum in the home. *Proceedings of the 1st ACM SIGCHI/SIGART Conference on Human-Robot Interaction*, 258–265. https://doi.org/10.1145/1121241.1121286

Fraser, M., Benford, S., Hindmarsh, J., & Heath, C. (1999). Supporting awareness and interaction through collaborative virtual interfaces. *Proceedings of the 12th Annual ACM Symposium on User Interface Software and Technology*, 27–36. https://doi.org/10.1145/320719.322580

Frid, E., & Bresin, R. (2022). Perceptual evaluation of blended sonification of mechanical robot sounds produced by emotionally expressive gestures: Augmenting consequential sounds to improve non-verbal robot communication. *International Journal of Social Robotics*, *14*(2), 357–372. https://doi.org/10.1007/s12369-021-00788-4

Frid, E., Bresin, R., & Alexandersson, S. (2018). Perception of mechanical sounds inherent to expressive gestures of a NAO robot - Implications for movement sonification of humanoids. *Proceedings of the 15th Sound and Music Computing Conference (SMC2018)*. https://doi.org/10.5281/zenodo.1422499

Frohlich, D., & Luff, P. (1990). Applying the technology of conversation to the technology for conversation. In P. Luff, N. Gilbert, & D. B. T. Frohlich (Eds.), *Computers and Conversation* (pp. 187–220). Academic Press. https://doi.org/https://doi.org/10.1016/B978-0-08-050264-9.50014-2

Fukuda, H., Yamazaki, K., Yamazaki, A., Saito, Y., Iiyama, E., Yamazaki, S., Kobayashi, Y., Kuno, Y., & Ikeda, K. (2018). Enhancing multiparty cooperative movements: A robotic wheelchair that assists in predicting next actions. *Proceedings of the 20th ACM International Conference on Multimodal Interaction*, 409–417. https://doi.org/10.1145/3242969.3242983

Gan, Y., Greiffenhagen, C., & Reeves, S. (2020). Connecting distributed families: Camera work for three-party mobile video calls. *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*, 1–12. https://doi.org/10.1145/3313831.3376704

Garfinkel, H. (1967). *Studies in Ethnomethodology*. Prentice-Hall.

Garfinkel, H. (Ed.). (1986). *Ethnomethodological Studies of Work*. Routledge.

Garfinkel, H. (1996). Ethnomethodology's program. *Social Psychology Quarterly*, *59*(1), 5. https://doi.org/10.2307/2787116

Garfinkel, H., & Sacks, H. (1970). On formal structures of practical actions. In J. C. McKinney & E. A. Tiryakian (Eds.), *Theoretical Sociology: Perspectives and Development* (pp. 337–366). Appleton-Century-Crofts.

Garfinkel, H., & Wieder, D. L. (1992). Two incommensurable, asymmetrically alternate technologies of social analysis. In G. Watson & R. M. Seiler

(Eds.), *Text in Context: Contributions to Ethnomethodology* (pp. 175–206). Sage.

Gaver, W., Blythe, M., Boucher, A., Jarvis, N., Bowers, J., & Wright, P. (2010). The prayer companion: Openness and specificity, materiality and spirituality. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2055–2064. https://doi.org/10.1145/1753326.1753640

Gaver, W. W. (1986). Auditory icons: Using sound in computer interfaces. *Human–Computer Interaction*, *2*(2), 167–177. https://doi.org/10.1207/s15327051hci0202_3

Gaver, W. W. (1993). What in the world do we hear?: An ecological approach to auditory event perception. *Ecological Psychology*, *5*(1), 1–29. https://doi.org/10.1207/s15326969eco0501_1

Gaver, W. W., & Norman, D. A. (1988). *Everyday Listening and Auditory Icons* (Doctoral dissertation) [AAI8908009]. University of California, San Diego.

Geertz, C. (1973). Thick description: Towards an interpretive theory of culture. In C. Geertz (Ed.), *The Interpretation of Cultures. Selected Essays by Clifford Geertz* (pp. 3–32). Basic Books.

Gehle, R., Pitsch, K., Dankert, T., & Wrede, S. (2017). How to open an interaction between robot and museum visitor? Strategies to establish a focused encounter in HRI. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*, 187–195. https://doi.org/10.1145/2909824.3020219

Gehle, R., Pitsch, K., & Wrede, S. (2014). Signaling trouble in robot-to-group interaction. Emerging visitor dynamics with a museum guide robot. *Proceedings of the Second International Conference on Human-Agent Interaction*, 361–368. https://doi.org/10.1145/2658861.2658887

Ghafurian, M., Reitter, D., & Ritter, F. E. (2020). Countdown timer speed. *ACM Transactions on Computer-Human Interaction*, *27*(2), 1–25. https://doi.org/10.1145/3380961

Goffman, E. (1971). *Relations in Public: Microstudies of the Public Order*. Basic Books.

Goffman, E. (1978). Response cries. *Language*, *54*(4), 787–815. https://doi.org/10.2307/413235

Goffman, E. (1981). *Forms of Talk*. University of Pennsylvania Press.

Goodwin, C. (1979). The interactive construction of a sentence in natural conversation. In G. Psathas (Ed.), *Everyday Language: Studies in Ethnomethodology* (pp. 97–121). Irvington Publishers.

Goodwin, C. (2018). *Co-Operative Action*. Cambridge University Press.

Goodwin, C., & Goodwin, M. H. (1996). Seeing as situated activity: Formulating planes. In Y. Engeström & D. Middleton (Eds.), *Cognition and Communication at Work* (pp. 61–95). Cambridge University Press.

Greenbaum, J., & Kyng, M. (Eds.). (1991). *Design at Work: Cooperative Design of Computer Systems*. CRC Press.

Greiffenhagen, C., & Sharrock, W. (2019). Tensions in Garfinkel's Ethnomethodological Studies of Work programme discussed through Livingston's studies of mathematics. *Human Studies*, *42*(2), 253–279. https://doi.org/10.1007/s10746-019-09509-3

Gutwin, C., Benford, S., Dyck, J., Fraser, M., Vaghi, I., & Greenhalgh, C. (2004). Revealing delay in collaborative environments. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 503–510. https://doi.org/10.1145/985692.985756

Haddington, P., & Rauniomaa, M. (2014). Interaction between road users: Offering space in traffic. *Space and Culture*, *17*(2), 176–190. https://doi.org/10.1177/1206331213508498

Hall, E. T. (1966). *The Hidden Dimension*. Doubleday.

Harjunpää, K., Mondada, L., & Svinhufvud, K. (2018). The coordinated entry into service encounters in food shops: Managing interactional space, availability, and service during openings. *Research on Language and Social Interaction*, *51*(3), 271–291. https://doi.org/10.1080/08351813.2018.1485231

Harrison, S., Minneman, S., Stults, B., & Weber, K. (1989). Video: A design medium. *ACM SIGCHI Bulletin*, *21*(2), 62–67. https://doi.org/10.1145/70609.70616

Harrison, S., Tatar, D., & Sengers, P. (2007). The three paradigms of HCI. *Proceedings of ALT.CHI Session at the SIGCHI Conference on Human Factors in Computing Systems*. https://people.cs.vt.edu/~srh/Downloads/TheThreeParadigmsofHCI.pdf

Heath, C. (1984). Participation in the medical consultation: The co-ordination of verbal and nonverbal behaviour between the doctor and patient. *Sociology of Health & Illness*, *6*(3), 311–388. https://doi.org/10.1111/1467-9566.ep10491964

Heath, C., Hindmarsh, J., & Luff, P. (2010). *Video in Qualitative Research: Analysing Social Interaction in Everyday Life*. Sage. https://doi.org/10.4135/9781526435385

Heath, C., Knoblauch, H., & Luff, P. (2000). Technology and social interaction: The emergence of "workplace studies". *British Journal of Sociology*, *51*(2), 299–320. https://doi.org/10.1111/j.1468-4446.2000.00299.x

Heath, C., & Luff, P. (1992). Collaboration and control: Crisis management and multimedia technology in london underground line control rooms. *Computer Supported Cooperative Work (CSCW)*, *1*(1-2), 69–94. https://doi.org/10.1007/BF00752451

Heath, C., & Luff, P. (2000). *Technology in Action*. Cambridge University Press. https://doi.org/10.1017/CBO9780511489839

Henderson, A. (1989). Video and design. *ACM SIGCHI Bulletin*, *21*(2), 104–107. https://doi.org/10.1145/70609.70627

Henschel, A., Laban, G., & Cross, E. S. (2021). What makes a robot social? a review of social robots from science fiction to a home or hospital near you. *Current Robotics Reports*, *2*(1), 9–19. https://doi.org/10.1007/s43154-020-00035-0

Heritage, J. (1984). *Garfinkel and Ethnomethodology*. Polity Press.

High-Level Expert Group on Artificial Intelligence. (2018). Draft ethics guidelines for trustworthy AI (European Commission, Ed.). https://ec.europa.eu/newsroom/dae/document.cfm?doc_id=57112

Hindmarsh, J., Fraser, M., Heath, C., Benford, S., & Greenhalgh, C. (2000). Object-focused interaction in collaborative virtual environments. *ACM Transactions on Computer-Human Interaction*, *7*(4), 477–509. https://doi.org/10.1145/365058.365088

Hoey, E. M. (2020). Waiting to inhale: On sniffing in conversation. *Research on Language and Social Interaction*, *53*(1), 118–139. https://doi.org/10.1080/08351813.2020.1712962

Hoffman, G., & Weinberg, G. (2010). Synchronization in human-robot musicianship. *19th International Symposium in Robot and Human Interactive Communication*, 718–724. https://doi.org/10.1109/ROMAN.2010.5598690

Hoffman, G., & Ju, W. (2014). Designing robots with movement in mind. *Journal of Human-Robot Interaction*, *3*(1), 91–122.

Hoffman, G., & Zhao, X. (2020). A primer for conducting experiments in human–robot interaction. *ACM Transactions on Human-Robot Interaction*, *10*(1). https://doi.org/10.1145/3412374

Hoffmann, E. (2003). *Der Sandmann*. Suhrkamp. (Original work published 1816)

Hofstetter, E. (2016). *Citizens Getting Help: Interactions at the Constituency Office* (Doctoral dissertation). Loughborough University. Loughborough. http://urn.kb.se/resolve?urn=urn:nbn:se:liu:diva-160942

Hofstetter, E. (2020). Thinking with the body: Embodying thinking as a practice in board games. In S. Wiggins & K. O. Cromdal (Eds.), *Discursive Psychology and Embodiment* (pp. 247–273). Springer. https://doi.org/10.1007/978-3-030-53709-8_10

Hofstetter, E. (2021). Analyzing the researcher-participant in EMCA. *Social Interaction. Video-Based Studies of Human Sociality*, *4*(2). https://doi.org/10.7146/si.v4i2.127185

Hofstetter, E., Keevallik, L., & Löfgren, A. (2021). Suspending syntax: Bodily strain and progressivity in talk. *Frontiers in Communication*, *6*. https://doi.org/https://doi.org/10.3389/fcomm.2021.663307

Hohenstein, J., Larson, L. E., Hou, Y. T.-Y., Harris, A. M., Schecter, A., Dechurch, L., Contractor, N., & Jung, M. F. (2022). Vero: A method for remotely studying human-AI collaboration. *Proceedings of the 55th Hawaii International Conference on System Sciences*. https://doi.org/10.24251/HICSS.2022.030

Hollan, J., Hutchins, E., & Kirsh, D. (2000). Distributed cognition. *ACM Transactions on Computer-Human Interaction*, *7*(2), 174–196. https://doi.org/10.1145/353485.353487

Höök, K. (2018). *Designing with the Body: Somaesthetic Interaction Design*. MIT Press.

Höök, K., & Löwgren, J. (2012). Strong concepts: Intermediate-level knowledge in interaction design research. *ACM Transactions on Computer-Human Interaction*, *19*(3). https://doi.org/10.1145/2362364.2362371

Hornecker, E., Krummheuer, A., Bischof, A., & Rehm, M. (2022). Beyond dyadic HRI: Building robots for society. *Interactions*, *29*(3), 48–53. https://doi.org/10.1145/3526119

Hutchby, I., & Wooffitt, R. (2008). *Conversation Analysis* (SSe). Polity Press.

Hutchins, E. (1995). *Cognition in the Wild*. MIT Press.

International Organization for Standardization. (2021). *Robotics — Vocabulary* (Standard). https://www.iso.org/obp/ui/#iso:std:iso:8373:ed-3:v1:en

Ivarsson, J., & Greiffenhagen, C. (2015). The organization of turn-taking in pool skate sessions. *Research on Language and Social Interaction*, *48*(4), 406–429. https://doi.org/10.1080/08351813.2015.1090114

Ivarsson, J., & Lindwall, O. (2023). Suspicious minds: The problem of trust and conversational agents. *Computer Supported Cooperative Work (CSCW)*. https://doi.org/10.1007/s10606-023-09465-8

Jaegher, H. D., Peräkylä, A., & Stevanovic, M. (2016). The co-creation of meaningful action: Bridging enaction and interactional sociology. *Philosophical Transactions of the Royal Society B: Biological Sciences*, *371*(1693), 20150378. https://doi.org/10.1098/rstb.2015.0378

Jakonen, T., & Jauni, H. (2021). Mediated learning materials: Visibility checks in telepresence robot mediated classroom interaction. *Classroom Discourse*, *12*(1-2), 121–145. https://doi.org/10.1080/19463014.2020.1808496

Jarske, S., Raudaskoski, S., & Kaipainen, K. (2020). The "social" of the socially interactive robot: Rethinking human-robot interaction through ethnomethodology. In M. Nørskov, J. Seibt, & O. S. Quick (Eds.), *Culturally Sustainable Social Robotics. Proceedings of Robophilosophy 2020* (pp. 194–203). IOS Press. https://doi.org/10.3233/FAIA200915

Jefferson, G. (1984). Notes on a systematic deployment of the acknowledgement tokens "yeah"; and "mm hm". *Paper in Linguistics*, *17*(2), 197–216. https://doi.org/10.1080/08351818409389201

Jefferson, G. (2004). Glossary of transcript symbols with an introduction. In G. H. Lerner (Ed.), *Conversation Analysis: Studies from the First Generation* (pp. 13–31). John Benjamins. https://doi.org/10.1075/pbns.125.02jef

Johannsen, G. (2001). Auditory displays in human–machine interfaces of mobile robots for non-speech communication with humans. *Journal of*

Intelligent and Robotic Systems, *32*(2), 161–169. https://doi.org/10.1023/A:1013953213049

Jordan, B., & Henderson, A. (1995). Interaction analysis: Foundations and practice. *The Journal of the Learning Sciences*, *4*(1), 39–103. http://www.jstor.org/stable/1466849

Ju, W. (2015). *The Design of Implicit Interactions*. Springer. https://doi.org/10.1007/978-3-031-02210-4

Jung, M., & Hinds, P. (2018). Robots in the wild: A time for more robust theories of human-robot interaction. *ACM Transactions on Human-Robot Interaction*, *7*(1). https://doi.org/10.1145/3208975

Jung, M. F. (2017). Affective grounding in human-robot interaction. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*, 263–273. https://doi.org/10.1145/2909824.3020224

Jung, M. F., Lee, J. J., DePalma, N., Adalgeirsson, S. O., Hinds, P. J., & Breazeal, C. (2013). Engaging robots: Easing complex human-robot teamwork using backchanneling. *Proceedings of the 2013 Conference on Computer Supported Cooperative Work*, 1555–1566. https://doi.org/https://doi.org/10.1145/2441776.2441954

Kahneman, D. (2011). *Thinking, Fast and Slow* (1st ed). Farrar, Straus; Giroux.

Kappas, A., Küster, D., Dente, P., & Basedow, C. (2015). Simply the B.E.S.T.! Creation and validation of the Bremen emotional sounds toolkit. https://www.researchgate.net/publication/273577331%5C_Simply%5C_the%5C_BEST%5C_Creation%5C_and%5C_Validation%5C_of%5C_the%5C_Bremen%5C_Emotional%5C_Sounds%5C_Toolkit

Kaptelinin, V., & Nardi, B. A. (2006). *Acting with Technology: Activity Theory and Interaction Design*. MIT Press.

Keevallik, L. (2013). The interdependence of bodily demonstrations and clausal syntax. *Research on Language and Social Interaction*, *46*(1), 1–21. https://doi.org/10.1080/08351813.2013.753710

Keevallik, L. (2015). Coordinating the temporalities of talk and dance. In A. Deppermann & S. Günthner (Eds.), *Temporality in Interaction* (pp. 309–336). John Benjamins. https://doi.org/10.1075/slsi.27.10kee

Keevallik, L. (2018). Sequence initiation or self-talk? Commenting on the surroundings while mucking out a sheep stable. *Research on Language and Social Interaction*, *51*(3), 313–328. https://doi.org/10.1080/08351813.2018.1485233

Keevallik, L. (2021). Vocalizations in dance classes teach body knowledge. *Linguistics Vanguard*, *7*(s4), 20200098. https://doi.org/doi:10.1515/lingvan-2020-0098

Keevallik, L. (2023). Strain grunts and the organization of participation. In L. Mondada & A. Peräkylä (Eds.), *Body, Participation, and the Self:*

New Perspectives on Goffman in Language and Interaction. Routledge.

Keevallik, L., & Ekström, A. (2019). How to take the floor as a couple: Turn-taking in Lindy Hop jam circles. *Visual Anthropology*, *32*(5), 423–444. https://doi.org/10.1080/08949468.2019.1671750

Keevallik, L., Hofstetter, E., Weatherall, A., & Wiggins, S. (2023). Sounding others' sensations in interaction. *Discourse Processes*, *60*(1), 1–19. https://doi.org/10.1080/0163853X.2023.2165027

Keevallik, L., & Ogden, R. (2020). Sounds on the margins of language at the heart of interaction. *Research on Language and Social Interaction*, *53*(1), 1–18. https://doi.org/10.1080/08351813.2020.1712961

Kendon, A. (1990). *Conducting Interaction: Patterns of Behavior in Focused Encounters*. Cambridge University Press.

Kerrison, A. (2018). *We're All Behind You: The Co-Construction of Turns and Sequences-at-Cheering* (Doctoral dissertation). Ulster University. https://pure.ulster.ac.uk/ws/portalfiles/portal/12689965/2018KerrisonATPhD.pdf

Kirsh, D. (2010). Thinking with external representations. *AI & Society*, *25*(4), 441–454. https://doi.org/10.1007/s00146-010-0272-8

Kontogiorgos, D., & Pelikan, H. R. M. (2020). Towards adaptive and least-collaborative-effort social robots. *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, 311–313. https://doi.org/10.1145/3371382.3378249

Kose-Bagci, H., Ferrari, E., Dautenhahn, K., Syrdal, D. S., & Nehaniv, C. L. (2009). Effects of embodiment and gestures on social interaction in drumming games with a humanoid robot. *Advanced Robotics*, *23*(14), 1951–1996. https://doi.org/10.1163/016918609X12518783330360

Kress-Gazit, H., Eder, K., Hoffman, G., Admoni, H., Argall, B., Ehlers, R., Heckman, C., Jansen, N., Knepper, R., Křetínský, J., Levy-Tzedek, S., Li, J., Murphey, T., Riek, L., & Sadigh, D. (2021). Formalizing and guaranteeing human-robot interaction. *Communications of the ACM*, *64*(9), 78–84. https://doi.org/10.1145/3433637

Krummheuer, A. (2010). *Interaktion mit Virtuellen Agenten* (J. R. Bergmann, S. Hirschauer, & H. Kalthoff, Eds.; Vol. 11). Lucius.

Krummheuer, A. (2015). Technical agency in practice: The enactment of artefacts as conversation partners, actants and opponents. *PsychNology Journal*, *13*(2–3), 179–202.

Krummheuer, A. L., Rehm, M., & Rodil, K. (2020). Triadic human-robot interaction. distributed agency and memory in robot assisted interactions. *Companion of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, 317–319. https://doi.org/10.1145/3371382.3378269

Kuno, Y., Sadazuka, K., Kawashima, M., Yamazaki, K., Yamazaki, A., & Kuzuoka, H. (2007). Museum guide robot based on sociological in-

teraction analysis. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1191–1194. https://doi.org/10.1145/1240624.1240804

Kuzuoka, H., Pitsch, K., Suzuki, Y., Kawaguchi, I., Yamazaki, K., Yamazaki, A., Kuno, Y., Luff, P., & Heath, C. (2008). Effect of restarts and pauses on achieving a state of mutual orientation between a human and a robot. *Proceedings of the 2008 ACM Conference on Computer Supported Cooperative Work*, 201–204. https://doi.org/10.1145/1460563.1460594

Lagström, T., & Lundgren, V. M. (2015). *AVIP - Autonomous Vehicles' Interaction with Pedestrians: An Investigation of Pedestrian-Driver Communication and Development of a Vehicle External Interface* (Master's thesis). Chalmers University of Technology.

Langeveld, L., Egmond, R. V., Jansen, R., & Özcan, E. (2013). Product sound design: Intentional and consequential sounds. In D. A. Coelho (Ed.), *Advances in Industrial Design Engineering*. IntechOpen. https://doi.org/10.5772/55274

Latupeirissa, A. B., Panariello, C., & Bresin, R. (2020). Exploring emotion perception in sonic HRI. *Proceedings of the 17th Sound and Music Computing Conference*. https://doi.org/https://doi.org/10.5281/zenodo.3898927

Latupeirissa, A. B., Panariello, C., & Bresin, R. (2023). Probing aesthetics strategies for robot sound: Complexity and materiality in movement sonification. *ACM Transactions on Human-Robot Interaction*, 3585277. https://doi.org/10.1145/3585277

Laurier, E. (2014). The graphic transcript: Poaching comic book grammar for inscribing the visual, spatial and temporal aspects of action. *Geography Compass*, *8*(4), 235–248. https://doi.org/10.1111/gec3.12123

Laurier, E., Muñoz, D., Miller, R., & Brown, B. (2020). A bip, a beeeep, and a beep beep: How horns are sounded in Chennai traffic. *Research on Language and Social Interaction*, *53*(3), 341–356. https://doi.org/10.1080/08351813.2020.1785775

Lee, H. R., Cheon, E., Lim, C., & Fischer, K. (2022). Configuring humans: What roles humans play in HRI research. *Proceedings of the 2022 ACM/IEEE International Conference on Human-Robot Interaction*, 478–492.

Lee, H. R., Šabanović, S., Chang, W.-L., Nagata, S., Piatt, J., Bennett, C., & Hakken, D. (2017). Steps toward participatory design of social robots: Mutual learning with older adults with depression. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*, 244–253. https://doi.org/10.1145/2909824.3020237

Levinson, S. C., & Enfield, N. J. (Eds.). (2006). *Roots of Human Sociality*. Routledge.

Licoppe, C., Luff, P. K., Heath, C., Kuzuoka, H., Yamashita, N., & Tuncer, S. (2017). Showing objects: Holding and manipulating artefacts in video-mediated collaborative settings. *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*, 5295–5306. https://doi.org/10.1145/3025453.3025848

Licoppe, C., & Rollet, N. (2020). "Je dois y aller". Analyses de séquences de clôtures entre humains et robot. *Réseaux*, *220-221*(2), 151–193. https://www.cairn.info/article.php?ID_ARTICLE=RES_220_0151

Linell, P. (2009). *Rethinking Language, Mind, and World Dialogically*. Information Age Publishing.

Linke, A. (2009). Wie man Wissen teilt - oder: Vom Glück der Kommunikation. In H.-U. Rüegger, M. Arioli, & H. Murer (Eds.), *Universitäres Wissen teilen. Forschende im Dialog* (pp. 33–46.). VDF. https://doi.org/https://doi.org/10.5167/uzh-24813

Löffler, D., Schmidt, N., & Tscharn, R. (2018). Multimodal expression of artificial emotion in social robots using color, motion and sound. *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction*, 334–343. https://doi.org/10.1145/3171221.3171261

Löwgren, J. (2014). Interaction design - brief intro. In *The Encyclopedia of Human-Computer Interaction* (2nd Edition). Interaction Design Foundation. https://www.interaction-design.org/literature/book/the-encyclopedia-of-human-computer-interaction-2nd-ed/interaction-design-brief-intro

Luff, P., Gilbert, N., & Frohlich, D. B. T. (Eds.). (1990). *Computers and Conversation*. Academic Press. https://doi.org/https://doi.org/10.1016/C2009-0-21641-2

Luff, P., & Heath, C. (2000). The collaborative production of computer commands in command and control. *International Journal of Human-Computer Studies*, *52*(4), 669–699. https://doi.org/10.1006/ijhc.1999.0354

Luff, P., Heath, C., Kuzuoka, H., Hindmarsh, J., Yamazaki, K., & Oyama, S. (2003). Fractured ecologies: Creating environments for collaboration. *Human–Computer Interaction*, *18*(1-2), 51–84. https://doi.org/10.1207/S15327051HCI1812_3

Lupetti, M. L., Zaga, C., & Cila, N. (2021). Designerly ways of knowing in HRI: Broadening the scope of design-oriented HRI through the concept of intermediate-level knowledge. *Proceedings of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 389–398. https://doi.org/10.1145/3434073.3444668

Luria, M., Zimmerman, J., & Forlizzi, J. (2019). Championing research through design in HRI. *Arxiv*. https://doi.org/https://doi.org/10.48550/arXiv.1908.07572

Mackay, W. (2002). *Video to Support Interaction Design* [DVD]. ACM. https://www.lri.fr/~mackay/VideoForDesign/

Mackay, W. E., & Fayard, A. L. (1999). Video brainstorming and prototyping: Techniques for participatory design. *CHI '99 Extended Abstracts on Human Factors in Computing Systems*, 118–119. https://doi.org/10.1145/632716.632790

Mackay, W. E., Ratzer, A. V., & Janecek, P. (2000). Video artifacts for design: Bridging the gap between abstraction and detail. *Proceedings of the 3rd Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques*, 72–82. https://doi.org/10.1145/347642.347666

Mahadevan, K., Somanath, S., & Sharlin, E. (2018). Can interfaces facilitate communication in autonomous vehicle-pedestrian interaction? *Proceedings of the 2018 ACM/IEEE International Conference on Human-Robot Interaction Companion*, 309–310. https://doi.org/10.1145/3173386.3176909

Matarić, M. J. (2007). *The Robotics Primer*. The MIT Press.

McIlvenny, P., Davidsen, J. G., Kovács, A. B., & Stein, A. (2022). *DOTE: Distributed Open Transcription Environment* [Computer program, Github.]. BigSoftVideo. https://github.com/BigSoftVideo/DOTE/

McMillan, D., Brown, B., Kawaguchi, I., Jaber, R., Belenguer, J. S., & Kuzuoka, H. (2019). Designing with gaze: Tama – A gaze activated smart-speaker. *Proceedings of the ACM on Human-Computer Interaction*, *3*(CSCW), 1–26. https://doi.org/10.1145/3359278

Merleau-Ponty, M. (1945). *Phénoménologie de la Perception*. Gallimard.

Miller, R. B. (1968). Response time in man-computer conversational transactions. *Proceedings of the December 9-11, 1968, Fall Joint Computer Conference, Part I*, 267–277. https://doi.org/10.1145/1476589.1476628

Minneman, S. L. (1991). *The Social Construction of a Technical Reality: Empirical Studies of Group Engineering Design Practice* (Doctoral dissertation). Stanford University.

Mollen, J., van der Putten, P., & Darling, K. (2023). Bonding with a couchsurfing robot: The impact of common locus on human-robot bonding in-the-wild. *ACM Transactions on Human-Robot Interaction*, *12*(1), 1–33. https://doi.org/10.1145/3563702

Mondada, L. (2007). Multimodal resources for turn-taking: Pointing and the emergence of possible next speakers. *Discourse Studies*, *9*(2), 194–225. https://doi.org/10.1177/1461445607075346

Mondada, L. (2014). Requesting immediate action in the surgical operating room: Time, embodied resources and praxeological embeddedness. In P. Drew & E. Couper-Kuhlen (Eds.), *Requesting in Social Interaction* (pp. 269–302). John Benjamins Publishing Company.

Mondada, L. (2019). Contemporary numbers in conversation analysis: Embodiment and materiality, multimodality and multisensoriality in social interaction. *Journal of Pragmatics*, *145*, 47–62. https://doi.org/10.1016/j.pragma.2019.01.016

Moore, D., Currano, R., & Sirkin, D. (2020). Sound decisions: How synthetic motor sounds improve autonomous vehicle-pedestrian interactions. *12th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 94–103. https://doi.org/10.1145/3409120.3410667

Moore, D., Currano, R., Strack, G. E., & Sirkin, D. (2019). The case for implicit external human-machine interfaces for autonomous vehicles. *Proceedings of the 11th International Conference on Automotive User Interfaces and Interactive Vehicular Applications*, 295–307. https://doi.org/10.1145/3342197.3345320

Moore, D., Tennent, H., Martelaro, N., & Ju, W. (2017). Making noise intentional: A study of servo sound perception. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*, 12–21. https://doi.org/10.1145/2909824.3020238

Moore, R. J., & Arar, R. (2019). *Conversational UX Design: A Practitioner's Guide to the Natural Conversation Framework*. Morgan & Claypool.

Moubayed, S. A., Baklouti, M., Chetouani, M., Dutoit, T., Mahdhaoui, A., Martin, J.-C., Ondas, S., Pelachaud, C., Urbain, J., & Yilmaz, M. (2009). Generating robot/agent backchannels during a storytelling experiment. *2009 IEEE International Conference on Robotics and Automation*, 3749–3754.

Musk, N. (2021). "How do you spell that?": Doing spelling in computer-assisted collaborative writing. In S. Kunitz, N. Markee, & O. Sert (Eds.), *Classroom-Based Conversation Analytic Research* (pp. 103–131). Springer. https://doi.org/https://doi.org/10.1007/978-3-030-52193-6_6

Mutlu, B., Kanda, T., Forlizzi, J., Hodgins, J., & Ishiguro, H. (2012). Conversational gaze mechanisms for humanlike robots. *ACM Transactions on Interactive Intelligent Systems*, *1*(2), 12:1–12:33. https://doi.org/10.1145/2070719.2070725

Nevile, M. (2015). The embodied turn in research on language and social interaction. *Research on Language and Social Interaction*, *48*(2), 121–151. https://doi.org/10.1080/08351813.2015.1025499

Nielsen, A. M. R. (2020). Co-constructing the video consultation-competent patient. *Social Interaction. Video-Based Studies of Human Sociality*, *3*(3). https://doi.org/10.7146/si.v3i3.122708

Norman, D. A. (2013). *The Design of Everyday Things* (Revised and expanded edition). Basic Books.

Norman, M., & Thomas, P. (1991). Informing HCI design through conversation analysis. *International Journal of Man-Machine Studies*, *35*(2), 235–250. https://doi.org/10.1016/S0020-7373(05)80150-6

Norman, M., & Thomas, P. (1990). The very idea. In P. Luff, N. Gilbert, & D. B. T. Frohlich (Eds.), *Computers and Conversation* (pp. 51–65). Elsevier. https://doi.org/10.1016/B978-0-08-050264-9.50008-7

Ochs, E. (1979). Transcription as theory. In E. Ochs & B. B. Schieffelin (Eds.), *Developmental Pragmatics* (pp. 43–72). Academic Press.

Ogden, R. (2020). Audibly not saying something with clicks. *Research on Language and Social Interaction*, *53*(1), 66–89. https://doi.org/10.1080/08351813.2020.1712960

Ohshima, N., Kimijima, K., Yamato, J., & Mukawa, N. (2015). A conversational robot with vocal and bodily fillers for recovering from awkward silence at turn-takings. *2015 24th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 325–330. https://doi.org/10.1109/ROMAN.2015.7333677

Ostrowski, A. K., Breazeal, C., & Park, H. W. (2021). Long-term co-design guidelines: Empowering older adults as co-designers of social robots. *2021 30th IEEE International Conference on Robot & Human Interactive Communication (RO-MAN)*, 1165–1172. https://doi.org/10.1109/RO-MAN50785.2021.9515559

Panariello, C., Sköld, M., Frid, E., & Bresin, R. (2019). From vocal sketching to sound models by means of a sound-based musical transcription system. *Proceedings of the SMC Conference 2019*. https://www.smc2019.uma.es/articles/S2/S2_05_SMC2019_paper.pdf

Paolo, E. A. D., Cuffari, E. C., & Jaegher, H. D. (2018). *Linguistic Bodies*. MIT Press.

Park, H. W., Gelsomini, M., Lee, J. J., & Breazeal, C. (2017). Telling stories to robots. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*, 100–108. https://doi.org/10.1145/2909824.3020245

Park, H. W., Gelsomini, M., Lee, J. J., Zhu, T., & Breazeal, C. (2017). Backchannel opportunity prediction for social robot listeners. *2017 IEEE International Conference on Robotics and Automation (ICRA)*, 2308–2314.

Pehkonen, S., Rauniomaa, M., & Siitonen, P. (2021). Participating researcher or researching participant?: On possible positions of the researcher in the collection (and analysis) of mobile video data. *Social Interaction. Video-Based Studies of Human Sociality*, *4*(2). https://doi.org/10.7146/si.v4i2.127267

Pelikan, H. (2019). "A stubborn child" - How robot sounds are oriented to in everyday situated interaction at home. *Mensch und Computer 2019 Workshopband*. https://doi.org/10.18420/muc2019-ws-659

Pelikan, H. (2024). Transcribing human-robot interaction: Methodological implications of participating machines. In P. Haddington, T. Eilittä, A. Kamunen, L. Kohonen-Aho, T. Oittinen, I. Rautiainen, & A. Vatanen (Eds.), *Ethnomethodological Conversation Analysis in Motion: Emerging Methods and New Technologies*. Routledge.

Pelikan, H., Broth, M., & Keevallik, L. (2022). When a robot comes to life: The interactional achievement of agency as a transient phenomenon. *Social Interaction. Video-Based Studies of Human Sociality*, *5*(3). https://doi.org/10.7146/si.v5i3.129915

Pelikan, H., Fu, J., Hou, Y. T.-Y., Hu, Q., & Jung, M. (submitted). Minimal sound for maximal interaction: Sequential timing of expressive robot behavior. *ACM Transactions on Human-Robot Interaction*.

Pelikan, H., & Hofstetter, E. (2022). Managing delays in human-robot interaction [Just Accepted]. *ACM Transactions on Computer-Human Interaction*. https://doi.org/10.1145/3569890

Pelikan, H., Keevallik, L., & Broth, M. (2020). Människa-robot-interaktion. In M. Broth & L. Keevallik (Eds.), *Multimodal Interaktionsanalys* (pp. 395–415). Studentlitteratur.

Pelikan, H., Porfirio, D., & Winkle, K. (2023). Designing better human-robot interactions through enactment, engagement, and reflection. *Proceedings of the CUI@HRI Workshop at the 2023 ACM/IEEE International Conference on Human-Robot Interaction (HRI '23)*. https://cui.acm.org/workshops/HRI2023/pdfs/HRCI23_paper_7.pdf

Pelikan, H., Robinson, F. A., Keevallik, L., Velonaki, M., Broth, M., & Bown, O. (2021). Sound in human-robot interaction. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 706–708. https://doi.org/10.1145/3434074.3444871

Pelikan, H. R. M. (2018). *"What's Going on There?" Negotiating Common Ground in Robotic vs. Open Surgery: A Comparison of Surgeon-Initiated Requests for Action in Open and Robotic Surgery* (Master's thesis). University of Twente. http://essay.utwente.nl/76805/

Pelikan, H. R. M. (2021). Why autonomous driving is so hard: The social dimension of traffic. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 81–85. https://doi.org/10.1145/3434074.3447133

Pelikan, H. R. M. (2023). Robot sound-in-interaction. *Companion of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, 754–756. https://doi.org/10.1145/3568294.3579975

Pelikan, H. R. M., & Broth, M. (2016). Why that Nao?: How humans adapt to a conventional humanoid robot in taking turns-at-talk. *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems*, 4921–4932. https://doi.org/10.1145/2858036.2858478

Pelikan, H. R. M., Broth, M., & Keevallik, L. (2020). "Are you sad, Cozmo?": How humans make sense of a home robot's emotion displays. *Proceed-*

ings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction, 461–470. https://doi.org/10.1145/3319502.3374814

Pelikan, H. R. M., Hou, Y. T.-Y., Fu, X. J., Keevallik, L., Broth, M., & Jung, M. F. (2022). Interaction prototyping with video: Bridging video interaction analysis & design. *CHI Conference on Human Factors in Computing Systems Extended Abstracts*, 1–4. https://doi.org/10.1145/3491101.3503765

Pelikan, H. R. M., & Jung, M. F. (2023). Designing robot sound-in-interaction: The case of autonomous public transport shuttle buses. *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, 172–182. https://doi.org/10.1145/3568162.3576979

Peräkylä, A., & Sorjonen, M.-L. (Eds.). (2012). *Emotion in Interaction*. Oxford University Press.

Pitsch, K. (2016). Limits and opportunities for mathematizing communicational conduct for social robotics in the real world? Toward enabling a robot to make use of the human's competences. *AI & SOCIETY*, *31*(4), 587–593. https://doi.org/10.1007/s00146-015-0629-0

Pitsch, K., & Koch, B. (2010). How infants perceive the toy robot Pleo: An exploratory case study on infant-robot-interaction. *Proceedings of the ASME International Design Engineering Technical Conferences and Computers and Information in Engineering Conference*, 80–87.

Pitsch, K., Kuzuoka, H., Suzuki, Y., Sussenbach, L., Luff, P., & Heath, C. (2009). The first five seconds: Contingent stepwise entry into an interaction as a means to secure sustained engagement in HRI. *RO-MAN 2009 - The 18th IEEE International Symposium on Robot and Human Interactive Communication*, 985–991. https://doi.org/10.1109/ROMAN.2009.5326167

Porcheron, M., Fischer, J. E., & Reeves, S. (2020). Pulling back the curtain on the Wizards of Oz. *Proceedings of the ACM on Human-Computer Interaction*, *4*(CSCW3). https://doi.org/10.1145/3432942

Porcheron, M., Fischer, J. E., Reeves, S., & Sharples, S. (2018). Voice interfaces in everyday life. *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*, 640:1–640:12. https://doi.org/10.1145/3173574.3174214

Randall, N., & Šabanović, S. (2023). A picture might be worth a thousand words, but it's not always enough to evaluate robots. *Proceedings of the 2023 ACM/IEEE International Conference on Human-Robot Interaction*, 437–445. https://doi.org/10.1145/3568162.3576970

Read, R., & Belpaeme, T. (2012). How to use non-linguistic utterances to convey emotion in child-robot interaction. *Proceedings of the Seventh Annual ACM/IEEE International Conference on Human-Robot Interaction (HRI'12)*, 219–220. https://doi.org/10.1145/2157689.2157764

Read, R., & Belpaeme, T. (2014). Situational context directs how people affectively interpret robotic non-linguistic utterances. *Proceedings of the 2014 ACM/IEEE International Conference on Human-Robot Interaction*, 41–48. https://doi.org/10.1145/2559636.2559680

Read, R., & Belpaeme, T. (2016). People interpret robotic non-linguistic utterances categorically. *International Journal of Social Robotics*, *8*(1), 31–50. https://doi.org/10.1007/s12369-015-0304-0

Read, R. G., & Belpaeme, T. (2010). Interpreting non-linguistic utterances by robots. *Proceedings of the 3rd International Workshop on Affective Interaction in Natural Environments (AFFINE'10)*, 65–70. https://doi.org/10.1145/1877826.1877843

Reber, E. (2012). *Affectivity in Interaction: Sound Objects in English*. John Benjamins.

Reeves, S. (2022). Navigating incommensurability between ethnomethodology, conversation analysis, and artificial intelligence. *Arxiv*. https://doi.org/10.48550/ARXIV.2206.11899

Reeves, S., & Brown, B. (2016). Embeddedness and sequentiality in social media. *Proceedings of the 19th ACM Conference on Computer-Supported Cooperative Work & Social Computing*, 1052–1064. https://doi.org/10.1145/2818048.2820008

Reeves, S., Porcheron, M., & Fischer, J. (2018). "This is not what we wanted": Designing for conversation with voice interfaces. *Interactions*, *26*(1), 46–51. https://doi.org/10.1145/3296699

Rehm, M., Krummheuer, A. L., Rodil, K., Nguyen, M., & Thorlacius, B. (2016). From social practices to social robots – User-driven robot development in elder care (A. Agah, J.-J. Cabibihan, A. M. Howard, M. A. Salichs, & H. He, Eds.). *Social Robotics*, *9979*, 692–701. https://doi.org/https://doi.org/10.1007/978-3-319-47437-3_68

Riek, L. D. (2012). Wizard of Oz studies in HRI: A systematic review and new reporting guidelines. *ACM Transactions on Human-Robot Interaction*, *1*(1), 119–136. https://doi.org/10.5898/JHRI.1.1.Riek

Roberts, F. (2012). Transcribing and transcription. In *The International Encyclopedia of Communication*. John Wiley & Sons, Ltd. https://doi.org/10.1002/9781405186407.wbiect056.pub2

Robins, B., Dickerson, P., Stribling, P., & Dautenhahn, K. (2004). Robot-mediated joint attention in children with autism: A case study in robot-human interaction. *Interaction Studies*, *5*(2), 161–198. https://doi.org/10.1075/is.5.2.02rob

Robinson, F. A., Bown, O., & Velonaki, M. (2022). Designing sound for social robots: Candidate design principles. *International Journal of Social Robotics*, *14*(6), 1507–1525. https://doi.org/10.1007/s12369-022-00891-0

Robinson, F. A., Velonaki, M., & Bown, O. (2021). Smooth operator: Tuning robot perception through artificial movement sound. *Proceedings*

of the 2021 ACM/IEEE International Conference on Human-Robot Interaction, 53–62. https://doi.org/10.1145/3434073.3444658

Robinson, F. A., Velonaki, M., & Bown, O. (2022). Crafting the language of robotic agents: A vision for electroacoustic music in human–robot interaction. *Organised Sound*, *27*(2), 1–13. https://doi.org/10.1017/S1355771822000358

Rocchesso, D., Lemaitre, G., Susini, P., Ternström, S., & Boussard, P. (2015). Sketching sound with voice and gesture. *Interactions*, *22*(1), 38–41. https://doi.org/10.1145/2685501

Rocchesso, D., Serafin, S., & Rinott, M. (2013). Pedagogical approaches and methods. In *Sonic Interaction Design* (pp. 125–150). MIT Press.

Rudaz, D., Tatarian, K., Stower, R., & Licoppe, C. (2023). From inanimate object to agent: Impact of pre-beginnings on the emergence of greetings with a robot. *ACM Transactions on Human-Robot Interaction*, 3575806. https://doi.org/10.1145/3575806

Russell, J. A., & Barrett, L. F. (1999). Core affect, prototypical emotional episodes, and other things called emotion: Dissecting the elephant. *Journal of Personality and Social Psychology*, *76*(5), 805.

Ryave, L., & Schenkein, J. N. (1974). Notes on the art of walking. In R. Turner (Ed.), *Ethnomethodology. Selected Readings* (pp. 265–274). Penguin Education.

Šabanović, S., Michalowski, M., & Simmons, R. (2006). Robots in the wild: Observing human-robot social interaction outside the lab. *9th IEEE International Workshop on Advanced Motion Control, 2006.*, 596–601. https://doi.org/10.1109/AMC.2006.1631758

Šabanović, S. (2010). Robots in society, society in robots. *International Journal of Social Robotics*, *2*(4), 439–450. https://doi.org/10.1007/s12369-010-0066-7

Šabanović, S., Reeder, S. M., & Kechavarzi, B. (2014). Designing robots in the wild: In situ prototype evaluation for a break management robot. *Journal of Human-Robot Interaction*, *3*(1), 70–88. https://doi.org/10.5898/JHRI.3.1.Sabanovic

Sacks, H. (1984). Notes on methodology. In J. Heritage & J. M. Atkinson (Eds.), *Structures of Social Action: Studies in Conversation Analysis* (pp. 21–27). Cambridge University Press.

Sacks, H. (1992). *Lectures on Conversation* (G. Jefferson, Ed.). Blackwell.

Sacks, H., Schegloff, E. A., & Jefferson, G. (1974). A simplest systematics for the organization of turn-taking for conversation. *Language*, *50*(4), 696. https://doi.org/10.2307/412243

Salem, M., Lakatos, G., Amirabdollahian, F., & Dautenhahn, K. (2015). Would you trust a (faulty) robot? Effects of error, task type and personality on human-robot cooperation and trust. *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction*, 141–148. https://doi.org/10.1145/2696454.2696497

Savery, R., Rose, R., & Weinberg, G. (2019). Establishing human-robot trust through music-driven robotic emotion prosody and gesture. *2019 28th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 1–7. https://doi.org/10.1109/RO-MAN46459.2019.8956386

Schadenberg, B. R., Heylen, D. K. J., & Evers, V. (2018). Affect bursts to constrain the meaning of the facial expressions of the humanoid robot Zeno. In C. Tsiourti, S. Hanke, & L. Santos (Eds.), *Joint Proceedings of the Workshop on Social Interaction and Multimodal Expression for Socially Intelligent Robots and the Workshop on the Barriers of Social Robotics Take-Up by Society* (pp. 30–39). CEUR. https://ris.utwente.nl/ws/portalfiles/portal/81520962/paper4.pdf

Schecter, A., Hohenstein, J., Larson, L., Harris, A., Hou, T.-Y., Lee, W.-Y., Lauharatanahirun, N., DeChurch, L., Contractor, N., & Jung, M. (2023). Vero: An accessible method for studying human-AI teamwork. *Computers in Human Behavior*, *141*, 107606. https://doi.org/10.1016/j.chb.2022.107606

Schegloff, E. A. (1968). Sequencing in conversational openings. *American Anthropologist*, *70*(6), 1075–1095. http://www.jstor.org/stable/669510

Schegloff, E. A. (1992). Repair after next turn: The last structurally provided defense of intersubjectivity in conversation. *American Journal of Sociology*, *97*(5), 1295–1345. https://doi.org/10.1086/229903

Schegloff, E. A. (1995). Discourse as an interactional achievement III: The omnirelevance of action. *Research on Language and Social Interaction*, *28*(3), 185–211. https://doi.org/10.1207/s15327973rlsi2803_2

Schegloff, E. A. (2007). *Sequence Organization in Interaction: A Primer in Conversation Analysis* (Vol. 1). Cambridge University Press. https://doi.org/10.1017/CBO9780511791208

Schegloff, E. A., & Sacks, H. (1973). Opening up closings. *Semiotica*, *8*(4), 289–327. https://doi.org/10.1515/semi.1973.8.4.289

Schneiders, E., Cheon, E., Kjeldskov, J., Rehm, M., & Skov, M. B. (2022). Non-dyadic interaction: A literature review of 15 years of human-robot interaction conference publications. *ACM Transactions on Human-Robot Interaction*, *11*(2). https://doi.org/10.1145/3488242

Schuetz, A. (1945). On multiple realities. *Philosophy and Phenomenological Research*, *5*(4), 533. https://doi.org/10.2307/2102818

Serholt, S., Ljungblad, S., & Bhroin, N. N. (2022). Introduction: Special issue—critical robotics research. *AI & SOCIETY*, *37*, 417–423. https://doi.org/10.1007/s00146-021-01224-x

Sharrock, W., & Button, G. (2003). Plans and situated action ten years on. *The Journal of the Learning Sciences*, *12*(2), 259–264. http://www.jstor.org/stable/1466895

Shiwa, T., Kanda, T., Imai, M., Ishiguro, H., & Hagita, N. (2008). How quickly should communication robots respond? *Proceedings of the 3rd Inter-*

national Conference on Human-Robot Interaction (HRI'08), 153–160. https://doi.org/10.1145/1349822.1349843

Shneiderman, B. (1984). Response time and display rate in human performance with computers. *ACM Computing Surveys*, *16*(3), 265–285. https://doi.org/10.1145/2514.2517

Sidnell, J., & Stivers, T. (Eds.). (2012). *The Handbook of Conversation Analysis*. Wiley-Blackwell. https://doi.org/10.1002/9781118325001

Sirkin, D., Baltodano, S., Mok, B., Rothenbücher, D., Gowda, N., Li, J., Martelaro, N., Miller, D., Sibi, S., & Ju, W. (2016). Embodied design improvisation for autonomous vehicles. In H. Plattner, C. Meinel, & L. Leifer (Eds.), *Design Thinking Research: Taking Breakthrough Innovation Home* (pp. 125–143). Springer. https://doi.org/10.1007/978-3-319-40382-3_9

Sirkin, D., & Ju, W. (2014). Using embodied design improvisation as a design research tool. *Proceedings of the International Conference on Human Behavior in Design (HBiD 2014)*. https://www.wendyju.com/publications/HBiD2014SirkinJu.pdf

Sirkin, D., Mok, B., Yang, S., & Ju, W. (2015). Mechanical ottoman: How robotic furniture offers and withdraws support. *Proceedings of the Tenth Annual ACM/IEEE International Conference on Human-Robot Interaction*, 11–18. https://doi.org/10.1145/2696454.2696461

Skantze, G., Johansson, M., & Beskow, J. (2015). Exploring turn-taking cues in multi-party human-robot discussions about objects. *Proceedings of the 2015 ACM on International Conference on Multimodal Interaction*, 67–74. https://doi.org/10.1145/2818346.2820749

Song, S., & Yamada, S. (2017). Expressing emotions through color, sound, and vibration with an appearance-constrained social robot. *Proceedings of the 2017 ACM/IEEE International Conference on Human-Robot Interaction*, 2–11. https://doi.org/10.1145/2909824.3020239

Stayton, E. L. (2020). *Humanizing Autonomy: Social Scientists' and Engineers' Futures for Robotic Cars* (Doctoral dissertation). Massachusetts Institute of Technology. https://dspace.mit.edu/bitstream/handle/1721.1/129050/1227095115-MIT.pdf

Stegner, L., Senft, E., & Mutlu, B. (2023). Situated participatory design: A method for in situ design of robotic interaction with older adults. *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*. https://doi.org/10.1145/3544548.3580893

Stivers, T., Enfield, N. J., Brown, P., Englert, C., Hayashi, M., Heinemann, T., Hoymann, G., Rossano, F., de Ruiter, J. P., Yoon, K.-E., & Levinson, S. C. (2009). Universals and cultural variation in turn-taking in conversation. *Proceedings of the National Academy of Sciences*, *106*(26), 10587–10592. https://doi.org/10.1073/pnas.0903616106

Stivers, T., & Robinson, J. D. (2006). A preference for progressivity in interaction. *Language in Society*, *35*(3), 367–392. https://doi.org/10.1017/S0047404506060179

Stommel, W., de Rijk, L., & Boumans, R. (2022). "Pepper, what do you mean?" Miscommunication and repair in robot-led survey interaction. *2022 31st IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 385–392. https://doi.org/10.1109/RO-MAN53752.2022.9900528

Streeck, J., Goodwin, C., & LeBaron, C. D. (Eds.). (2011). *Embodied Interaction: Language and Body in the Material World*. Cambridge University Press.

Suchman, L. (1993). Technologies of accountability: Of lizards and airplanes. In G. Button (Ed.), *Technology in Working Order: Studies of Work, Interaction and Technology* (pp. 113–126). Routledge.

Suchman, L. A. (1987). *Plans and Situated Actions: The Problem of Human-Machine Communication*. Cambridge University Press.

Suchman, L. A. (2007). *Human-Machine Reconfigurations: Plans and Situated Actions* (2nd Edition). Cambridge University Press.

Suchman, L. A., & Trigg, R. H. (1992). Understanding practice: Video as a medium for reflection and design. In *Design at Work: Cooperative Design of Computer Systems* (pp. 65–90).

Suguitan, M., & Hoffman, G. (2019). Blossom: A handcrafted open-source robot. *ACM Transactions on Human-Robot Interaction*, *8*(1). https://doi.org/10.1145/3310356

Svanaes, D., & Solheim, M. (2016). Wag your tail and flap your ears: The kinesthetic user experience of extending your body. *Proceedings of the 2016 CHI Conference Extended Abstracts on Human Factors in Computing Systems*, 3778–3779. https://doi.org/10.1145/2851581.2890268

Syrdal, D. S., Otero, N., & Dautenhahn, K. (2008). Video prototyping in human-robot interaction: Results from a qualitative study. *Proceedings of the 15th European Conference on Cognitive Ergonomics: The Ergonomics of Cool Interaction*. https://doi.org/10.1145/1473018.1473055

Takayama, L. (2022). Putting human-robot interaction research into design practice. *Proceedings of the 2022 17th ACM/IEEE International Conference on Human-Robot Interaction*, 1–1. https://doi.org/10.1109/HRI53351.2022.9889659

Takayama, L., Dooley, D., & Ju, W. (2011). Expressing thought: Improving robot readability with animation principles. *Proceedings of the 6th International Conference on Human-Robot Interaction*, 69–76. https://doi.org/10.1145/1957656.1957674

Tang, J. C. (1989). *Toward an Understanding of the Use of Shared Workspaces by Design Teams* (Doctoral dissertation). Stanford University.

ten Have, P. (2002a). The notion of member is the heart of the matter: On the role of membership knowledge in ethnomethodological inquiry. *Forum: Qualitative Social Research*, *3*(3), 53 paragraphs. https://doi.org/https://doi.org/10.17169/fqs-3.3.834

ten Have, P. (2002b). Reflections on transcription. *Cahiers de Praxématique*, (39), 21–43. https://doi.org/10.4000/praxematique.1833

ten Have, P. (2007). *Doing Conversation Analysis*. Sage. https://doi.org/10.4135/9781849208895

Tennent, H., Moore, D., Jung, M., & Ju, W. (2017). Good vibrations: How consequential sounds affect perception of robotic arms. *2017 26th IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)*, 928–935. https://doi.org/10.1109/ROMAN.2017.8172414

Tennent, H., Shen, S., & Jung, M. (2019). Micbot: A peripheral robotic object to shape conversational dynamics and team performance. *Proceedings of the 14th ACM/IEEE International Conference on Human-Robot Interaction*, 133–142.

Tolmeijer, S., Weiss, A., Hanheide, M., Lindner, F., Powers, T. M., Dixon, C., & Tielman, M. L. (2020). Taxonomy of trust-relevant failures and mitigation strategies. *Proceedings of the 2020 ACM/IEEE International Conference on Human-Robot Interaction*, 3–12. https://doi.org/10.1145/3319502.3374793

Torre, I., Latupeirissa, A. B., & McGinn, C. (2020). How context shapes the appropriateness of a robot's voice. *2020 29th IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)*, 215–222. https://doi.org/10.1109/RO-MAN47096.2020.9223449

Tuncer, S. (2015). Walking away: An embodied resource to close informal encounters in offices. *Journal of Pragmatics*, *76*, 101–116. https://doi.org/https://doi.org/10.1016/j.pragma.2014.11.012

Tuncer, S., Gillet, S., & Leite, I. (2022). Robot-mediated inclusive processes in groups of children: From gaze aversion to mutual smiling gaze. *Frontiers in Robotics and AI*, *9*. https://doi.org/10.3389/frobt.2022.729146

Tuncer, S., Licoppe, C., Luff, P., & Heath, C. (2023). Recipient design in human–robot interaction: The emergent assessment of a robot's competence. *AI & SOCIETY*. https://doi.org/10.1007/s00146-022-01608-7

Velkovska, J. (2019). When an emotional robot meets real customers exploring HRI in a customer relationship setting. *Mensch und Computer 2019 Workshopband*. https://doi.org/https://dx.doi.org/10.18420/muc2019-ws-646

Vertelney, L. (1989). Using video to prototype user interfaces. *ACM SIGCHI Bulletin*, *21*(2), 57–61. https://doi.org/10.1145/70609.70615

Vetenskapsrådet. (2017). Good research practice. https://www.vr.se/download/18.5639980c162791bbfe697882/1555334908942/Good-Research-Practice_VR_2017.pdf

Vinkhuyzen, E., & Cefkin, M. (2016). Developing socially acceptable autonomous vehicles. *Ethnographic Praxis in Industry Conference Proceedings*, *2016*(1), 522–534. https://doi.org/https://doi.org/10.1111/1559-8918.2016.01108

Weiss, A., & Spiel, K. (2022). Robots beyond science fiction: Mutual learning in human–robot interaction on the way to participatory approaches. *AI & SOCIETY*, *37*(2), 501–515. https://doi.org/10.1007/s00146-021-01209-w

Westerkamp, H. (1974). Soundwalking [Revised 2001]. *Sound Heritage*, *3*(4). https://www.hildegardwesterkamp.ca/writings/writingsby/?post_id=13&title=soundwalking

Wiggins, S. (2002). Talking with your mouth full: Gustatory Mmms and the embodiment of pleasure. *Research on Language and Social Interaction*, *35*(3), 311–336. https://doi.org/10.1207/S15327973RLSI3503_3

Wiggins, S., & Keevallik, L. (2021). Parental lip-smacks during infant mealtimes: Multimodal features and social functions. *Interactional Linguistics*, *1*(2), 241–272. https://doi.org/https://doi.org/10.1075/il.21006.wig

Winkle, K., Lagerstedt, E., Torre, I., & Offenwanger, A. (2022). 15 years of (who)man robot interaction: Reviewing the H in human-robot interaction [Just Accepted]. *ACM Transactions on Human-Robot Interaction*. https://doi.org/10.1145/3571718

Winkle, K., Senft, E., & Lemaignan, S. (2021). LEADOR: A method for end-to-end participatory design of autonomous social robots. *Frontiers in Robotics and AI*, *8*. https://doi.org/10.3389/frobt.2021.704119

Wittenburg, P., Brugman, H., Russel, A., Klassmann, A., & Sloetjes, H. (2006). ELAN: A professional framework for multimodality research. *5th International Conference on Language Resources and Evaluation (LREC 2006)*, 1556–1559.

Woodruff, A., & Aoki, P. M. (2004). Conversation analysis and the user experience. *Digital Creativity*, *15*(4), 232–238. https://doi.org/10.1080/1462626048520184

Woodruff, A., Szymanski, M. H., Grinter, R. E., & Aoki, P. M. (2002). Practical strategies for integrating a conversation analyst in an iterative design process. *Proceedings of the 4th Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques*, 255–264. https://doi.org/10.1145/778712.778748

Wundt, W. M. (1900). *Völkerpsychologie: Die Sprache.* (Vol. 1). W. Englemann.

Yamazaki, A., Yamazaki, K., Kuno, Y., Burdelski, M., Kawashima, M., & Kuzuoka, H. (2008). Precision timing in human-robot interaction: Coordination of head movement and utterance. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 131–140. https://doi.org/10.1145/1357054.1357077

Yamazaki, K., Yamazaki, A., Okada, M., Kuno, Y., Kobayashi, Y., Hoshi, Y., Pitsch, K., Luff, P., vom Lehn, D., & Heath, C. (2009). Revealing gauguin: Engaging visitors in robot guide's explanation in an art museum. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 1437–1446. https://doi.org/10.1145/1518701.1518919

Yilmazyildiz, S., Read, R., Belpeame, T., & Verhelst, W. (2016). Review of semantic-free utterances in social human–robot interaction. *International Journal of Human-Computer Interaction*, *32*(1), 63–85. https://doi.org/10.1080/10447318.2015.1093856

Ylirisku, S., & Buur, J. (2007). *Designing with Video: Focusing the User-Centred Design Process*. Springer London. https://doi.org/10.1007/978-1-84628-961-3_1

Zaga, C. (2021). *The Design of Robothings: Non-Anthropomorphic and Non-Verbal Robots to Promote Children's Collaboration Through Play* (Doctoral dissertation). University of Twente. https://doi.org/10.3990/1.9789036551380

Zamfirescu-Pereira, J., Sirkin, D., Goedicke, D., Lc, R., Friedman, N., Mandel, I., Martelaro, N., & Ju, W. (2021). Fake it to make it: Exploratory prototyping in HRI. *Companion of the 2021 ACM/IEEE International Conference on Human-Robot Interaction*, 19–28.

Zhang, B. J., & Fitter, N. T. (2023). Nonverbal sound in human-robot interaction: A systematic review [Just Accepted]. *ACM Transactions on Human-Robot Interaction*. https://doi.org/10.1145/3583743

Zhang, B. J., Stargu, N., Brimhall, S., Chan, L., Fick, J., & Fitter, N. T. (2021). Bringing WALL-E out of the silver screen: Understanding how transformative robot sound affects human perception. *2021 IEEE International Conference on Robotics and Automation (ICRA)*, 3801–3807. https://doi.org/10.1109/ICRA48506.2021.9562082

Zimmerman, J., & Forlizzi, J. (2014). Research through Design in HCI. In J. S. Olson & W. A. Kellogg (Eds.), *Ways of Knowing in HCI* (pp. 167–189). Springer New York. https://doi.org/10.1007/978-1-4939-0378-8_8

Zimmerman, J., Forlizzi, J., & Evenson, S. (2007). Research through Design as a method for interaction design research in HCI. *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 493–502. https://doi.org/10.1145/1240624.1240704

A Consent Form for German Cozmo Study

Einverständniserklärung

Dieses Forschungsprojekt befasst sich mit der Frage, wie Menschen mit Robotern interagieren. In meiner Doktorarbeit konzentriere ich mich auf feine Aspekte in der Interaktion, die ich durch detaillierte Videoanalyse sichtbar mache. Dafür würde ich Sie/Dich gerne bei der Interaktion mit dem Cozmo-Roboter filmen. Folgendes sollten Sie/solltest Du wissen:

- Die Teilnahme ist freiwillig und kann jederzeit ohne Angabe von Gründen beendet werden.
- Sie haben/Du hast das Recht, unmittelbar nach der Interaktion mit dem Roboter das Löschen von Videomaterial zu erbitten in dem Sie/Du aufgezeichnet sind/bist.
- Die Daten werden passwortgeschützt gespeichert und nur zu Forschungszwecken genutzt.
- Bei Präsentationen der Daten außerhalb der Forschungsgruppe werden Pseudonyme anstelle Ihres/Deines echten Namens benutzt.

Bei Fragen können Sie sich/kannst Du dich jederzeit an die verantwortlichen Forscher wenden:

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☐ Hiermit erkläre ich, dass ich die oben angegebenen Informationen zur Kenntnis genommen habe und damit einverstanden bin, dass Videoaufnahmen von mir gemacht werden, während ich mit dem Cozmo Roboter interagiere.

Meine Videodaten dürfen folgendermaßen verwendet werden (bitte Zutreffendes ankreuzen)

☐ in wissenschaftlichen Publikationen (z.B. als Screenshot)
☐ bei Treffen mit anderen Wissenschaftlern (z.B. Workshops, Konferenzen)
☐ in der Lehre (z.B. Vorlesung)
☐ bei öffentlichen Präsentationen vor nicht-wissenschaftlichem Publikum (z.B. öffentliche Vorträge, Radio, Fernsehen)
☐ kurzen Videosequenzen dürfen auf öffentlich zugängliche Websites hochgeladen werden (z.B. Institutswebsite, Illustration von Online-Publikationen)

Datum Unterschrift

Bei eventuellen Rückfragen darf die Forschungsgruppe mich kontaktieren:

☐ Ja ☐ Nein

Ich möchte über Ergebnisse der Forschung informiert werden:

☐ Ja ☐ Nein

Falls Sie/Du eine der Fragen mit *ja* beantwortet haben/hast, geben Sie/gib hier bitte Ihre/Deine Kontaktdaten an:

Name

Email-Adresse oder Telefonnummer

B Consent Form and Information for Swedish Cozmo Study

19-04-17
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Information om studien ”Interaktion med Cozmo-roboten i hemmiljö”

Studien ”Interaktion med Cozmo-roboten i hemmiljö” syftar till att kartlägga hur människor interagerar med en robot som är byggd för hemmabruk och, specifikt, hur de tolkar Cozmo’s ljud tillsammans med dess rörelser. Studien är del av ett större projekt om olika ljuds funktioner i människa-robot-interaktion.

För att kunna undersöka detta behöver jag inspelningar av lek och annat umgänge kring roboten. Det som sägs och görs kommer sedan att bli återgivet i text (transkriptioner). Deltagare döps om i transkriptioner, presentationer och i publikationer. Eventuella bilder och filmklipp används enbart i enlighet med ditt medgivande nedan. För att anonymisera bilder och filmklipp kan man exempelvis göra dem suddiga, rutiga, eller förvandla dem till teckningar/tecknade filmer.

Roboten som används i den här studien kan inte spela in ljud och enligt tillverkaren bearbetas bilddata bara lokalt i tillhörande Cozmo appen. Robotens funktion att dela aggregerade data (t ex spelresultat, användning av funktioner) med tillverkaren och andra intressenter kommer att stängas av vid installation av appen.

Videoinspelningen kommer endast att användas i forskningssyfte. Inspelningen förvaras på universitetets server och lösenordskyddade portabla hårddiskar som bara jag och mina handledare kommer att ha tillgång till. Data sparas så länge de bedöms relevanta för analys, varefter de raderas.

Att delta är helt frivilligt. Du kan närsomhelst avsluta ditt deltagande i inspelningen utan att behöva ange någon särskild anledning. Du meddelar bara mig eller min handledare att du inte längre vill delta.

Lämna din mailadress om du vill ta del av resultatet av studien (vilket är frivilligt). Då skickas en kopia när arbetet är färdigställt.

Du kan alltid kontakta mig eller min huvudhandledare:

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Medgivande att delta i studie

☐ Efter att ha tagit del av informationen om studien ”Interaktion med Cozmo-roboten i hemmiljö” som är del av ett större forskningsprojekt om olika ljuds funktioner i människa-robot-interaktion samtycker jag till att delta i studien.

Användning av inspelning

Jag ger mitt samtycke till att de video- och ljudinspelningar som görs av **mig och mina barn** kan användas i följande syften (jag kryssar i den användning som jag godkänner):

JA?

☐ inspelningen används som **data till vetenskapliga arbeten** så länge som de är relevanta eller som längst 5 år.
☐ anonymiserade filmklipp får analyseras med en liten grupp **forskare som är relaterade till projektet**.
☐ oredigerade filmklipp får analyseras med en liten grupp **forskare som är relaterade till projektet**.
☐ **transkriptioner** (tal och aktion omskriven till text) får användas i forskningssyfte, t ex i vetenskapliga texter i pappers- såväl som elektronisk form.
☐ **anonymiserade bilder** får publiceras i forskningssyfte, t ex i vetenskapliga texter i pappers- såväl som elektronisk form.
☐ **oredigerade bilder** får publiceras i forskningssyfte, t ex i vetenskapliga texter i pappers- såväl som elektronisk form.
☐ **anonymiserade filmklipp** får visas i vetenskapliga sammanhang.
☐ **oredigerade filmklipp** får visas i vetenskapliga sammanhang.
☐ filmklipp får visas i **undervisning** av studenter och forskare.
☐ filmklipp får publiceras i **multimediala vetenskapliga verk.**
☐ filmklipp får visas i **allmän presentation**, t ex radio, teve.
☐ korta filmklipp får visas på **allmänt tillgängliga webbplatser,** t ex universitets webbplats.
☐ dessa tillstånd gäller även **vetenskapliga arbeten som går bortom** det aktuella projektet, av interaktionsanalytisk eller etnografisk karaktär.

OBS! Du får ändra dig när som helst under eller omedelbart efter inspelningen.

Datum Namnteckning

19-04-17
Hannah Pelikan
hannah.pelikan@liu.se

JA?

☐ Forskarna får kontakta mig med eventuella följdfrågor.

☐ Jag önskar bli informerad av resultatet.

Vänligen lämna din kontaktinformation nedan om du önskar bli kontaktad angående forskningsresultaten.

Namn E-postadress eller telefonnummer

Instructions for Videotaping

Ongoing consent:

- Always make sure that the people you record are okay with being videotaped in the specific situation (e.g. ask "is it okay if I videotape you *now*?")
- Don't videotape people who (whose parents) have not agreed to being videotaped

Everyday interaction

- Do as you normally do as a family (the goal is to get video material that is not staged)
- Try to first switch on the camera and then Cozmo

Instances that you are uncomfortable with

- If you feel you recorded a scene that should not be on video note down the time and date that this occurred at (and roughly how many minutes). I will then delete the respective scene before starting to work on the data.

Switching the Camera on

1) Open the display (camera switches on automatically)
2) Open the lens cover with the switch labelled "LENSE COVER OPEN/CLOSE" on front right-hand side of camera
3) Press the button labelled "START/STOP" on the back right-hand side of the camera to start recording
   Text "STBY" in display should switch to "REC" (flashes two times before changing)

Switching the Camera off

1) Press the button labelled "START/STOP" on the back right-hand side of the camera to stop recording
   Text "REC" in display should switch to "STBY" (flashes two times before changing)
2) Close the lens cover with switch labelled "LENSE COVER OPEN/CLOSE"
3) Close the display (camera switches off automatically)

Switching Cozmo on

1) Put Cozmo on docking station
2) Connect to Cozmo's Wifi
3) Open the Cozmo app and wait until it connects
4) In the app you can choose to let Cozmo roam freely, play games, program Cozmo etc.

Switching Cozmo off

1) Put Cozmo on docking station
2) Close the app and/or disconnect Wifi to fully shut Cozmo down

C Consent Form for Bus Study

20-03-23
Hannah Pelikan
hannah.pelikan@liu.se

Information om studien ”Trafikinteraktion med autonoma bussar”

Studien "Interaktion mellan autonoma bussar och människor i trafik” syftar till att kartlägga hur trafikanter hanterar bussens ljud. Studien är del av ett större projekt om olika ljuds funktioner i människa-robot-interaktion.

Jag är bland annat intresserad av ”oskrivna regler” i trafiken. För att kunna undersöka det professionella perspektivet på detta (hur bussförare observerar och förutser vad som ska hända i olika trafiksituationer) behövs inspelningar av säkerhetsförarnas beteende inne i bussen. Det som sägs och görs kommer sedan att bli återgivet i text (transkriptioner). Förarna anonymiseras (dvs. ges andra namn än de riktiga) i transkriptioner, presentationer och i publikationer. Eventuella bilder och filmklipp används enbart i enlighet med ditt medgivande nedan. För att anonymisera bilder och filmklipp kan man exempelvis göra dem suddiga, rutiga, eller förvandla dem till teckningar/tecknade filmer.

Videoinspelningarna kommer endast att användas i forsknings- och utbildningssyfte. Inspelningarna förvaras på universitetets server och lösenordskyddade externa hårddiskar som bara jag och mina forskarkollegor kommer att ha tillgång till. Inga data kommer att lämnas till din arbetsgivare. Data sparas så länge de bedöms relevanta för analys, varefter de raderas.

Det är helt frivilligt att låta sig spelas in. Du kan närsomhelst avbryta ditt deltagande i inspelningen och utan att behöva ange någon särskild anledning. Du meddelar bara mig eller någon av mina handledare att du inte längre vill delta.

Du kan alltid kontakta mig eller mina handledare:

Hannah Pelikan	Leelo Keevallik	Mathias Broth
Doktorand	Professor (Hannahs handledare)	Professor (Hannahs bihandledare)
Kultur och samhälle (IKOS)	Kultur och samhälle (IKOS)	Kultur och samhälle (IKOS)
Linköpings Universitet	Linköpings Universitet	Linköpings Universitet
hannah.pelikan@liu.se	leelo.keevallik@liu.se	mathias.broth@liu.se
013-28 58 47	013-28 18 15	013-28 18 51

Medgivande att delta i studie

- ☐ Efter att ha tagit del av informationen om studien ”interaktion mellan autonoma bussar och människa i trafik” som är del av ett större forskningsprojekt om olika ljuds funktioner i människa-robot-interaktion samtycker jag till att delta i studien.

20-03-23
Hannah Pelikan
hannah.pelikan@liu.se

Användning av inspelning

Jag ger mitt samtycke till att de video- och ljudinspelningar som görs av mig kan användas i följande syften:

Inspelningen används som **data till vetenskapliga arbeten**

☐ ja ☐ nej

Filmklipp får analyseras av **forskare som har anknytning till projektet**

☐ ja, oredigerade ☐ ja, anonymiserade ☐ nej

Transkriptioner (tal och övrigt beteende omskrivet till text) får användas i forskningssyfte, t.ex. i vetenskapliga texter i såväl pappers- som elektronisk form

☐ ja ☐ nej

Bilder får publiceras i forskningssyfte, t ex i vetenskapliga texter i pappers- såväl som elektronisk form

☐ ja, oredigerade ☐ ja, anonymiserade ☐ nej

Filmklipp får visas i vetenskapliga sammanhang och (presentation av) artiklar

☐ ja, oredigerade ☐ ja, anonymiserade ☐ nej

Filmklipp får visas i **undervisning** av studenter och forskare

☐ ja, oredigerade ☐ ja, anonymiserade ☐ nej

Filmklipp får visas i **allmän presentation**, t ex radio, teve, universitets webbplats, forskningskanaler på YouTube

☐ ja, oredigerade ☐ ja, anonymiserade ☐ nej

Dessa tillstånd gäller även **vetenskapliga arbeten som går bortom** det aktuella projektet, av interaktionsanalytisk eller etnografisk karaktär

☐ ja ☐ nej

Tack för dina svar! Notera att du får ändra dina val när som helst under eller omedelbart efter inspelningen. Du kan också när som helst välja att avbryta ditt deltagande.

Datum Namnteckning

☐ Forskarna får kontakta mig med eventuella följdfrågor.
☐ Jag önskar bli informerad av resultatet.

Om du medger att projektets forskare ska kunna komma i kontakt med dig vid senare tillfälle, vänligen lämna din kontaktinformation nedan (frivilligt).

Namn E-postadress eller telefonnummer

Articles

The articles associated with this thesis have been removed for copyright reasons. For more details about these see:

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At the Faculty of Arts and Sciences at Linköping University, research and doctoral studies are carried out within interdisciplinary research environments, often addressing broad problem areas. Linköping Studies in Arts and Sciences is the Faculty's own series for publishing research. This thesis comes from the graduate school in Language and Culture, Division of Language, Culture and Interaction at the Department of Culture and Society.

